# State of EDR in the US
# CDR Update Oct. 2019

## CDR Software Level 19.1.1




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# Introduction -Main Messages

1. EDR's improve recon **accuracy** & get to **causation** – use with **all other** available scene evidence –EDR is also PRIMARY EVIDENCE.
2. EDR strength is speed **PRIOR** TO *BRAKING* OR *LOSS OF CONTROL, &* DRIVER BEHAVIOR on gas and brake pedal. Modern ABS brakes TOOK AWAY your tire marks, only EDR can "give you back" speed prior to braking. Newer data elements give us EVEN MORE INSIGHT into precrash behavior.
3. Event Data Recording capability and data accessibility ***varies widely*** by **manufacturer, model *and* model year.** Coverage keeps growing with MORE AND BETTER DATA available with each new model released.

# Main Messages

4. NHTSA Part 563 EDR rule Sept 2012 was a game changer – required **Minimum** Data in US vehicles equipped with EDR
5. EDR not required but 99% of US 2019 vehicles have one.
6. EDR's will NEVER replace human beings in figuring out what really happened.
7. US Driver Privacy Act Dec 2015 requires owner permission or a warrant to access.
8. AEB (auto braking) + other driver assist tech= fewer crashes + more data
9. Budget for new CDR 900 interface $3800 and $2500 new cables for 2019

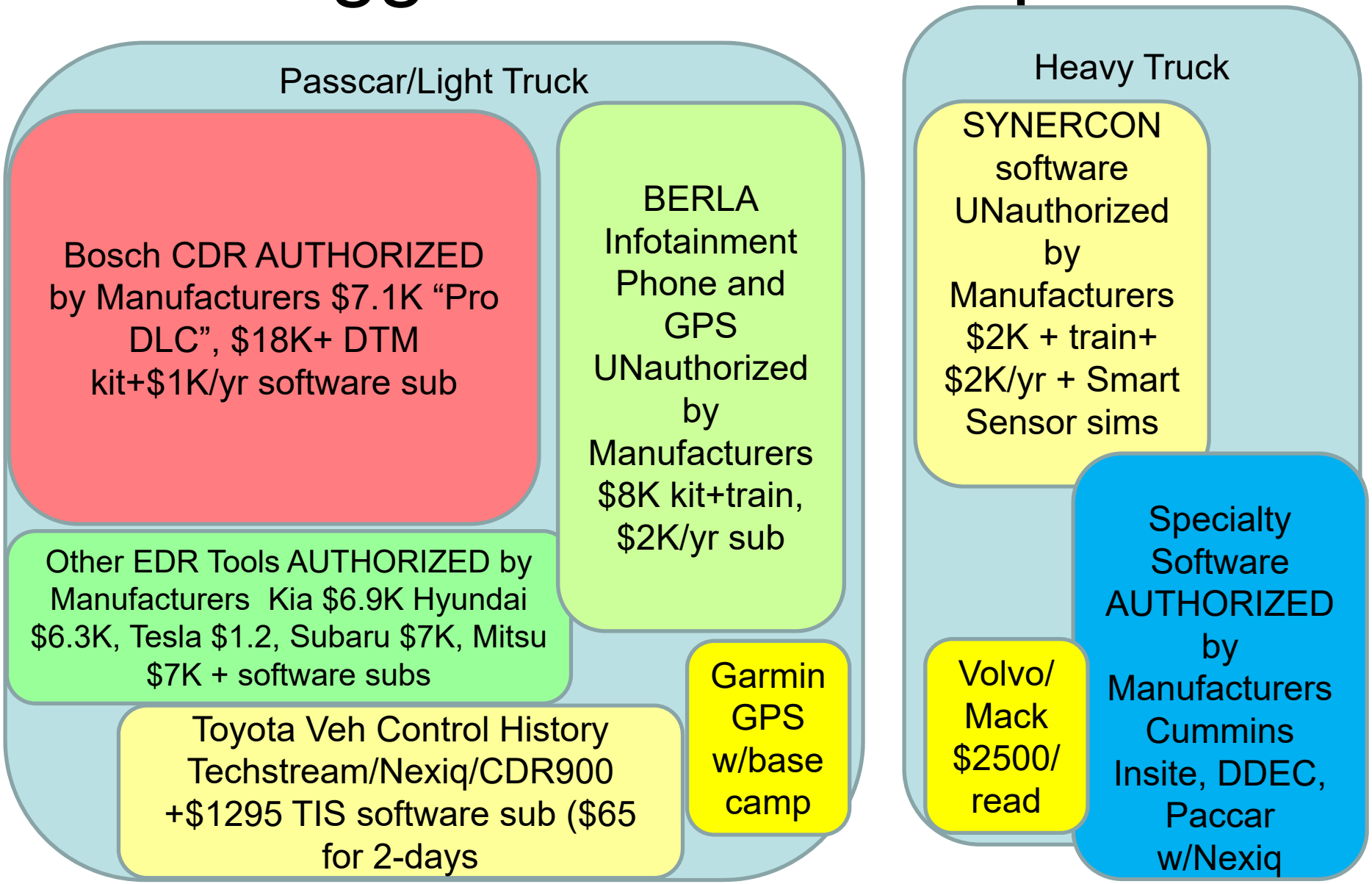
# Keeping EDR's in Perspective

- There are 305 million vehicles registered in the US. **59%** have an EDR with data accessible by the Bosch CDR system
- 87+% of 2019's are supported by Bosch.
- Another 12% have EDR accessible by other tools. **99%** have EDR.
- With *advanced* analysis techniques, we can get ***speed at impact*** for both vehicles even if only one has a 563 EDR.

# Scope of EDR Investigators

- 2000+ CDR Software subscriptions.
- Over half Law Enforcement 53%
- Private reconstructionists 41%
- Insurance 5+%, Govt, Fleets rest
- **New EU reg 3/2022** new design, 3/2024 old
- Europe and Australia – mostly Toyota, Volvo, FCA, Holden, 2018+ VW.
- New Korean reg Dec 2015.
- Japan 563 type reg, Tokyo police 50 kits.
- China making EDR reg now for 2021/23.

# The bigger vehicle data picture



# US Lt. Veh EDR & Tool Used

U.S. EDR PRESENCE AND TOOL BY MANUFACTURER

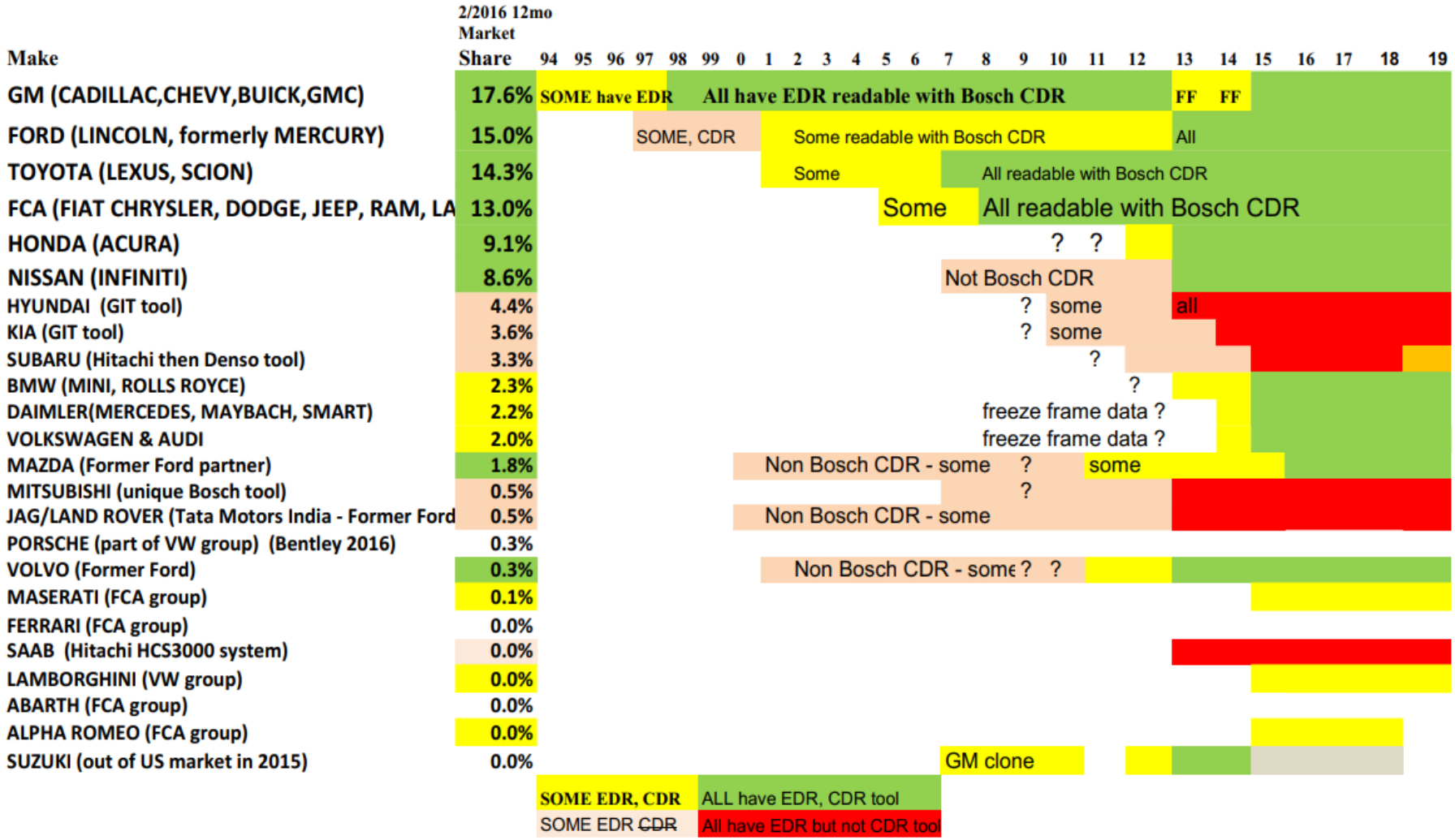
| Manufacturer | 12 months Sales end 10/16 US | 1Q 2018 Market Share | EDR First Known Use | Tool? |
|---|---|---|---|---|
| GM (Chevy, Buick, GMC, Cadillac) | 2,990,549 | 17.4% | 1994 | Bosch CDR |
| Ford (Ford, Lincoln) | 2,589,480 | 14.5% | 1997 | Bosch CDR 2001+ |
| Toyota (includes Lexus, Scion) | 2,436,580 | 13.9% | 2001 | Bosch CDR |
| Fiat Chrysler (includes Dodge, Ram, Jeep, Lancia | 2,292,735 | 12.6% | 2005 | Bosch CDR |
| Honda (includes Acura) | 1,620,875 | 8.8% | 2012 | Bosch CDR 2012+ |
| Nissan (includes Infinity) | 1,542,927 | 10.1% | 2006 | Bosch CDR 2013+ |
| Hyundai (Includes Genesis) | 773,708 | 3.6% | 2010 | GIT tool co. |
| Kia | 640,535 | 3.1% | 2010 | GIT tool co. |
| Subaru | 602,991 | 3.6% | 2012 | Denso +SSM, Bosch 2019+ |
| BMW (incudes Mini, Rolls Royce) | 373,096 | 2.1% | 2013 | Bosch CDR |
| Mercedes-Benz (includes Maybach, Smart) | 382,598 | 2.1% | 2014 | Bosch CDR |
| Volkswagen (Includes Bentley) | 310,885 | 2.0% | 2015 | Bosch CDR |
| Mazda | 299,004 | 2.0% | YES | Bosch CDR 2011+ |
| Audi | 206,999 | 1.2% | 2015 | Bosch CDR |
| Mitsubishi | 96,647 | 0.9% | 2006 | OTC/Bosch SPX** |
| Tata (Jaguar, Land Rover) | 100,733 | 0.8% | early | OTC/Bosch SPX |
| Geely (Volvo) | 80,984 | 0.5% | early | Bosch CDR 2011+ |
| Porsche | 53,138 | 0.3% | NO | N/A |
| Tesla | | 0.3% | 2012 | Tesla |
| Ferrari | 2,500 | 0.0% | NO | N/A |
| Industry Total | 17,394,464 | 100.0% | 99.5% | Bosch CDR 87.2% of new |
| | | | of new | Non CDR 12.3% of new |

** May be converting to Bosch CDR [redacted] 2019

Source: WSJ 3/19, Edmunds https://www.edmunds.com/industry-center/data/sales-by-make.html

# EDR Phase-in by Mfr

Data is incomplete for smaller mfrs with no public tool



# 2019 Recording Capability

**EDR Major Data Elements by Manufacturer** (All Mfrs with EDR have the required 15 data elements, all are not listed)

Information is taken from Bosch CDR sample files, Ruth cases, and 2012 MY practice projected into 2019 model year

| | Part 563 Table 1 | | | | | Part 563 Table 2 | | | | | | Beyond 563 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Mfr | Event Complete | Key Cycles | Long. ΔV | Speed/Brake/ Throttle or Accel | Multi Event | Lateral ΔV | RPM | Steering Angle | ACM Acceleration | ABS on/off | Roll Angle | ESC Long Accel | ESC Lat Accel | Yaw Rate | Tire Pressure | Panic Brake Assist | Wheel Speeds | Roll Rate | Brake Pressure | OTHER |
| GM | X | X | X | 5@0.5 TA | X | X | X | - | X, Y, Z 300@2 | X | | - | in roll | | LO | - | - | 1 @ 10ms | | 2016 ASCM w supercruise |
| Ford | X | X | X | 5@0.5 A | X | X | X | 5@0.1 | - | X | | 5@0.1 | | 5@0.1 | LO | | - | 5@0.1 | ** | 2017 AB12 precrash 0.2 sec |
| Chrysler | X | X | X | 5@0.1 TA | X | X | X | 5@0.1 | - | X | | - | | 5@0.1 | PSI | X | X | | 2018 | ACC set speed 2018, Avoidance |
| Toyota | X | X | X | 5@0.5 TA | X | Side | X | 5@0.5 | - | | Peak&atTRG | 5 | in roll | 5@0.5 | - | - | - | In Roll | 5@0.5 | Ped 2018, Veh history rpt |
| Honda | X | X | X | 5@0.5 TA | X | X | X | 5@0.5 | X, Y, Z 250@10 | X | -1+5@0.1 | - | | - | - | - | - | | | AEmBrk. Lanekeep |
| Nissan | X | X | X | 5@0.5 A | X | X | X | 5@0.5 | X, Y 250@10 | - | - | - | | - | - | - | ? | | | |
| Mazda | X | X | X | 5@0.5 T | X | X | X | | X, Y, Z 250@10 | - | | | | | | | - | | | |
| Volvo | X | X | X | 5@0.5T | X | 2016 XC90 | - | 2016 XC90 | X,Y,Z 250@10 | 2016 XC90 | -1+0.3@0.1 | | | | | | | | | |
| BMW | X | X | X | 5@0.5A | X | | X | 5@0.5 | X, Y 300@10ms | X | - | | | | | | | | | |
| Mercedes | X | X | X | 5@0.5A | X | 2017 E class adds | | | | | | | | | | | | | | |
| VW | X | X | X | 5@0.5A | X | X | | | | | | | | | | | | | | Time of Crash |
| Hyundai/Kia | X | X | X | 5@0.5 | X | X | X | 5@0.5 | X,Y,Z 250@10 | X | -1+5@0.1 | - | | - | - | - | - | | | |
| Subaru | X | X | X | 5@0.5A | X | X | X | 5@0.5 | | X | | | | | | | | -1 to+1.5 | | Eyesight |
| Mitsubishi | X | X | X | 5@0.5A | X | X | X | 5@0.5 | X, Y 250@10 | | -1+5@0.1 | | | | | | | | | |

# Bosch CDR Hardware Update

- *New interface module CDR-900 kit with new communication protocols needed in newer vehicles (*CAN-HD and Ethernet) Bigger 19-pin connector needed (vs 15).
- *New cables are being made with 19 pins, you will NEED a CDR 900 to read new manufacturers and models.*
- *The CDR900 is a derivative of a proven auto industry Bosch scan tool*
- *CDR 900 must be registered to your laptop (network device) before use.*



**CDR 900 Upgrade Kit**

**$3800**

# CDR900

- Initially for NEW vehicles
- Will eventually handle <u>most</u> legacy vehicles too…..

# CDR900 Upgrade Kit



# During Changeover

- *Many* “legacy” vehicles will EVENTUALLY become supported by the CDR-900. CDR-900 comes with an adapter to the old 15-pin cables so you don’t have to buy all new ones.
- Legacy coverage will take more than 1.5 years to implement, test, and release – so the CAN+ interface will still be needed through 2020.
- The help file clearly identifies which interface(s) will work with each vehicle

# New users need **both** CAN+ and CDR900



## Bosch CDR Pro Tool Kit

The CDR Pro Tool Kit is the heart and soul of the Bosch CDR Tool. This package contains all of the hardware required to perform a DLC/OBD retrieval of EDR data in Bosch supported vehicles. It includes both the CDR 900 and CANplus vehicle interface modules. This package also comes with a 1-year CDR software subscription. NOTE: this package does NOT contain any hardware to perform a "direct-to-module (D2M)" retrieval of EDR data. Please see the accessory cable/adapter packages below for D2M options.

MSRP: $7,100.00

plus shipping & handling

Add to Cart ›

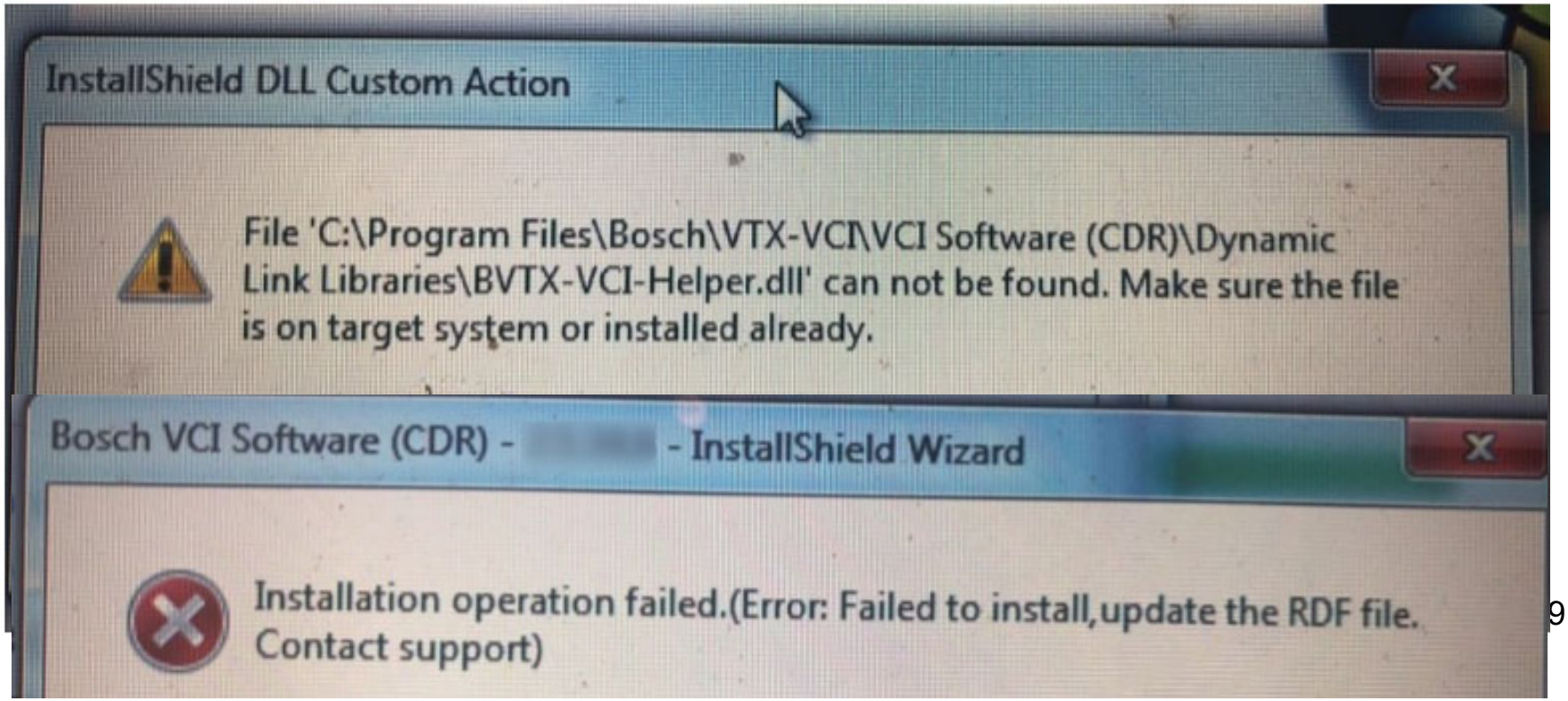
- CDR 900 vehicle interface module
- CDR 900 power cable
- CDR 900/CDR CAN*plus* power supply cable
- CDR 900/CDR CAN*plus* CDR cigarette power adapter
- CDR 900 DLC/OBD cable
- CDR 900 wireless dongles (2)
- CDR 900 extension cable
- CDR 900 legacy cable adapter
- CDR 900 USB cable
- CDR CAN*plus* vehicle interface module
- CDR CAN*plus* DLC/OBD cable
- CDR CAN*plus* GM DLC/OBD cable
- CDR CAN*plus* DLC/OBD adapters (3)
- CDR CAN*plus* PCM adapter
- CDR CAN*plus* USB to Serial port cable
- Deluxe repowering rig
- 1-year CDR software subscription
- Custom carrying case
- Coupon for **FREE online class**: How to Use the Bosch CDR Tool

# CDR 900 details

- CDR900 integrates power management (no external relays/adapters).
- CDR900 uses the existing 2.5 amp current limited power supply, but new ACM's may drive bigger supply later.
- Includes WIRELESS laptop to interface
- CDR900 does NOT integrate the current CDR500 Flex Ray adapter – still needed!
- CDR 900 firmware must be flashed initially

# Bosch TSB on 17.8

- Installation fails caused by not having the latest Windows service packs needed by CDR900 (you get install error even if you are not yet using CDR900)



# New Bugs being worked out

- OK and AR police report that they could not install 17.9 with their NetMotion VPN (Virtual Private Network) software used for police communications.
- They removed the VPN software and were then successful.
- CDR900 is designed to look like a network device, VPN's are designed to keep out "intruder" network devices

# HARDWARE CHECKLIST 19.1

| Part Number | Description | Released |
|---|---|---|
| 1699200596 | Cable, ID828 CDR Nissan ACM | 17.5 |
| 1699200618 | Cable, ID829 CDR FCA ACM | 17.5 |
| 1699200553 | Cable, ID826 CDR Audi ACM (requires CDR 500 Adapter) | 17.6 |
| 1699200646 | Cable, ID830 CDR Volvo ACM (requires CDR 500 Adapter) | 17.7 |
| 1699200630 | Kit, CDR 900 Upgrade, with Wireless | 17.8 |
| 1699200696 | Cable, ID831 CDR Mazda ACM | 17.8 |
| 1699200697 | Cable, ID832 CDR Daimler ACM | 17.8 |
| 1699200778 | Cable, ID835, CDR Subaru ACM 1 | 19.0 |
| 1699200779 | Cable, ID836, CDR Subaru ACM 2 | 19.0 |
| 1699200783 | Cable, ID837, CDR Mitsubishi ACM 1 | 19.0 |
| 1699200784 | Cable, ID838, CDR Mitsubishi ACM 2 | 19.0 |
| 1699200785 | Cable, ID839, CDR Mitsubishi ACM 3 | 19.0 |
| 1699200786 | Cable, ID840, CDR Mitsubishi ACM 4 | 19.0 |
| 1699200776 | Cable, ID833, CDR GM ASCM 1 | 19.1 |
| 1699200777 | Cable, ID834, CDR GM ASCM 2 | 19.1 |
| 1699200797 | Cable, ID842, CDR VW Group ACM 4 | 19.1 |
| 1699200800 | Cable, ID843, CDR Mazda ACM | 19.1 |
| 1699200838 | Cable, ID845, CDR GM ACM | 19.1 |

New 3-digit Part ID

Gen 4 Pop Up Hood

2019

software to use them not out yet

Available end Sept or early Oct

# New Back Powering Gear Needed

- Manufacturers are replacing mini and micro fuses with Micro 2 and Micro 3 fuses to pack more fuses into smaller spaces. The complete back power kit must now have several different yellow clips to fit all the different vehicles on the road. GM started it, Ford followed, more to come. Mini fits into sub mini, don't need sub mini.




- Buy General Technologies Corp GTC063 set of 6 (including 2 bigger ones you don't really need) at the site below for $19.95

https://www.amazon.com/General-Technologies-Corp-GTC063-Connector/dp/B0787JKMTH?ref_=bl_dp_s_web_9003703011

# EMERGING TRENDS

- More manufacturers are becoming concerned about cyber security, of hackers trying to take over vehicle control systems.
- Many are installing “gate-way” modules or features to protect ECU’s from hacking.
- For CDR DLC users, “back powering” may have to power ACM fuse AND cyber security (gateway) module fuse. Of course Direct to module is still an alternative.
- Recently needed on 2017 RAM 3500

# Recent CDR Software Updates

- 17.8 – July 31, 2018 – **CDR900** + more 19's
- 17.9 – Sept 18, 2018 – DLC GM active safety control module, Freightliner, EU privacy
- 17.10 Dec. 20, 2018 – 2018-19 VW/Audi Worldwide, more 2019's, some GM 2020's
- 18.0 Feb 20, 2019 **Subaru** 19 Forstr/Acsendr
- 18.0.1 & 2 – minor patches
- 19.0 June 6, 2019 **Mitsubishi** Outlanders
- 19.1 Sept. 20, 2019, 15+ Subaru Outback, GM ASCM reports, more 2020's
- 19.1.1 Patch for Mitsubishi Accel Pedal

# The Bosch CDR Story

- New CDR 900 interface – old interface lasted 13 years – technology has changed!
- New GM active safety control module gives access to semi-autonomous vehicle data
- VW/Audi going worldwide with EDR
- The covered vehicles on the road with CDR is 59% and climbing about 3%/yr. The total available including mfrs with non-CDR tools is much more!!
- Subaru new models in CDR, legacy to follow, Mitsubishi added June 2019.

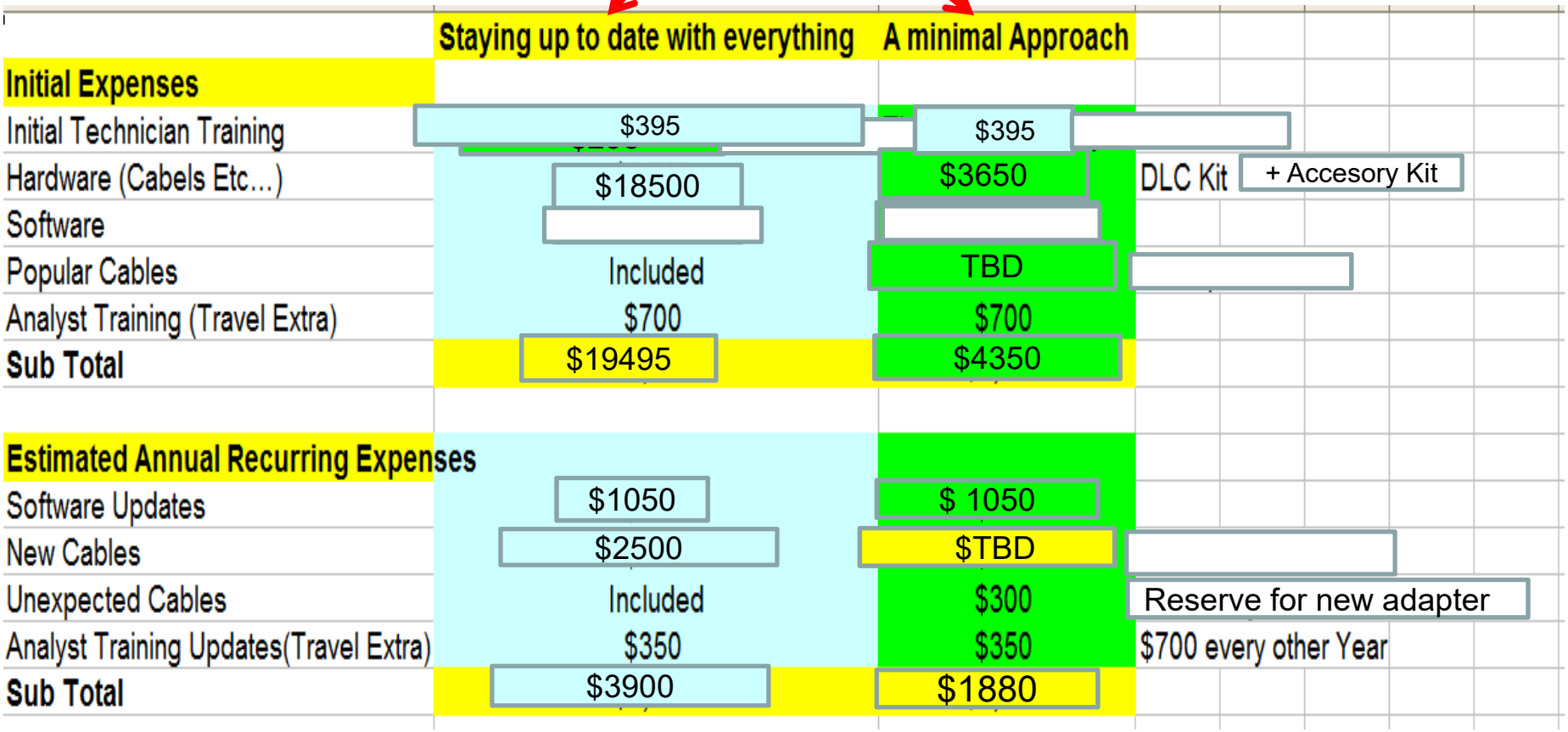
# What's Coming in next 12 mo?

- One software release per quarter (or more)
- Subaru, Mitsubishi, GM ASCM & new cables **only** be compatible with **CDR900**
- **15 new DTM cables in 2019  4 Subaru, 4 Mitsubishi, Daimler, VW, 2 GM, Mazda**
- **GM back model semi autonomous modules (not ACM's).**

# CDR Budgeting

## Two approaches to a budget

Add one time new CDR900
$3,800



| | Staying up to date with everything | A minimal Approach | |
|---|---|---|---|
| **Initial Expenses** | | | |
| Initial Technician Training | $395 | $395 | |
| Hardware (Cabels Etc...) | $18500 | $3650 | DLC Kit + Accesory Kit |
| Software | | | |
| Popular Cables | Included | TBD | |
| Analyst Training (Travel Extra) | $700 | $700 | |
| **Sub Total** | $19495 | $4350 | |
| | | | |
| **Estimated Annual Recurring Expenses** | | | |
| Software Updates | $1050 | $ 1050 | |
| New Cables | $2500 | $TBD | |
| Unexpected Cables | Included | $300 | Reserve for new adapter |
| Analyst Training Updates(Travel Extra) | $350 | $350 | $700 every other Year |
| **Sub Total** | $3900 | $1880 | |

# CDR/EDR Training

- Bosch Training Provider Page lists IPTM (Rick Ruth+), Crash Data Specialists (Brad Muir+), CSI (Rusty Haight), CDR-training.com (CSI tech 1&2 trainers) and Northwestern (thru CSI). SAE (Rick R) not listed, teaches analysis only.
- There is no “certification” – your qualifications depend on your training and experience
- IPTM has ONLINE technician training $395
- New online tech basic from Crash Data Group
- All trainers cover basics, but each has unique style and additional content -try them all!

# *Additional EDR training available*

- IPTM has a “Level 2” analyst class which goes into greater depth on how to use the law of sines and cosines to solve Velocity Vector triangles, offset collisions w/EMR, ground forces & new data elements.
- IPTM EDR Analyst 1 now ONLINE as of April 15, 2019.

# Analysis Update: Speed Data

- Accuracy: We typically apply +/-4% to speed data based on 1999 GM paper
- The NHTSA part 563 regulation specifies speed data MUST be accurate to within **+/-1 kph** (0.6 mph)
- This is for new as sold vehicles, may not include tire wear which can add up to 2% to reported speed.

# Analysis Update: Delta V Usage

## VEHICLE WITH EDR

- Inline w/departure

  **V1 = V3 - ΔVx**

- Angled with Departure

  **V1=V3cosβ - ΔVx**

**Note no assumptions about whether collision was central or vehicles reached a common velocity**

## VEHICLE WITHOUT EDR

- Inversely Proportional ΔV at centers of mass $\Delta V_1 = -\Delta V_2 \frac{W_2}{W_1}$
- Closing Speed Central
- $\text{Closing Speed} = \left[\frac{1}{1+e}\right]\left[|\Delta V_1| + |\Delta V_2|\right]$ w/common v
- Closing Speed Offset w/common v

  $\text{ClosingSpeed} = \left[\frac{1}{1+e}\right]\left[\frac{|\Delta V_1|}{\gamma_1} + \frac{|\Delta V_2|}{\gamma_2}\right]$
- Intersection inline approximation (new)
- V1/V3/ΔV Triangle Vel Vectors
- V1/V2/Closing Speed Triangle

  (relatively new)

# TRIANGULAR VELOCITY VECTOR METHOD

APPOACH VELOCITY = V1

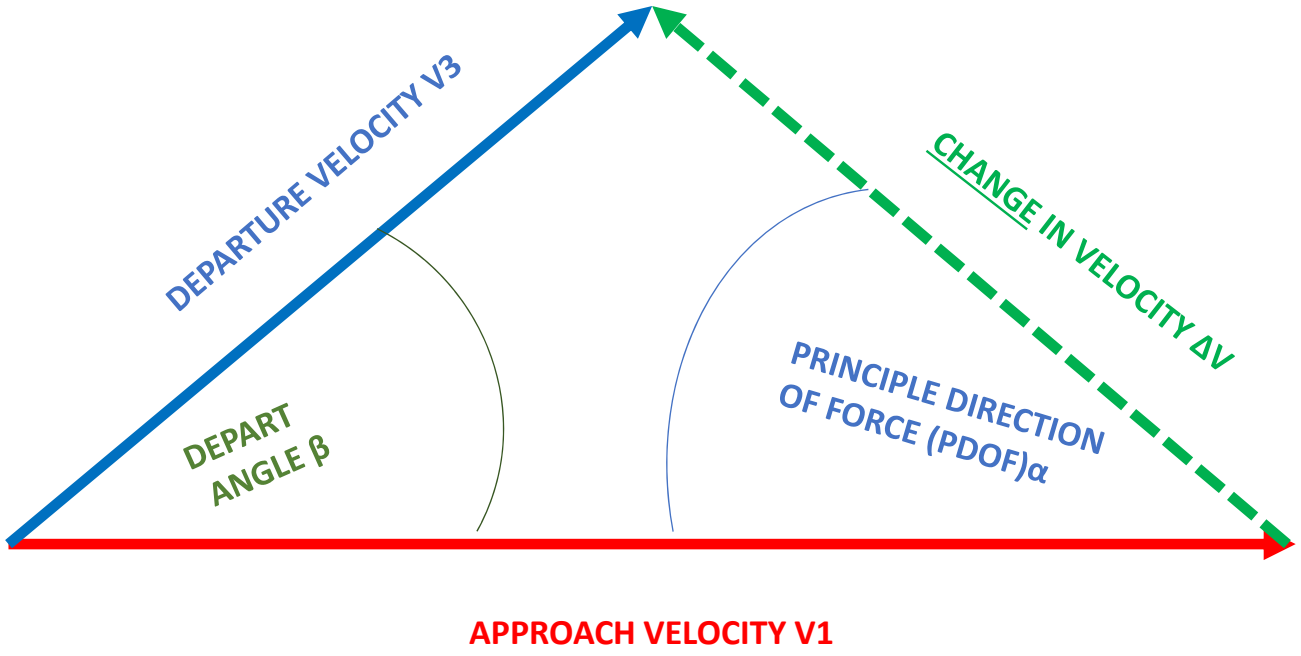
**DEPARTURE VELOCITY = V3**

**CHANGE IN VELOCITY = ΔV**

**DEPARTURE ANGLE = β**

**PDOF = α**

**SIDES AND OPPOSITE ANGLES (PAIRS) ARE SHOWN IN THE SAME COLOR**

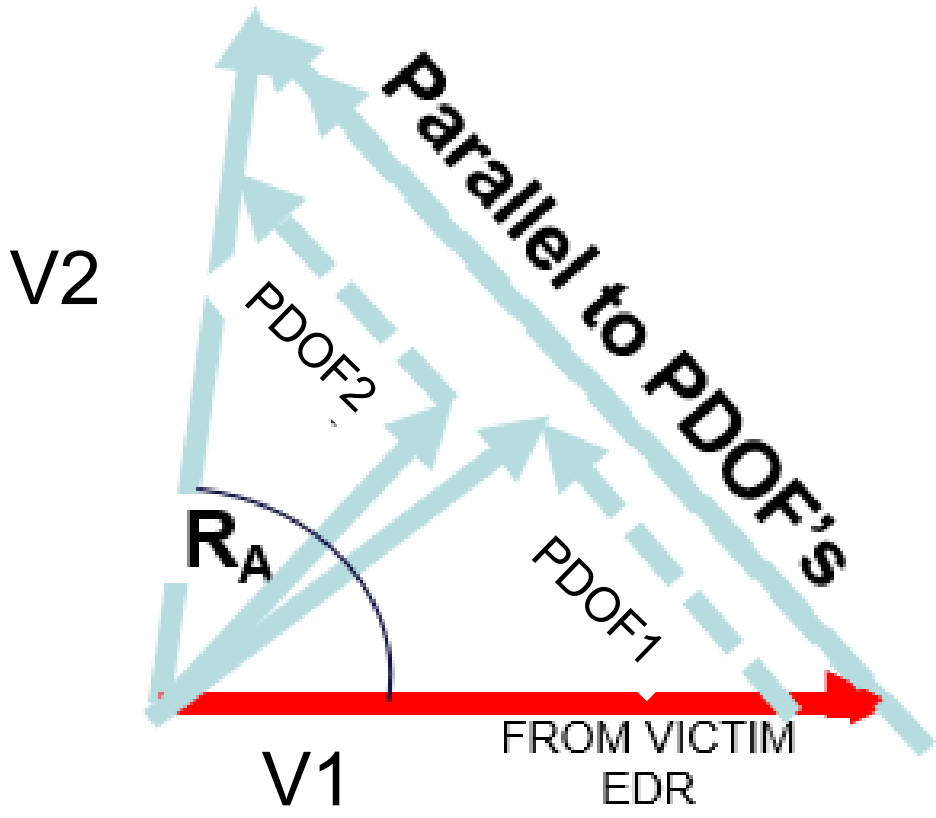


# PDOF Triangle -2nd Triangle

- **Must know relative approach $R_A$**
- **Get PDOF1 from Vic EDR**
- **PDOF2=180- $R_A$ -PDOF1**
- **ΔV2 = -ΔV1(W1/W2)**
- **Must have departure speed or angle (3rd piece) to solve**
- Use law of sines

Note small inner triangles have a parallel edge

Delta V's. Outer triangle is the PDOF triangle.



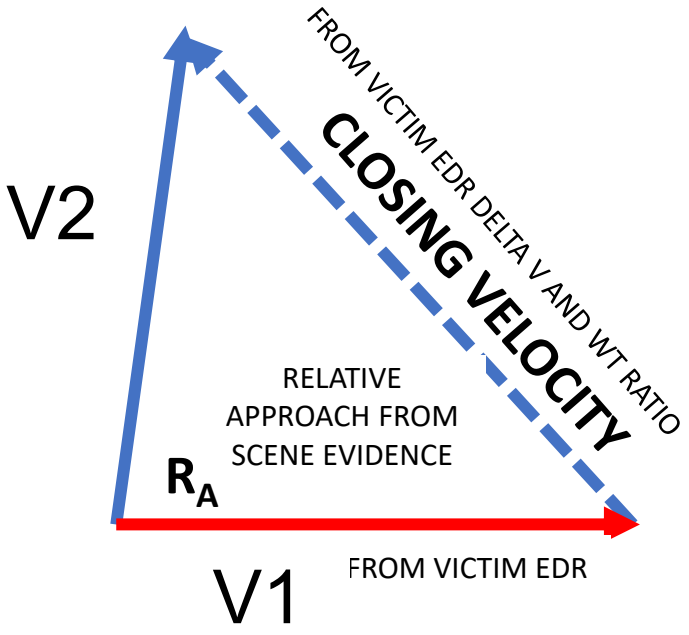
# Closing Speed Triangle

ANOTHER WAY TO FIND V1 AT IMPACT

- The definition closing speed is the difference between the V1 and V2 approach velocity vectors
- From **$R_A$, V1, and CS** find V2
- Use law of sines

Note angles in other corners

Are NOT the PDOF's



# Analysis Updates

- Closing speed adjustment for Offset collisions assumes ACM is on center of gravity.
- Additional adjustment may be required for ACM location away from CG.
- Look for pending publications from Bob Scurlock and Andy Rich

# Analysis Update: Delta V Data

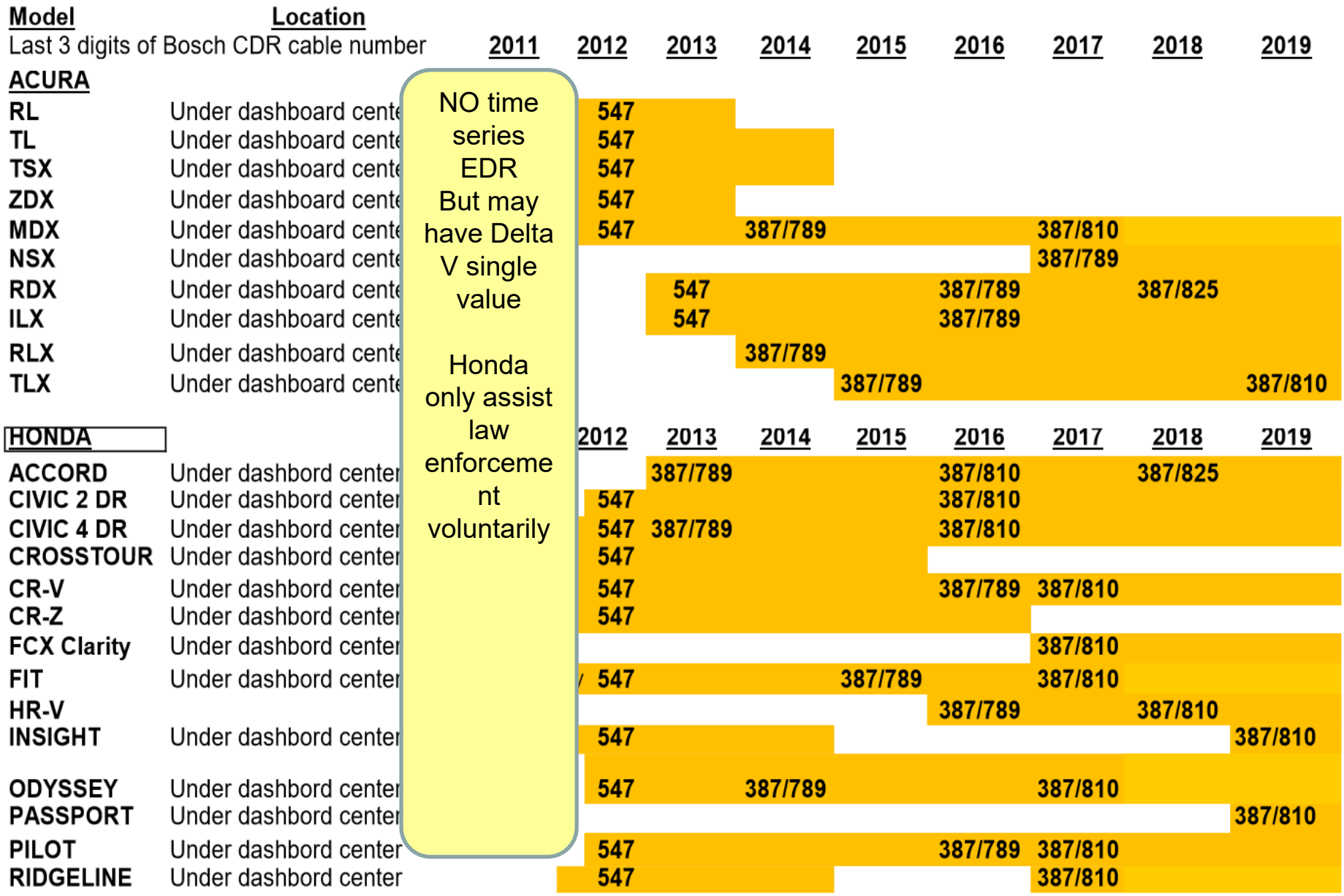
- **Ground Forces were typically ignored in reconstruction**
- **In small ΔV's (ped/mc) ground forces become significant**
- **If car locks up brakes before hitting ped, Long. ΔV of ped hit is overstated**
- **If MC t-bones car, Lat ΔV understated**
- **ΔV is multiplied by big weight ratio, Closing speed error gets bigger**

# NEW NHTSA CISS DATABASE

- https://crashviewer.nhtsa.dot.gov/CISS/SearchIndex
- In 2018 NHTSA loaded a sample of 125 2016 Crash Investigation Sampling System cases. 2,068 2017 cases were uploaded late Sept, 2019.
- Searchable by manufacturer, model year, model, EDR data available, driver assist technology available, etc
- Great source of info on what to expect in your vehicle's EDR, confirmation of EDR polarity, etc.

# Details by Manufacturer

# Honda/Acura CDR 17.10 Covr'g 40%



# Honda EDR Evolution

- One set of DL's thru 2015
  - Roll rate polarity varies by supplier – polarity added under graph
  - Accel data instantaneous
- 2nd set of data limitations for recent models
- 2016-17 adds "Honda sensing" & Cruise
- 2018 – 3rd Gen to include "ig switch status"
  - Airbag deploys even if ign off until car stops
  - Markets outside US/Japan do not get precrash data

# Honda Gen 2 EDR

- If equipped may have semi autonomous vehicle data elements

**Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1) - Table 2 of 3**

| Time Stamp (sec) | PCM Derived Accelerator Pedal Position, % full | Forward Collision Warning (Not Warning/ Warning) | Collision Mitigation Braking System (Not Engaged/ Engaged) | Collision Mitigation Braking System, Forward Collision Warning (On/Off) | Lane Departure Warning (Not Warning/ Warning) | Road Departure Mitigation (Not Engaged/ Engaged) | Road Departure Mitigation, Lane Departure Warning (On/Off) |
|---|---|---|---|---|---|---|---|
| -5.0 | 16 | Not warning | Not engaged | On | Not warning | Not engaged | Off |
| -4.5 | 8 | Not warning | Not engaged | On | Not warning | Not engaged | Off |
| -4.0 | 11 | Not warning | Not engaged | On | Not warning | Not engaged | Off |
| -3.5 | 13 | Not warning | Not engaged | On | Not warning | Not engaged | Off |
| -3.0 | 18 | Not warning | Not engaged | On | Not warning | Not engaged | Off |
| -2.5 | 18 | Not warning | Not engaged | On | Not warning | Not engaged | Off |
| -2.0 | 17 | Not warning | Not engaged | On | Not warning | Not engaged | Off |
| -1.5 | 0 | Warning | Engaged | On | Not warning | Not engaged | Off |
| -1.0 | 0 | Warning | Engaged | On | Not warning | Not engaged | Off |
| -0.5 | 0 | Warning | Engaged | On | Not warning | Not engaged | Off |
| 0.0 | 0 | Warning | Engaged | On | Not warning | Not engaged | Off |

# Honda Gen 2 Semi Auton cont'd

**Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1) - Table 3 of 3**

| Time Stamp (sec) | Adaptive Cruise Control (Not Engaged/ Engaged) | Adaptive Cruise Control (On/Off) | Lane Keeping Assist (Not Engaged/ Engaged) | Lane Keeping Assist (On/Off) | Cruise Control (Not Engaged/ Engaged) | Cruise Control (On/Off) |
|---|---|---|---|---|---|---|
| -5.0 | Not engaged | On | Not engaged | On | Not Engaged | On |
| -4.5 | Not engaged | On | Not engaged | On | Not Engaged | On |
| -4.0 | Not engaged | On | Not engaged | On | Not Engaged | On |
| -3.5 | Not engaged | On | Not engaged | On | Not Engaged | On |
| -3.0 | Not engaged | On | Not engaged | On | Not Engaged | On |
| -2.5 | Not engaged | On | Not engaged | On | Not Engaged | On |
| -2.0 | Not engaged | On | Not engaged | On | Not Engaged | On |
| -1.5 | Not engaged | On | Not engaged | On | Not Engaged | On |
| -1.0 | Not engaged | On | Not engaged | On | Not Engaged | On |
| -0.5 | Not engaged | On | Not engaged | On | Not Engaged | On |
| 0.0 | Not engaged | On | Not engaged | On | Not Engaged | On |

# *Reminders: UNIQUE* TO HONDA

- Most manufactures record the speed calculated in the powertrain control module as it is being sent **TO** the speedometer on the CAN bus.
- Honda records the speed being displayed **ON** the speedometer, after the PCM calculated speed has been dampened or “smoothed”.
- The last data point before impact may be delayed more than it is from other manufacturers.

# New Defense Allegation

- NHTSA has opened an “EA” investigation on frontal airbags not deploying in crashes due to electrical interference overloading an ASIC (chip) handling front sensor inputs in TRW supplied modules.
- Kia, Hyundai, and Chrysler have recalled ACM’s for SOME affected units. Toyota has 2 reported incidents. Mitsubishi and Honda also use TRW.
- Defense claims all TRW EDR’s 2010 to 2019 are bad. It’s bullshit.

# Chevrolet Recent Changes 86% 563

1 animation

| | | Gen 3 | Gen 3.5 - 3 Events | | Part 563 Intent | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Make | Model | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 |
| Chevrolet | Avalanche | 3320 ROS | 454 | 454 | 454 | 454 | * | | | | | | |
| Chevrolet | Aveo | 3320 | 3320 | 3320 | | 454 | * | | | | 454 | 454 | |
| Chevrolet | Beat (India) | | | | | | 454 | 454 | | | | Blazer 454 | |
| Chevrolet | Bolt EV | | | | | | | | | | 454 | 454 | |
| Chevrolet | Camaro | | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | |
| Chevrolet | Caprice | | | 452 | **454** | police | police | police | police | police | police | ? | |
| Chevrolet | Captiva | | 454 | 454 | 454 | 454 | * | Sport | Sport | | | | |
| Chevrolet | City Express | | | | | | | 780 | 780 | 780 | 780 | 780 | |
| Chevrolet | Cobalt | 3321 | 3321 | | | | | | | | | | |
| Chevrolet | Colorado | 3003 | 3003 | 3003 | 3003 | | | 454 | 454 | 454 | 454 | 454 | 454 |
| Chevrolet | Corvette | 3003 | 3003 | 3003 | 3003 | 3003 | 454 | 454 | 454 | 454 | 454 | 454 | |
| Chevrolet | Cruze | | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 |
| Chevrolet | Equinox | 3320 ROS | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 |
| Chevrolet | Express | 3320 ROS | 3320 ROS | 3320 ROS | 3320 ROS | 3320 ROS | 3320 ROS | 3320 ROS | 454 | 454 | 454 | 454 | |
| Chevrolet | HHR | 3321 | 3321 | 3321 | | | | | | | | | |
| Chevrolet | Impala | 3320 | 3320 | 3320 | **454** | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 |
| Chevrolet | Kodiak | 2829 | | | | | | | | | | | |
| Chevrolet | Malibu | 3321 | 3321 | 3321 | 3321 | 3321 (2012 | 454 | 454 | 454 | 454 | 454 | 454 | 454 |
| Chevrolet | Matiz | | | | 454 | | | | | | | | |
| Chevrolet | Orlando | | | | 454 | 454 | 454 | | | | | | |
| Chevrolet | Silverado | 3320 ROS | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 |
| Chevrolet | Silverado HD | | | | | | | | | | | 454 | |
| Chevrolet | Sonic | | | | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 |
| Chevrolet | Spark | | | | | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 |
| Chevrolet | SSR | | | | | | | SS 454 | 454 | 454 | | | |
| Chevrolet | Suburban | 3320 ROS | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | |
| Chevrolet | Tahoe | 3320 ROS | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | |
| Chevrolet | Tracker | | | | | | | | | | 454 | 454 | 454 |
| Chevrolet | TrailBlazer | 3293 | | | | | | | | | | | |
| Chevrolet | Traverse | 3320 ROS | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 |
| Chevrolet | Trax | | | | | 454 | 454 | 454 | 454 | 454 | 454 | 454 | 454 |
| Chevrolet | Uplander | 3293 w/AW9 | | | | | | | | | | | |
| Chevrolet | Volt | | | 454 | | | | | | | 454 | 454 | 454 |

older family

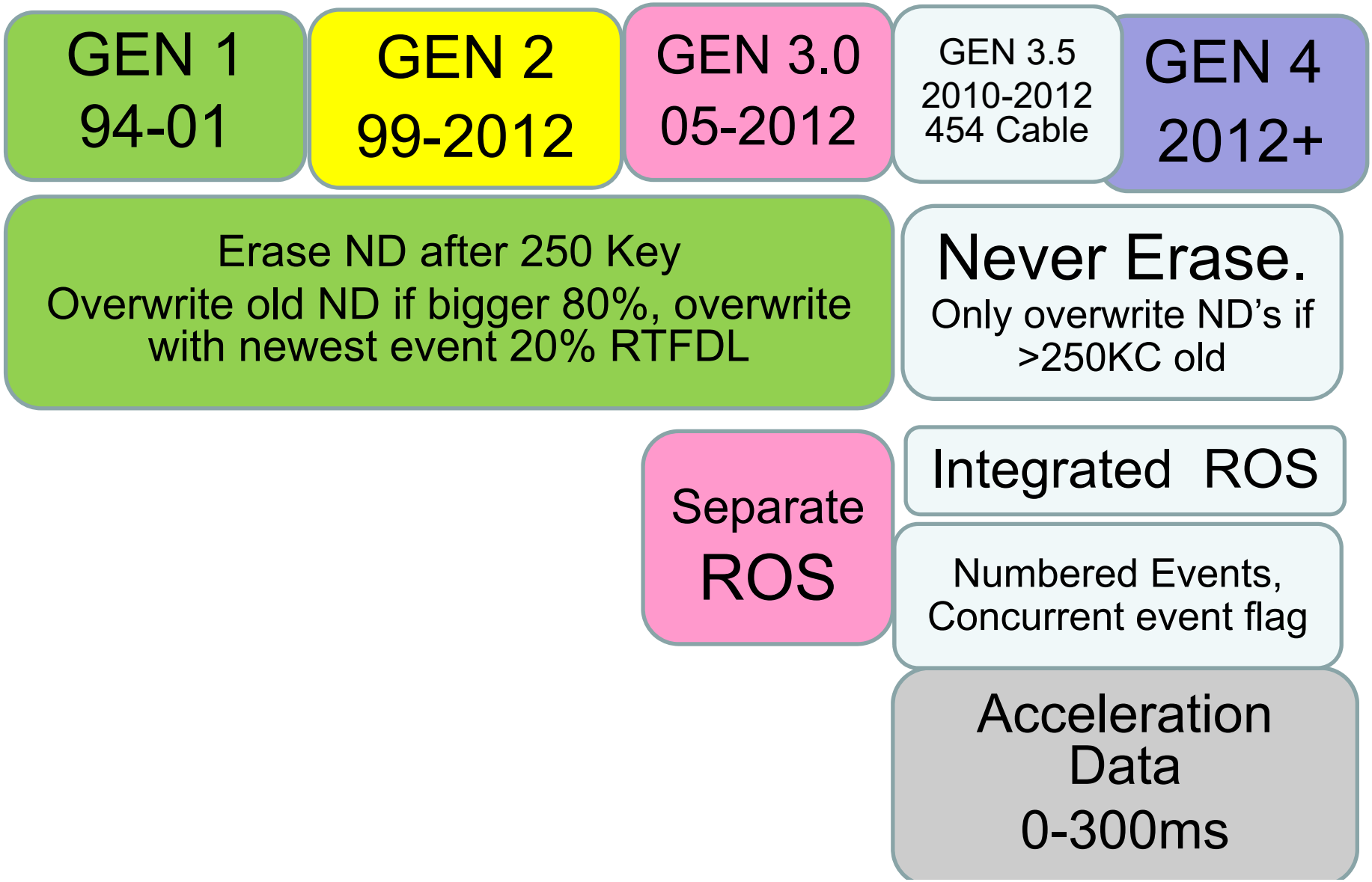
Freeze Frame

No Precrash in 2013 Express/Savanna

# GM EDR Evolution

| GEN 1<br>94-01 | GEN 2<br>99-2012 | GEN 3.0<br>05-2012 | GEN 3.5<br>2010-2012<br>454 Cable | GEN 4<br>2012+ |
|---|---|---|---|---|
| No Precrash | Precrash 5 @ 1 sec for BOTH events | Precrash 5 @ 0.5 or 1 sec<br>ONE SET of precrash | Precrash 5 @ 0.5 or 1 sec<br>For each event | Precrash 10 @ 0.5 sec |
| 2 EVENTS - FIRST D, One ND/ DLE, replace SDM after D | | | 3 Events<br>ND's or D's, replace after 3D's | |
| Record on frontal airbag algo wakeup | | Record on 5mph X or Y DV, | | |
| 300ms XDV from wakeup, must be neg | 150ms XDV, must be neg, 50ms before D 100 after | 300ms X+Y DV, Neg or Pos, some D's -70 to +220, some 300ms after wakeup | | |

# GM EDR Evolution #2



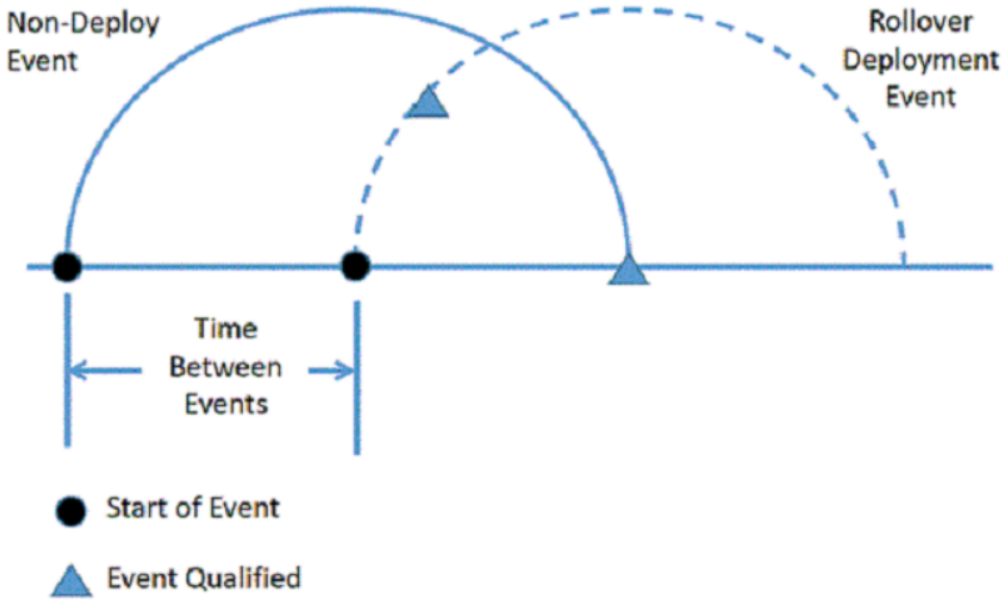
# GM Generation 3.5 Features

- 2010 454 cable SDM's do not require replacement after a D (until 3 D's).
- Events & deployments all numbered vs. labeled D/ND.
- Events may not be in order they started, they are in the order they qualified for recording (see graphic next page)
- **Align precrash data for all events to see relationship**
- Black plastic SDM housings
- Accelerometer max values listed in DL's

# GM Data Limitations

- Many updated in 17.9 – new timing graphic added for gen 3.5 and higher

Example of a Concurrent Event:
A Non-Deployment event begins. Before the Non-Deployment event is qualified, a Rollover Deployment event begins and is qualified. Sometime after the Rollover event is qualified, the Non-Deployment event is qualified. The Rollover event will be recorded in the first open record even though the Non-Deployment event enabled before the Rollover event. The Non-Deployment event will be recorded in the next open record. The "Concurrent Event Flag Set" parameter will indicate "Yes" for the Non-Deployment event. The "Time Between Events" parameter will indicate the time from the start of the Non-Deployment event to the start of the Rollover event.




| Event Record #1 | Event Record #2 |
|---|---|
| Event record Type = Rollover | Non-deployment |
| Concurrent Event Flag = No | Concurrent Event Flag = Yes |
| Time Between Events = N/A | Time Between Events = XX seconds |

# Other New GM News

- Some product liability data elements not relevant to reconstruction ***moved*** from CDR main report into **E**ngineering **T**ranslation **R**eports. Other elements do not display if not equipped (used to say "if equipped")
- Euro Pedestrian protection coming to US
- Some 2011-16 Autoliv had single stage airbags (severity status NO for stage 2)
- Known "Features" noted in Data Lims
- Occupant classification only child vs adult (no midsize or large adult classification)

# Important Aug 2016 Bug Fix

- **Some 2010-2012 GM (specifically all Autoliv SDM 10 & 11) interpreted *before* 17.0 misinterpreted some data**
- Brake status was reported ***backwards***
- Precrash was reported ***out of proper time order***
- **Fixed** in Software version 17.0
- As ALWAYS, be SURE to ***reinterpret data in the latest version of software!!!!!***

**Specific models affected by brake and time order:**

- **2010: SRX, Equinox, Terrain, Enclave, Captiva Sport, Traverse, Acadia, Outlook**
- **2011: SRX, Equinox, Terrain, Cruze, Volt, 9-4x, Enclave, Captiva Sport, Traverse, Acadia**
- **2012: Caprice, Impala, *Enclave, Captiva Sport, *Traverse, *Acadia**



- **Check closed cases????**

# Important Aug 2016 Bug Fix 2

- In SDM 10 and 11 Autoliv, plus a few additional Autoliv models, the Roll Rate polarity if interpreted before 17.0 may be the opposite of SAE polarity
- **Affected models include Autoliv SDM-10/10i, SDM-10P, SDM-11, SDM-11e, SDM-11P and SDM-30.**

# Important Dec 2017 Bug Fix

- SDM-30 modules read in 17.6 did not read properly.
- Modules must be re-read in 17.6.1 or higher
- 17.6 files attempting to be reopened will warn that the file is invalid (the whole file)
- Think about securing all ACMs in evidence…

# Important Bug Fix 4 & Other GM News

- One module family Conti SDM-30 comes out misinterpreted – data is scrambled
- Must send to Conti for manual readout and interp. (ask instructors for how)
- Foreign coverage – China since 99 (but only Gov't/NPS can use CDR), Australia/Korea/Thailand since 07
- South America not covered or spoofable

# GM Auto Emergency Braking

- AEB will prevent many crashes and reduce the severity of crashes still happening
- If there is still a crash, the AEB system may store up to 3 camera images:
  1. 4 sec before event
  2. At event
  3. 4 sec after event
- Not accessible with CDR – must be requested with proper legal authority from GM

# GM Active Safety Control Module

- First installed in 2016 Cadillac CT6
- Only with GM “Super Cruise” opt UKL
- Previously only downloadable by GM, released in CDR 17.9 to get raw data file for future interpretation.
- Interpretation capability added in 19.1
- REQUIRES CDR 900 interface!
- DLC at first, DTM cables 833/834 available “Late Sept/Oct”

# GM Active Safety Control Module Help File Callout

| Mkt | Year/Make | Model | Module | | Vehicle Interface | OBD/DLC Connect Adapter/Cable | D2M Connect Adapter/Cable | Module Location |
|---|---|---|---|---|---|---|---|---|
| 4 | 2018 Cadillac | ATS (note 6) | ACM | Data | CANplus | CANplus - F00K108287 | CANplus - F00K108454 | Center tunnel |
| 4 | 2018 Cadillac | CT6 (note 6) | ACM | Data | CANplus or CDR 900 | CANplus - F00K108287 CDR 900 - click here | CANplus - F00K108454 CDR 900 - click here (use listed cable above) | Center tunnel |
| | | | ASCM (with RPO UKL) | HEX Only | CDR 900 | CDR 900 - click here | CDR 900 - Cable# 834 | In the luggage compartment, right center, mounted to the right side in the battery well |

# GM Active Safety Module

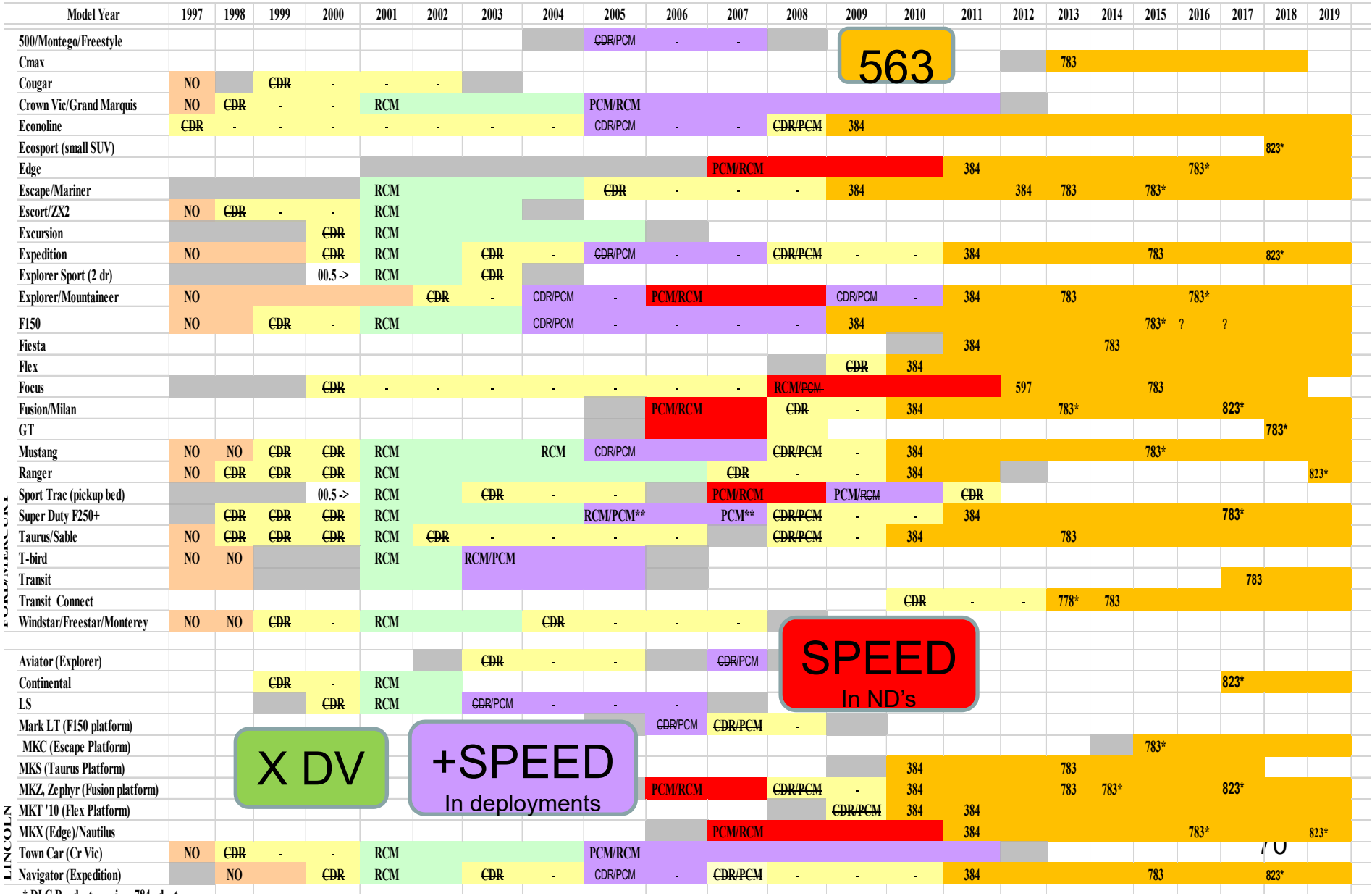



- Must be downloaded separately from ACM
- Icon will appear after VIN input for any vehicle that MIGHT have Super Cruise
- Module /data will only be present if option “UKL” (Super Cruise) has been purchased for the subject vehicle.

# GM Active Safety Module

- Data can be captured every 80 milliseconds and can record up to 50 records for a total of 4 seconds of data. The data recording window can be either 4 seconds before the end of an auto braking event or 4 seconds before an airbag deployment. ONE EVENT ONLY
- Data is recorded in non-volatile memory at the next proper power-down of the module. Data may **not** be stored if the battery power is lost before the power-down.

# Ford CDR 17.8 Coverage 65%



| | Model Year | 1997 | 1998 | 1999 | 2000 | 2001 | 2002 | 2003 | 2004 | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FORD/MERCURY | 500/Montego/Freestyle | | | | | | | | | ~~CDR~~/PCM | - | - | | | | | | | | | | | | |
| | Cmax | | | | | | | | | | | | | | | | | 783 | | | | | | |
| | Cougar | NO | | ~~CDR~~ | - | - | - | | | | | | | | | | | | | | | | | |
| | Crown Vic/Grand Marquis | NO | ~~CDR~~ | - | - | RCM | | | | PCM/RCM | | | | | | | | | | | | | | |
| | Econoline | ~~CDR~~ | - | - | - | - | - | - | - | ~~CDR~~/PCM | - | - | ~~CDR/PCM~~ | 384 | | | | | | | | | | |
| | Ecosport (small SUV) | | | | | | | | | | | | | | | | | | | | | | 823* | |
| | Edge | | | | | | | | | | | PCM/RCM | | | | 384 | | | | | 783* | | | |
| | Escape/Mariner | | | | | RCM | | | | ~~CDR~~ | - | - | - | 384 | | | 384 | 783 | | 783* | | | | |
| | Escort/ZX2 | NO | ~~CDR~~ | - | - | RCM | | | | | | | | | | | | | | | | | | |
| | Excursion | | | | ~~CDR~~ | RCM | | | | | | | | | | | | | | | | | | |
| | Expedition | NO | | | ~~CDR~~ | RCM | | ~~CDR~~ | - | ~~CDR~~/PCM | - | - | ~~CDR/PCM~~ | - | - | 384 | | | | 783 | | | 823* | |
| | Explorer Sport (2 dr) | | | | 00.5 -> | RCM | | ~~CDR~~ | | | | | | | | | | | | | | | | |
| | Explorer/Mountaineer | NO | | | | | ~~CDR~~ | - | ~~CDR~~/PCM | - | PCM/RCM | | | ~~CDR~~/PCM | - | 384 | | 783 | | | 783* | | | |
| | F150 | NO | | ~~CDR~~ | - | RCM | | | ~~CDR~~/PCM | - | - | - | - | 384 | | | | | | 783* | ? | ? | | | |
| | Fiesta | | | | | | | | | | | | | | | 384 | | | 783 | | | | | |
| | Flex | | | | | | | | | | | | | ~~CDR~~ | 384 | | | | | | | | | |
| | Focus | | | | ~~CDR~~ | - | - | - | - | - | - | - | RCM/~~PCM~~ | | | | 597 | | | 783 | | | | |
| | Fusion/Milan | | | | | | | | | | PCM/RCM | | ~~CDR~~ | - | 384 | | | 783* | | | | 823* | | |
| | GT | | | | | | | | | | | | | | | | | | | | | | 783* | |
| | Mustang | NO | NO | ~~CDR~~ | ~~CDR~~ | RCM | | | RCM | ~~CDR~~/PCM | | | ~~CDR/PCM~~ | - | 384 | | | | | 783* | | | | |
| | Ranger | NO | ~~CDR~~ | ~~CDR~~ | ~~CDR~~ | RCM | | | | | | ~~CDR~~ | - | - | 384 | | | | | | | | | 823* |
| | Sport Trac (pickup bed) | | | | 00.5 -> | RCM | | ~~CDR~~ | - | - | | PCM/RCM | | PCM/~~RCM~~ | | ~~CDR~~ | | | | | | | | |
| | Super Duty F250+ | | ~~CDR~~ | ~~CDR~~ | ~~CDR~~ | RCM | | | | RCM/PCM** | | PCM** | ~~CDR/PCM~~ | - | - | 384 | | | | | | 783* | | |
| | Taurus/Sable | NO | ~~CDR~~ | ~~CDR~~ | ~~CDR~~ | RCM | ~~CDR~~ | - | - | - | - | | ~~CDR/PCM~~ | - | 384 | | | 783 | | | | | | |
| | T-bird | NO | NO | | | RCM | | RCM/PCM | | | | | | | | | | | | | | | | |
| | Transit | | | | | | | | | | | | | | | | | | | | | 783 | | |
| | Transit Connect | | | | | | | | | | | | | | ~~CDR~~ | - | - | 778* | 783 | | | | | |
| | Windstar/Freestar/Monterey | NO | NO | ~~CDR~~ | - | RCM | | | ~~CDR~~ | - | - | - | | | | | | | | | | | | |
| LINCOLN | Aviator (Explorer) | | | | | | | ~~CDR~~ | - | - | | ~~CDR~~/PCM | | | | | | | | | | | | |
| | Continental | | | ~~CDR~~ | - | RCM | | | | | | | | | | | | | | | | 823* | | |
| | LS | | | | ~~CDR~~ | RCM | | ~~CDR~~/PCM | - | - | - | | | | | | | | | | | | | |
| | Mark LT (F150 platform) | | | | | | | | | | ~~CDR~~/PCM | ~~CDR/PCM~~ | - | | | | | | | | | | | |
| | MKC (Escape Platform) | | | | | | | | | | | | | | | | | | | 783* | | | | |
| | MKS (Taurus Platform) | | | | | | | | | | | | | | 384 | | | 783 | | | | | | |
| | MKZ, Zephyr (Fusion platform) | | | | | | | | | | PCM/RCM | | ~~CDR/PCM~~ | - | 384 | | | 783 | 783* | | | 823* | | |
| | MKT '10 (Flex Platform) | | | | | | | | | | | | | ~~CDR/PCM~~ | 384 | 384 | | | | | | | | |
| | MKX (Edge)/Nautilus | | | | | | | | | | | PCM/RCM | | | | 384 | | | | | 783* | | | 823* |
| | Town Car (Cr Vic) | NO | ~~CDR~~ | - | - | RCM | | | | PCM/RCM | | | | | | | | | | | | | | |
| | Navigator (Expedition) | | NO | | ~~CDR~~ | RCM | | ~~CDR~~ | - | ~~CDR~~/PCM | - | ~~CDR/PCM~~ | - | - | - | 384 | | | | 783 | | | 823* | |

563

SPEED
In ND's

X DV

+SPEED
In deployments

70

# Ford Part 563 EDR Subfamilies

Autoliv RC6
2009 Escape
384 Cable
+RC6_2011
(RC6X)

Bosch AB10
2012 Focus
597 Cable

Autoliv RC7
2013 Taurus
783 Cable
+RC7P upgrade

Bosch
AB10P
2015
Focus
783 Cable

RC7P

Bosch AB12
2017 Fusion
823 Cable

Precrash 5 sec @0.5 sec intervals, 250msec X&Y DV

2nd DV gra ph

Precrash
5 sec
@ *0.2* sec

Stability Control Sys Data 5 sec @0.1 sec intervals

RC6 last speed data
0.5 sec plus delay
RC6_2011 0.5 sec

Last data
point within
0.5 sec

Last Data
Point within
0.1 Sec

?

?

?

Incomplete Data Limitations

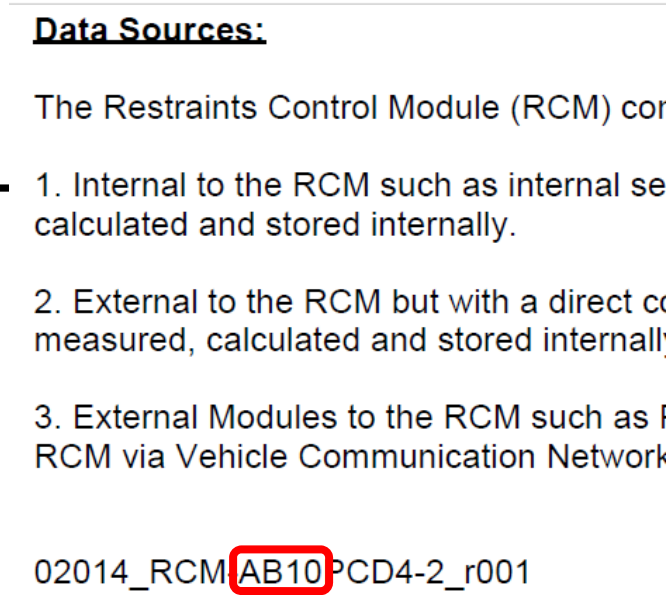
Fairly Complete Data Limitations

# Ford EDR evolution

- Autoliv/Veoneer RC5(06)-RC6(09)-RC6X(11)-RC7(13), RC7P
- Bosch AB9(05)-AB10(12)-AB10P, AB12(17)
- Chrysler had stability control yaw rate and steering earlier, but Ford upped the game in 2009 with RC6 longitudinal and lateral accel data at 0.1 second intervals pre-crash
- Fords have rich data, but are skimpy on data limitations until the 2013 Taurus Autoliv RC7.
- Critical info only found in training materials

# Identifying Ford EDR families

- At the end of the data limitations section, just below the last paragraph, on the left, Bosch puts what version of data limitations is being printed. For Ford, most data limitations by family.
- No difference between DL
  2009 RC6 and
  2011 RC6X.
- Open cdrx copy in notepad, see RC6_2011



# Polarity on Stability Control 09+

- The steering and yaw rate are opposite the normal right hand rule as documented in SAE 2014-01-0504.
- The paper reported lateral acceleration may be OK, but subsequent empirical data showed lateral acceleration is **also** backwards. In a hard right swerve, all 3 are negative.
- New RC7 and AB12 DL's show the polarity as stated above, all 3 backwards

# 09+ Ford Stability Control Data

**Pre-Crash Data -5 to 0 sec [10 samples/sec] (First Record)**

| Times (sec) | Stability Control Lateral Acceleration (g) | Stability Control Longitudinal Acceleration (g) | Stability Control Yaw Rate (deg/sec) |
|---|---|---|---|
| - 1.4 | 0.086 | -0.045 | 0.12 |
| - 1.3 | | -0.031 | 1.5 |
| - 1.2 | | -0.647 | -1.5 |
| - 1.1 | | -0.75 | -2.5 |
| - 1.0 | | -0.709 | -1.5 |
| - 0.9 | | -0.763 | -1.5 |
| - 0.8 | | -0.826 | -1.5 |
| - 0.7 | | -0.792 | -0.75 |
| - 0.6 | | -0.895 | 1.25 |
| - 0.5 | 0.014 | -0.83 | 1.87 |
| - 0.4 | 0.03 | -0.835 | 0.75 |
| - 0.3 | 0.022 | -0.607 | 0.37 |
| - 0.2 | 0.107 | -0.653 | 0.75 |
| - 0.1 | 0.05 | -0.823 | -3.37 |
| 0.0 | -0.004 | -0.811 | -3.75 |

Reaction

Integrate lat accel - get velocity – integrate again – get displacement

Drag Factor!

Integrate yaw – get **approach angle change**!! Take tangent of cumulative angle, multiply by speed, get **sideways movement** – how far out of lane vehicle went!!

# Ford Occupant Size Classification

not in data limitations

**Pre-Crash Data -1 sec (First Record)**

| | |
|---|---|
| Ignition cycle, crash | 359 |
| Frontal air bag warning lamp, on/off | OFF |
| Occupant size classification, front passenger (Child size Yes/No [Hex value]) | No [$08] |
| Safety belt status, driver | Driver Buckled |

- The Hex Value provides additional information that just child yes/no
- The Decoder is different for Ford's 2 main suppliers, Autoliv and Bosch
- Look at end of data limitations for module type – Autoliv=RC5, RC6, RC7; Bosch=AB9, AB10

02011_RCM-AB10P_r001

02007_RCM-RC6_r002

New Second Decoder for Bosch

| Autoliv | Bosch | |
|---|---|---|
| $01 | $00 | Empty |
| $02 | $01 | Child (clearly 55 lbs or less) |
| $04 | $05 | Indeterminate (gray zone somewhere between 55 to 110 lbs) |
| $08 | $02 | Small Adult (not a child and always registers for 110 lb 5th female adult) |
| $10 | $03 | Medium adult (not a 5th female and always registers for what is probably a 50th male adult of 170 lbs) |
| $20 | $04 | Large Adult (next bigger threshold than above, 95th male , 223 lbs) |

# Autoliv RC6 Anomalies (09-10)

- There are timing delays in reporting the 0.5 sec interval precrash data in the early Ford Autoliv RC6 modules (09 F150-Escape-Econoline and many 2010). Identify by lag between stability control longitudinal (braking) g's and brake on time and speed lost. Use Stability Control System long. accel for timing of accel release and brake on.
- Fixed by "RC6X", identifiable only by opening COPY of .cdrx in notepad, observe "RC6_2011" in header (and by agreement between stability control and 0.5 sec interval precrash data).

# 2012-2015 AB10 Anomaly

- The “Bosch AB10” ACM found in 2012+ Focus and 2015 Escape and other vehicles **produced before Nov. 2015**, which have a **second event**, incorrectly label pre-crash **time** to impact in the second event.
- Time between events is accurate – use it
- Second event precrash (both speed and stability control sections) include extra points at irregular intervals labeled as if they occurred at regular intervals.

# AB10 Anomaly cont'd

- If you can see the first event in the second event's data, it will lead you to think the time between events is longer than it really is.
- The data in the second event did occur, just not at the time it said it did.
- The second event last precrash speed is still within 0.5 seconds of the second event.
- There is no software fix for EDR's produced before Nov. 2015. Fix implemented 11/2015.
- No mention in DL's yet.

# Evolution: Getting Even Better

- RC-7 module that has larger RAM memory and buffers speed data every 0.1 seconds.
- Takes the most recent value and labels it “0.0”. Values at 0.5 sec intervals are picked from buffer relative to 0.0.
- Data point are within 0.1 seconds of what they are labeled
- Good set of data limitations
- Some driver assistance data elements
- Global event timer was 10X, getting fixed

# Bosch AB10P Unique Info

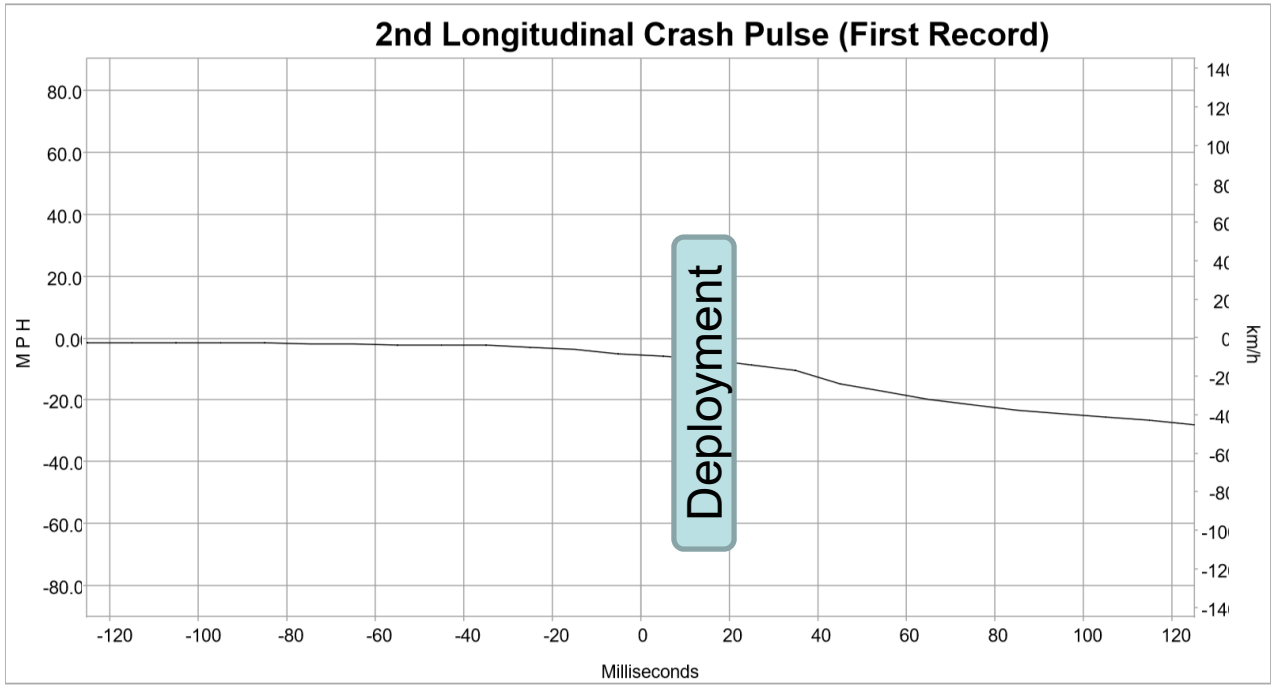
- Longitudinal Accel Stability Control Info is positive for slowing down, negative for speeding up – *opposite of what data limitations say*.
- Lateral Accel data is positive right, negative left – different than the other Ford modules.
- Steering and Yaw remain positive left and are identified as such in data limitations.

# Bosch AB12 ACM - 2017 MKZ

- Precrash data ***every 0.2 seconds*** instead of every 0.5 seconds
- Stability control data steering/yaw are on a different page than lat/long accel are (need to cut/paste columns to use current analysis template)
- Some Driver Assistance data elements

# New RC7P Feature- 2017 F250

- Ford has added a second graph to capture the deployment decision +/-125ms. This is important in hit-curb-then-hit-tree cases where the deployment was previously off the end of the mandated 250ms after wakeup graph



# Steering Angle Glitch

- Ford stores one value for steering relative to original zero position set at factory
- If vehicle steering is serviced, original zero may be lost and system learns a new one from average of all new driving.
- EDR may write down ORIGINAL position relative to zero – so a FEW models serviced may appear to have WRONG value for zero.
- No warning in DL's about this yet

# For Ford EDR **not supported** by Bosch CDR
# See the document on my website
# "Unsupported Fords"

Many requests can go directly to Ford's suppliers.

Other requests for assistance to Ford legal:

Email to EDRFORD@FORD.COM

(this is an address for a Bowman and Brooke outside counsel that handles these requests for Ford)

Contact Rick Ruth for additional information prior to making request

As of this writing, Veoneer (former Autoliv) charges $2000 to all, Bosch and Conti are no charge to law enforcement but $1500 to private parties.

Veoneer now says it cannot download modules >15 years old

**Recent 05 Freestar RCM sent to Takata was returned with Hex Data only and no guide to interpretation (no charge, no value!)**

Available on my website

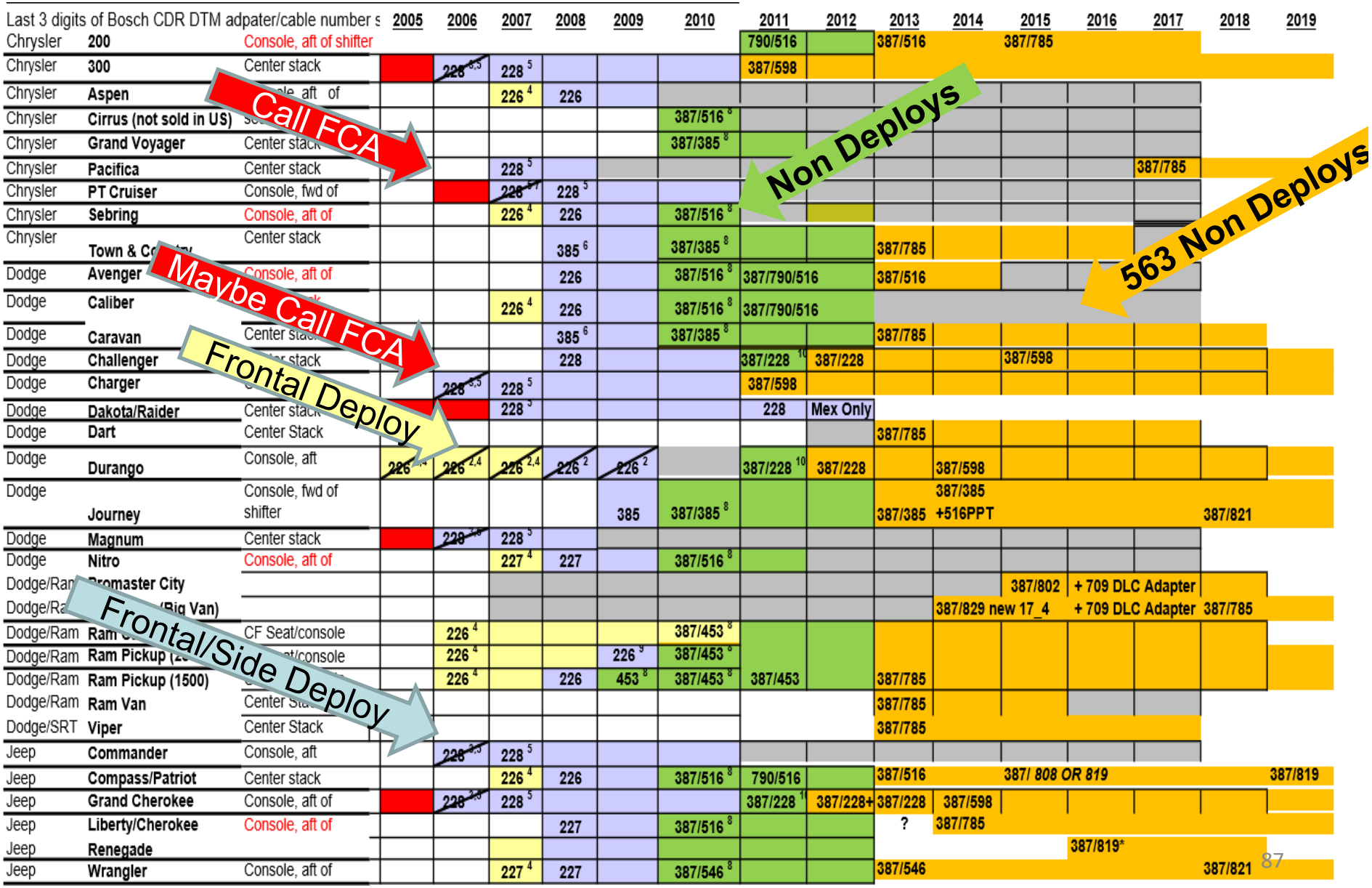
# Getting EDR Data from unsupported Fords

Richard R. Ruth, P.E.

(this copy has been updated 9-10-2018, after the original presentation at the Jan 2016 CDR Summit)



# FCA 18.0 CDR Coverage 60%




| Last 3 digits of Bosch CDR DTM adpater/cable number s | | | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Chrysler | 200 | Console, aft of shifter | | | | | | | 790/516 | | 387/516 | | 387/785 | | | | |
| Chrysler | 300 | Center stack | | ~~228~~ 3,5 | 228 5 | | | | 387/598 | | | | | | | | |
| Chrysler | Aspen | [illegible] aft of | | | 226 4 | 226 | | | | | | | | | | | |
| Chrysler | Cirrus (not sold in US) | [illegible] | | | | | | 387/516 8 | | | | | | | | | |
| Chrysler | Grand Voyager | Center stack | | | | | | 387/385 8 | | | | | | | | | |
| Chrysler | Pacifica | Center stack | | | 228 5 | | | | | | | | | | 387/785 | | |
| Chrysler | PT Cruiser | Console, fwd of | | | ~~228~~ 3,7 | 228 5 | | | | | | | | | | | |
| Chrysler | Sebring | Console, aft of | | | 226 4 | 226 | | 387/516 8 | | | | | | | | | |
| Chrysler | Town & Country | Center stack | | | | 385 6 | | 387/385 8 | | | 387/785 | | | | | | |
| Dodge | Avenger | Console, aft of | | | | 226 | | 387/516 8 | 387/790/516 | | 387/516 | | | | | | |
| Dodge | Caliber | [illegible] | | | 226 4 | 226 | | 387/516 8 | 387/790/516 | | | | | | | | |
| Dodge | Caravan | Center stack | | | | 385 6 | | 387/385 8 | | | 387/785 | | | | | | |
| Dodge | Challenger | [illegible] stack | | | | 228 | | | 387/228 10 | 387/228 | | | 387/598 | | | | |
| Dodge | Charger | [illegible] | | ~~228~~ 3,5 | 228 5 | | | | 387/598 | | | | | | | | |
| Dodge | Dakota/Raider | Center stack | | | 228 3 | | | | 228 | Mex Only | | | | | | | |
| Dodge | Dart | Center Stack | | | | | | | | | 387/785 | | | | | | |
| Dodge | Durango | Console, aft | ~~226~~ 1,4 | ~~226~~ 2,4 | ~~226~~ 2,4 | ~~226~~ 2 | ~~226~~ 2 | | 387/228 10 | 387/228 | | 387/598 | | | | | |
| Dodge | Journey | Console, fwd of shifter | | | | | 385 | 387/385 8 | | | 387/385 | 387/385 +516PPT | | | | 387/821 | |
| Dodge | Magnum | Center stack | | ~~228~~ 3,5 | 228 5 | | | | | | | | | | | | |
| Dodge | Nitro | Console, aft of | | | 227 4 | 227 | | 387/516 8 | | | | | | | | | |
| Dodge/Ram | Promaster City | | | | | | | | | | | | 387/802 | + 709 DLC Adapter | | | |
| Dodge/Ram | (Big Van) | | | | | | | | | | | 387/829 new 17_4 | | + 709 DLC Adapter | | 387/785 | |
| Dodge/Ram | Ram | CF Seat/console | | 226 4 | | | | 387/453 8 | | | | | | | | | |
| Dodge/Ram | Ram Pickup | [illegible]t/console | | 226 4 | | | 226 9 | 387/453 9 | | | | | | | | | |
| Dodge/Ram | Ram Pickup (1500) | | | 226 4 | | 226 | 453 8 | 387/453 8 | 387/453 | | 387/785 | | | | | | |
| Dodge/Ram | Ram Van | Center Stack | | | | | | | | | 387/785 | | | | | | |
| Dodge/SRT | Viper | Center Stack | | | | | | | | | 387/785 | | | | | | |
| Jeep | Commander | Console, aft | | ~~228~~ 3,5 | 228 5 | | | | | | | | | | | | |
| Jeep | Compass/Patriot | Center stack | | | 226 4 | 226 | | 387/516 8 | 790/516 | | 387/516 | | 387/ *808 OR 819* | | | | 387/819 |
| Jeep | Grand Cherokee | Console, aft of | | ~~228~~ 3,5 | 228 5 | | | | 387/228 11 | 387/228+ | 387/228 | 387/598 | | | | | |
| Jeep | Liberty/Cherokee | Console, aft of | | | | 227 | | 387/516 8 | | | ? | 387/785 | | | | | |
| Jeep | Renegade | | | | | | | | | | | | | 387/819* | | | |
| Jeep | Wrangler | Console, aft of | | | 227 4 | 227 | | 387/546 8 | | | 387/546 | | | | | 387/821 | |

# Chrysler Recent Changes

- **Added Full Size Promaster Van Coverage for 2014-2017 (Sprinter?)**
- **Fixed issue where 100% accel pedal and throttle reported at 77% in 17.7.1**
- **Tire Pressure data at 0.25 sec intervals instead of 0.1 sec to make room for other data to be recorded**
- **Four wheel speeds *in kph* instead of rpm – see non driven vs driven wheels**
- **Brake master cylinder pressure added to some 2018+ models**
- **Gear Position added (changes at end of shift)**

# Output New Look –

Removed multiple brake elements, added ABS/SC, moved strg to P1, hi res

**Pre-Crash Data [10 samples/sec] (Most Recent Event - table 1 of 2)**

(the most recent sampled values are recorded prior to the event)

| Time Stamp (sec) | Pre-Crash Recorder Status | Speed, Vehicle Indicated (MPH [km/h]) | Accelerator Pedal, % Full | Service Brake | Engine RPM | ABS Activity | Stability Control | Steering Input (deg) |
|---|---|---|---|---|---|---|---|---|
| -5.0 | Complete | 96 [155] | 0.00 | Off | 3,878 | No | On | 1 |
| -4.9 | Complete | 96 [154] | 0.00 | Off | 3,835 | No | On | -1 |
| -4.8 | Complete | 96 [154] | 0.00 | Off | 3,726 | No | On | -5 |
| -4.7 | Complete | 96 [154] | 0.00 | Off | 3,621 | No | On | -10 |
| -4.6 | Complete | 96 [154] | 0.00 | Off | 3,489 | No | On | -18 |
| -4.5 | Complete | 95 [154] | 0.00 | Off | 3,344 | No | On | -21 |
| -4.4 | Complete | 95 [153] | 0.00 | Off | 3,210 | No | On | -20 |
| -4.3 | Complete | 95 [153] | 0.00 | Off | 3,109 | No | On | -19 |
| -4.2 | Complete | 95 [153] | 0.00 | Off | 3,068 | No | On | -19 |
| -4.1 | Complete | 95 [152] | 0.00 | Off | 3,108 | No | On | -18 |
| -4.0 | Complete | 95 [152] | 0.00 | Off | 3,079 | No | On | -18 |

# Wheel RPM replaced by speed

See driven vs non driven, left vs right, ABS working…

**Pre-Crash Data [10 samples/sec] (Most Recent Event - table 2 of 2)**

(the most recent sampled values are recorded prior to the event)

| Time Stamp (sec) | ABS MIL | Yaw Rate (deg/sec) | Wheel Speed LF (km/h) | Wheel Speed RF (km/h) | Wheel Speed LR (km/h) | Wheel Speed RR (km/h) |
|---|---|---|---|---|---|---|
| -5.0 | Off | 0.40 | 154.37 | 154.52 | 154.55 | 154.52 |
| -4.9 | Off | -0.48 | 154.28 | 154.16 | 154.25 | 154.25 |
| -4.8 | Off | -1.76 | 154.40 | 154.09 | 153.88 | 153.77 |
| -4.7 | Off | -2.80 | 154.28 | 153.95 | 153.88 | 153.67 |
| -4.6 | Off | -4.24 | 154.01 | 153.64 | 153.85 | 153.55 |
| -4.5 | Off | -6.80 | 153.98 | 153.16 | 153.80 | 153.19 |
| -4.4 | Off | -7.12 | 153.80 | 152.88 | 153.70 | 152.80 |
| -4.3 | Off | -5.84 | 153.49 | 152.67 | 153.37 | 152.62 |
| -4.2 | Off | -5.12 | 153.13 | 152.19 | 153.25 | 152.38 |
| -4.1 | Off | -4.64 | 152.80 | 152.04 | 152.91 | 151.91 |
| -4.0 | Off | -4.64 | 152.49 | 151.73 | 152.46 | 151.65 |
| -3.9 | Off | -4.00 | 152.13 | 151.55 | 152.13 | 151.31 |
| -3.8 | Off | -3.68 | 151.83 | 151.25 | 151.95 | 151.13 |

# Current Gear & More @ 0.25 sec

**Pre-Crash Data [4 samples/sec] (Most Recent Event - table 1 of 2)**

(the most recent sampled values are recorded prior to the event)

NEW

Changes at END of shift

| Time Stamp (sec) | Pre-Crash Recorder Status | Raw Manifold Pressure (kPa) | PCM MIL | ETC Lamp | Current Gear | Reverse Gear Status |
|---|---|---|---|---|---|---|
| -5.00 | Complete | 20.00 | Off | Off | D5 | Not_Reverse |
| -4.75 | Complete | 20.00 | Off | Off | D5 | Not_Reverse |
| -4.50 | Complete | 21.60 | Off | Off | D5 | Not_Reverse |
| -4.25 | Complete | 23.20 | Off | Off | D5 | Not_Reverse |
| -4.00 | Complete | 23.20 | Off | Off | D6 | Not_Reverse |
| -3.75 | Complete | 23.20 | Off | Off | D6 | Not_Reverse |
| -3.50 | Complete | 23.20 | Off | Off | D6 | Not_Reverse |
| -3.25 | Complete | 23.20 | Off | Off | D6 | Not_Reverse |
| -3.00 | Complete | 22.40 | Off | Off | D6 | Not_Reverse |
| -2.75 | Complete | 21.60 | Off | Off | D6 | Not_Reverse |
| -2.50 | Complete | 21.60 | Off | Off | D7 | Not_Reverse |
| -2.25 | Complete | 21.60 | Off | Off | D7 | Not_Reverse |
| -2.00 | Complete | 21.60 | Off | Off | D7 | Not_Reverse |
| -1.75 | Complete | 21.60 | Off | Off | D7 | Not_Reverse |

10.63 in

# Tire pressure @ 0.25 sec intervals

**Pre-Crash Data [4 samples/sec] (Most Recent Event - table 2 of 2)**

(the most recent sampled values are recorded prior to the event)

NEW

| Time Stamp (sec) | Tire Pressure Position | Tire Pressure Status | Tire Pressure Value (PSI) | Cruise Control Lamp Status | Cruise Control Engaged Status | ACC On/Off | ACC Set Speed (MPH [km/h]) |
|---|---|---|---|---|---|---|---|
| -5.00 | LHF | Normal | 34 | Off | Not_Engaged | Off | 0 [0.0] |
| -4.75 | RHR | Normal | 32 | Off | Not_Engaged | Off | 0 [0.0] |
| -4.50 | RHR | Normal | 32 | Off | Not_Engaged | Off | 0 [0.0] |
| -4.25 | RHR | Normal | 32 | Off | Not_Engaged | Off | 0 [0.0] |
| -4.00 | RHR | Normal | 32 | Off | Not_Engaged | Off | 0 [0.0] |
| -3.75 | LHR | Normal | 33 | Off | Not_Engaged | Off | 0 [0.0] |
| -3.50 | LHR | Normal | 33 | Off | Not_Engaged | Off | 0 [0.0] |
| -3.25 | LHR | Normal | 33 | Off | Not_Engaged | Off | 0 [0.0] |
| -3.00 | LHR | Normal | 33 | Off | Not_Engaged | Off | 0 [0.0] |
| -2.75 | RHF | Normal | 33 | Off | Not_Engaged | Off | 0 [0.0] |
| -2.50 | RHF | Normal | 33 | Off | Not_Engaged | Off | 0 [0.0] |

# Integrating FCA Yaw Rate

From 08 charger case

| | |
|---|---|
| - 3 deg/sec * 0.1 sec = | -0.3 degrees |
| -11 deg/sec * 0.1 sec = | -1.1 degrees |
| -18 deg/sec * 0.1 sec = | -1.8 degrees |
| -15 deg/sec * 0.1 sec = | -1.5 degrees |
| - 9 deg/sec * 0.1 sec = | -0.9 degrees |
| - 4 deg/sec * 0.1 sec = | -0.4 degrees |
| - 1 deg/sec * 0.1 sec = | -0.1 degrees |
| Total Swerve Right | 6.1 degrees |

**Note 1: this is how far nose of car is pointing right – *CG* motion may *lag***

***Note 2: Polarity reversed in this module***

# Pedestrian Protection

- Starting 2012 Fiat Freemont & Dodge Journey, ***In Europe only***, FCA has added a pedestrian protection feature to deploy a feature that pops the hood up slightly to make the angle better and less injurious to pedestrians.
- It is a separate module with separate data and must be imaged separately thru the DLC or via cable 516.
- Widespread in Europe, may come to US

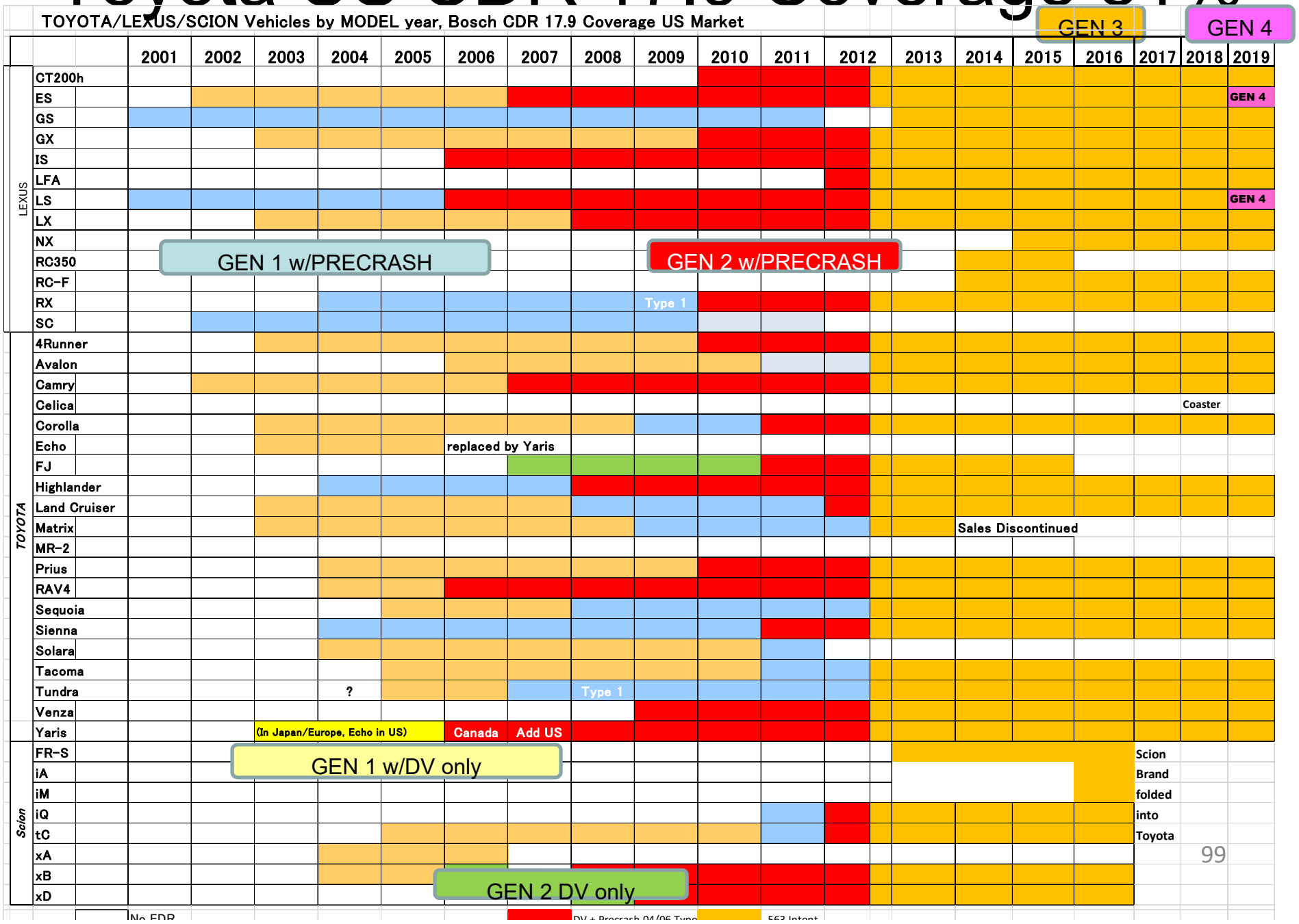
# Other FCA News

- Revised reporting of deploy times (was stage 1 to stage 2 vs AE to stage 2).
- 17.2 fixed some WOT reporting as 77%, rest fixed in 17.7.1
- 10-12 Journey + Minivan – all ND's display as "interrupted" recording
- 2013 TRW – key cycle counters may not match perfectly (different sources)
- Recording threshold intent 5mph DV, but some models may report a 4.9 DV.
- Most "features" noted in data lim's.

# New Chrysler Info

- Frontal sensors are 100G at 45 degrees (100*cos 45$^{o}$ = 71g straight ahead).
- Replaced clipping flag with “Sensor Design Range Exceeded” line on graph
- Buckle switch NOT normally used to determine if pre-tensioner fires (Exceptions on some non US built vehicles)
- Fiat Spider = Mazda Miata, threshold <5mph
- Master Cylinder Pressure on some 2018!
- Driver assistance data elements coming in

# Toyota US CDR 17.9 Coverage 81%



# Toyota EDR Evolution

| GEN 1<br>00,02 EDR | GEN 2<br>04,06 EDR | GEN 3<br>12,13,15 EDR | GEN 4<br>17 EDR |
|---|---|---|---|
| 3 front<br>1 side<br>1 roll | 2 front<br>2 side<br>2 roll | | 4 front<br>4 side<br>6 precrash |
| Record on 2g wakeup | | Record on 5mph XDV, side 3.5g wake | Record on 5 X or Y DV |
| 150ms XDV<br>5PC@1 + TRG | 200ms XDV<br>5PC@1+TRG | 250ms XDV<br>10PC@0.5 + TRG | 250ms XY<br>10PC@0.5+TRG |

# Toyota EDR Evolution cont'd

| GEN 1<br>00,02 EDR | GEN 2<br>04,06 EDR | GEN 3<br>12,13,15 EDR | GEN 4<br>17 EDR |
|---|---|---|---|
| 70ms<br>YDV@ACM+sats | | 12 YDV 70ms<br>sats only<br>13 250ms ACM | 250ms<br>YDV |
| REFRESH<br>500 MS SPEED | | REFRESH<br>24 MS SPEED | |
| 78 MPH TOP,<br>2kph res | 75 MPH TOP.<br>2kph res | 124 MPH TOP<br>1 kph res | |
| Accel Pedal Volts | | Accel Pedal % & Throttle<br>% | |

# Toyota EDR Evolution cont'd

GEN 1
00,02 EDR

GEN 2
04,06 EDR

GEN 3
12,13,15 EDR

GEN 4
17 EDR

RPM to next lower 400

RPM to 100

No Key Cycles

Key Cycles

Time Between Events < 5 sec

Time Between Events <16 sec

Brk Oil Pres, Stab. Cont. Long Accel

Strg & Yaw

# New “17 EDR”

- “Front/Rear” events also triggered by 5mph Y DV or side deployment
- Y Delta V from ACM added to frontal events for recon
- More precrash registers (6 vs 2 prior)
- More front/rear & side registers (4 vs 2)
- Separate side events still reported, but do NOT have precrash associated with them. Side events have detail of side satellite sensors (generally ***not used*** by Recons).
- Odometer data element added

# Same Data, New Look to Precrash

**Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 1 of 4**

| Time (sec) | Vehicle Speed (MPH [km/h]) | Accelerator Pedal, % Full (%) | Percentage of Engine Throttle (%) | Fuel Injection Quantity (mm^3/st) | Engine RPM (RPM) | Motor RPM (RPM) | Service Brake, ON/OFF |
|---|---|---|---|---|---|---|---|
| -5.00 | 29.8 [48] | 49.5 | 24.5 | Invalid | 2,400 | Invalid | OFF |
| -4.50 | 32.3 [52] | 47.0 | 24.5 | Invalid | 2,500 | Invalid | OFF |
| -4.00 | 34.2 [55] | 38.0 | 20.0 | Invalid | 2,700 | Invalid | OFF |
| -3.50 | 35.4 [57] | 32.5 | 16.0 | Invalid | 2,800 | Invalid | OFF |
| -3.00 | 36.7 [59] | 26.5 | 13.5 | Invalid | 2,400 | Invalid | OFF |
| -2.50 | 37.3 [60] | 22.5 | 11.0 | Invalid | 2,400 | Invalid | OFF |
| -2.00 | 37.3 [60] | 19.0 | 8.5 | Invalid | 2,200 | Invalid | OFF |
| -1.50 | 38.5 [62] | 19.0 | 6.0 | Invalid | 1,900 | Invalid | OFF |
| -1.00 | 37.9 [61] | 18.0 | 5.5 | Invalid | 1,900 | Invalid | OFF |
| -0.50 | 37.9 [61] | 18.0 | 5.0 | Invalid | 1,800 | Invalid | OFF |
| TRG(0) | 36.0 [58] | 100.0 | 26.0 | Invalid | 1,700 | Invalid | OFF |

**Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 2 of 4**

| Time (sec) | ABS Control Status | BOS Control Status | Brake Oil Pressure (Mpa) | Longitudinal Acceleration , VSC Sensor (m/s^2) | Yaw Rate (deg/s) | Steering Input (degrees) | Shift Position |
|---|---|---|---|---|---|---|---|
| -5.00 | OFF | OFF | 0.00 | 1.579 | -0.98 | -4.5 | D |
| -4.50 | OFF | OFF | 0.00 | 1.795 | 0.00 | -1.5 | D |
| -4.00 | OFF | OFF | 0.00 | 1.292 | 0.49 | 0.0 | D |
| -3.50 | OFF | OFF | 0.00 | 0.861 | 0.00 | -1.5 | D |
| -3.00 | OFF | OFF | 0.00 | 0.718 | 0.00 | -1.5 | D |
| -2.50 | OFF | OFF | 0.00 | -0.072 | 0.00 | -1.5 | D |
| -2.00 | OFF | OFF | 0.00 | 0.000 | 0.00 | -1.5 | D |
| -1.50 | OFF | OFF | 0.00 | 0.000 | -0.49 | -1.5 | D |
| -1.00 | OFF | OFF | 0.00 | 0.287 | 0.00 | -1.5 | D |
| -0.50 | OFF | OFF | 0.00 | -0.144 | 0.00 | -1.5 | D |
| TRG(0) | OFF | OFF | 0.48 | -1.507 | 61.00 | -19.5 | D |

**Pre-Crash Data -5 to 0 Seconds (1st Prior Event, TRG 2) - Table 3 of 4**

| Time (sec) | Sequential Shift Range | Cruise Control Status | VSC Control Status | READY Signal | Drive Mode, Power Train | Drive Mode, Snow | Drive Mode, EV |
|---|---|---|---|---|---|---|---|
| -5.00 | Undetermined | OFF | ON (enable) | Invalid | Normal | OFF | Invalid |

# Pedestrian Protection Module

## System Status at Pedestrian Event (1st Prior Event, TRG 1)

| | |
|---|---|
| TRG Count (times) | 1 |
| TRG Count not for Pedestrian (times) | SNA |
| Ignition Cycle, Crash (times) | 353 |
| Odometer signal (miles [km]) | 1,195 [1,923] |
| Trip count (times) | 392 |
| Time count (msec) | 1,283,900 |
| Time count input system | Normal |
| Airbag Warning Lamp, On/Off | OFF |
| Pedestrian Warning Lamp, On/Off | OFF |
| Time from Previous TRG (msec) | 32767 or greater |
| Time from Pre-Crash to TRG (msec) | 300 |
| Time between TRG not for Pedestrian and TRG for Pedestrian (msec) | SNA |
| TRG Establishment not for Pedestrian, within 5 sec. | ON |
| Freeze Signal | ON |
| Recording Status, Pre-Crash | Complete |
| Recording Status, Crash Info. | Complete |
| Recording Status, EDR Correlation Info. | Complete |

## Deployment Command Data (1st Prior Event, TRG 1)

| | |
|---|---|
| Front Pop-Up Hood, Time to Deploy (msec) | 3.0 |
| Rear Pop-Up Hood, Time to Deploy (msec) | 3.0 |
| Airbag for Pedestrian Deployment, Time to Deploy (msec) | SNA |

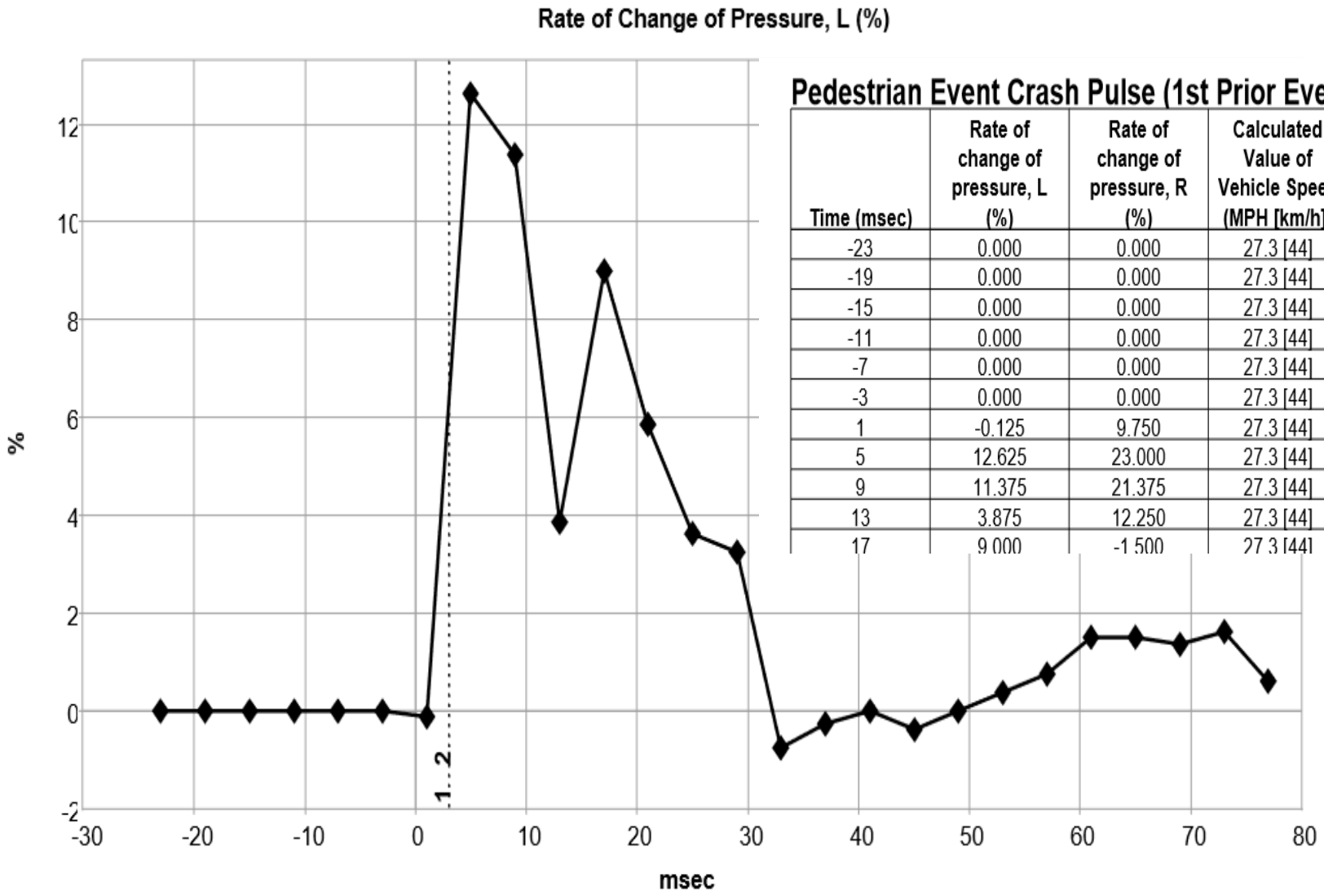
## DTCs Present at Time of Event (1st Prior Event, TRG 1)

| | |
|---|---|
| Recording Status, Diagnostic | Complete |
| Ignition Cycle Since DTC was Set (times) | 0 |
| Warning ON Time Since DTC was Set, for Pedestrian (min) | 0 |
| Diagnostic Trouble Codes | None |

## Pedestrian Event Crash Pulse (1st Prior Event, TRG 1)

| | |
|---|---|
| Recording Status, Time Series Data | Complete |
| Time from TRG to Next Sample (msec) | 1 |

# Data from L/R bumper pressure sensors




**Pedestrian Event Crash Pulse (1st Prior Event, TRG 1)**

| Time (msec) | Rate of change of pressure, L (%) | Rate of change of pressure, R (%) | Calculated Value of Vehicle Speed (MPH [km/h]) |
|---|---|---|---|
| -23 | 0.000 | 0.000 | 27.3 [44] |
| -19 | 0.000 | 0.000 | 27.3 [44] |
| -15 | 0.000 | 0.000 | 27.3 [44] |
| -11 | 0.000 | 0.000 | 27.3 [44] |
| -7 | 0.000 | 0.000 | 27.3 [44] |
| -3 | 0.000 | 0.000 | 27.3 [44] |
| 1 | -0.125 | 9.750 | 27.3 [44] |
| 5 | 12.625 | 23.000 | 27.3 [44] |
| 9 | 11.375 | 21.375 | 27.3 [44] |
| 13 | 3.875 | 12.250 | 27.3 [44] |
| 17 | 9.000 | -1.500 | 27.3 [44] |

# Toyota Help File Organization

## Supported Toyota Vehicles

Use this section of the help file to determine if a vehicle is supported by the CDR tool and to look up the correct cable (and in some cases, adapter) to use for EDR data directly from the ACM. It is important to note that even if the vehicle is listed in this section, CDR must also support the part number on the ACM. Cli list of supported ACM part numbers.

Click here for tips on how to use this *Supported Vehicles* help topic

This Started in Nov 2015

**IMPORTANT:** To prevent damage to Toyota ECUs when imaging EDR data directly from the ECU, you must always use the appropriate cable called out in this the help file.

**General Toyota Application Notes:**

The CDR Tool software will accept most Toyota VINs entered by the user and the *Collect Airbag Control Module Data* icon will no longer be grayed out. **This is n indicator that the vehicle is supported.** Only the vehicles listed in the help file below are supported as indicated by the list and any associated coverage not

All supported vehicles listed can be imaged through the vehicle's DLC connector using the CDR OBDII DLC Cable (F00K108287) unless otherwise stated.

All supported vehicles require the vehicle's ignition key to be switched on prior to and during EDR record imaging through the vehicle's DLC connector.

**Vehicle Specific Application Notes:**

Click here to view all vehicle specific Toyota coverage notes

Vehicles Are Listed Alphabetically
Not by Model Year

Toyota Models begin with: 4 8 9 A B C D E F H I K L M N P R S T V W X Y Z

**4Runner (Toyota) 2003 and later Market: 7**

| Year | EDR Generation | CDR Cable & Adapter | Module | Module Location |
|---|---|---|---|---|
| 2002 - 2008 | Gen 1 (02EDR) | F00K108617 (no adapter) | ACM | Under Center Stack |
| 2009 - 2011 | Gen 2 (06EDR) | F00K108616 & F00K108387 | ACM | Under Center Stack |
| 2012 and later | Gen 3 (12EDR/13EDR) | F00K108616 & F00K108387 | ACM | Under Center Stack |

Help file may be CALENDAR year not model year,
EDR may not yet be present in Model Year of first year listed

# Toyota Vehicle Lookup

**Camry (Toyota)** **2001 and later** **Market: 7**

All Markets the Vehicle is Sold

| Year | EDR Generation | CDR Cable & Adapter | M |
|---|---|---|---|
| 2001 - 2004<br>2005 (US/Canada) | Gen 1 (00EDR/02EDR) | F00K108615 (no adapter)<br>F00K108617 (no adapter)<br>Click here [?] to match the ACM part number to the correct cable | A |
| 2005 - 2011 | Gen 2 (04EDR/06EDR) | F00K108613 (no adapter)<br>F00K108614 & F00K108387<br>F00K108615 (no adapter)<br>F00K108616 & F00K108387<br>Click here [?] to match the ACM part number to the correct cable | A |
| 2012 and later | Gen 3 (12EDR/13EDR) | F00K108616 & F00K108387 | A |

Note CAMRY got EDR in 02 Model Year in US

Note CAMRY got 04 EDR in 07 Model Year in US

< Models

Note CAMRY got 12 EDR as running change in 12 model year, early models have 04/06 EDR in US

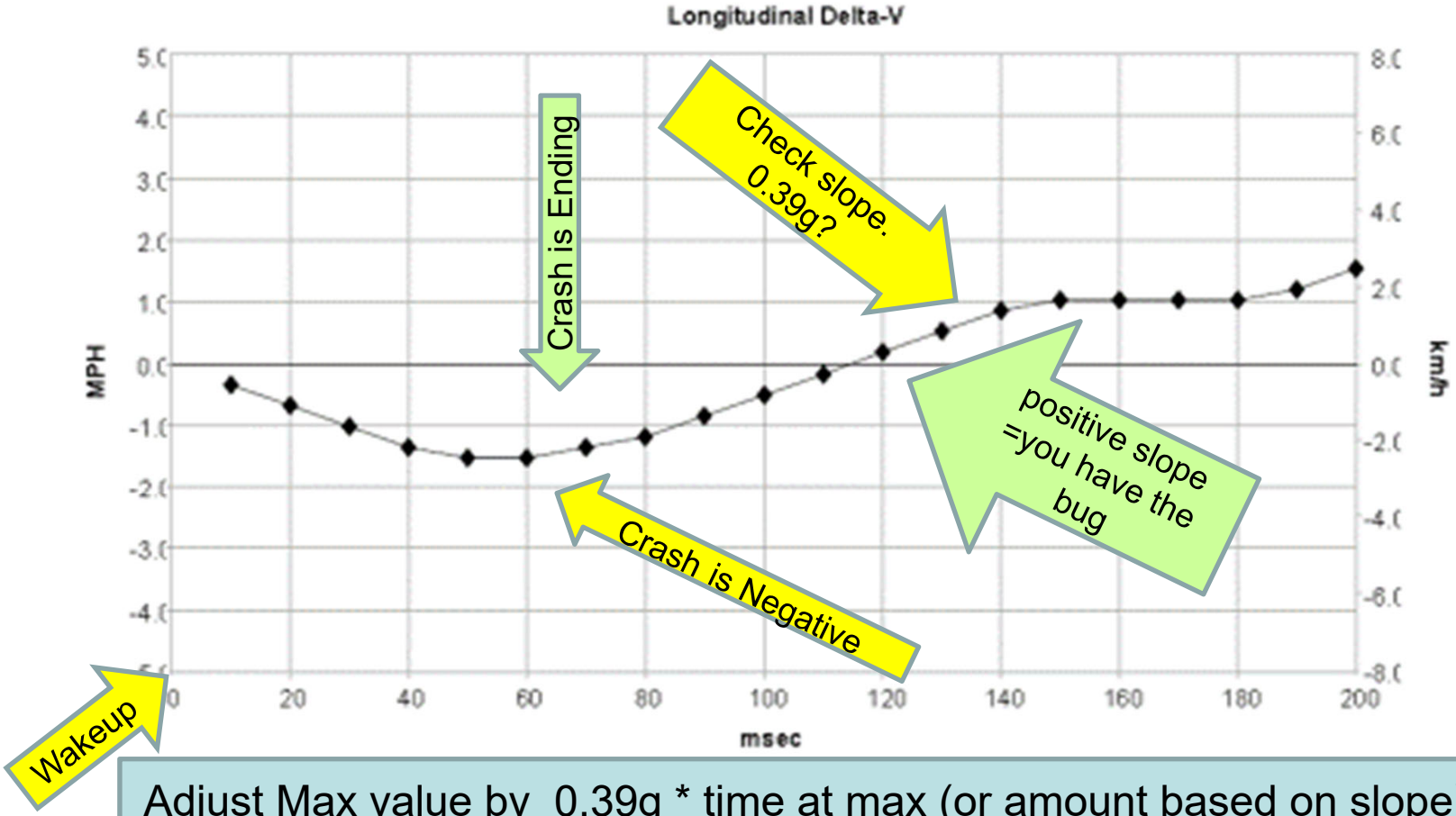
# Known Bug in Gen 1&2 Delta V

- Empirical research indicates in all Gen 1's tested to date and SOME Gen 2 models
- ***Under*** reports *frontals* slightly
- ***Over*** reports *rears* slightly
- Sources in Toyota indicate the zero-offset applied when converting analog to digital accelerometer output may be off by 1 count, creating an error of 0.39G.
- 0.39g*150msec*32.2/1.466 = 1.3 mph
- Empirical data suggests 2 counts on some

# How to Identify you have the bug

**Longitudinal Crash Pulse (Most Recent Event, TRG 3 - table 1 of 2)**

| | |
|---|---|
| Recording Status, Time Series Data | Complete |
| Max Longitudinal Delta-V (MPH [km/h]) | 1.5 [2.5] |




Adjust Max value by 0.39g * time at max (or amount based on slope)

No positive slope = no adjustment needed

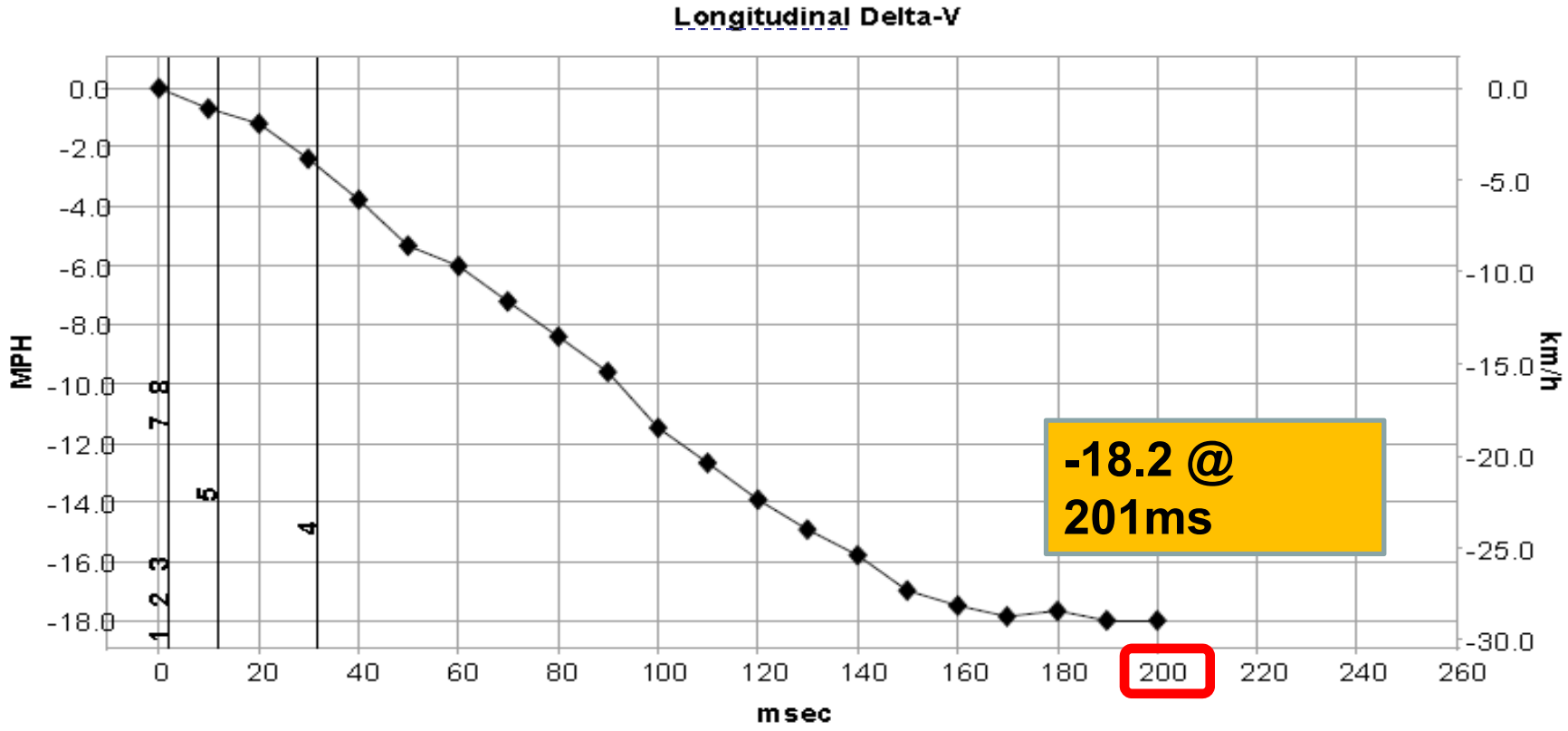
# Toyota 12 EDR Drawback

- Lateral acceleration is only recorded in a side algorithm wakeup.
- When it is recorded, it is only for 68ms
- It is only recorded for the side satellites, not for the central control module
- If trying to calculate a PDOF you must be very aware of the different recording timing and sensor location
- Fixed in “13EDR” – get 250ms of Y accel

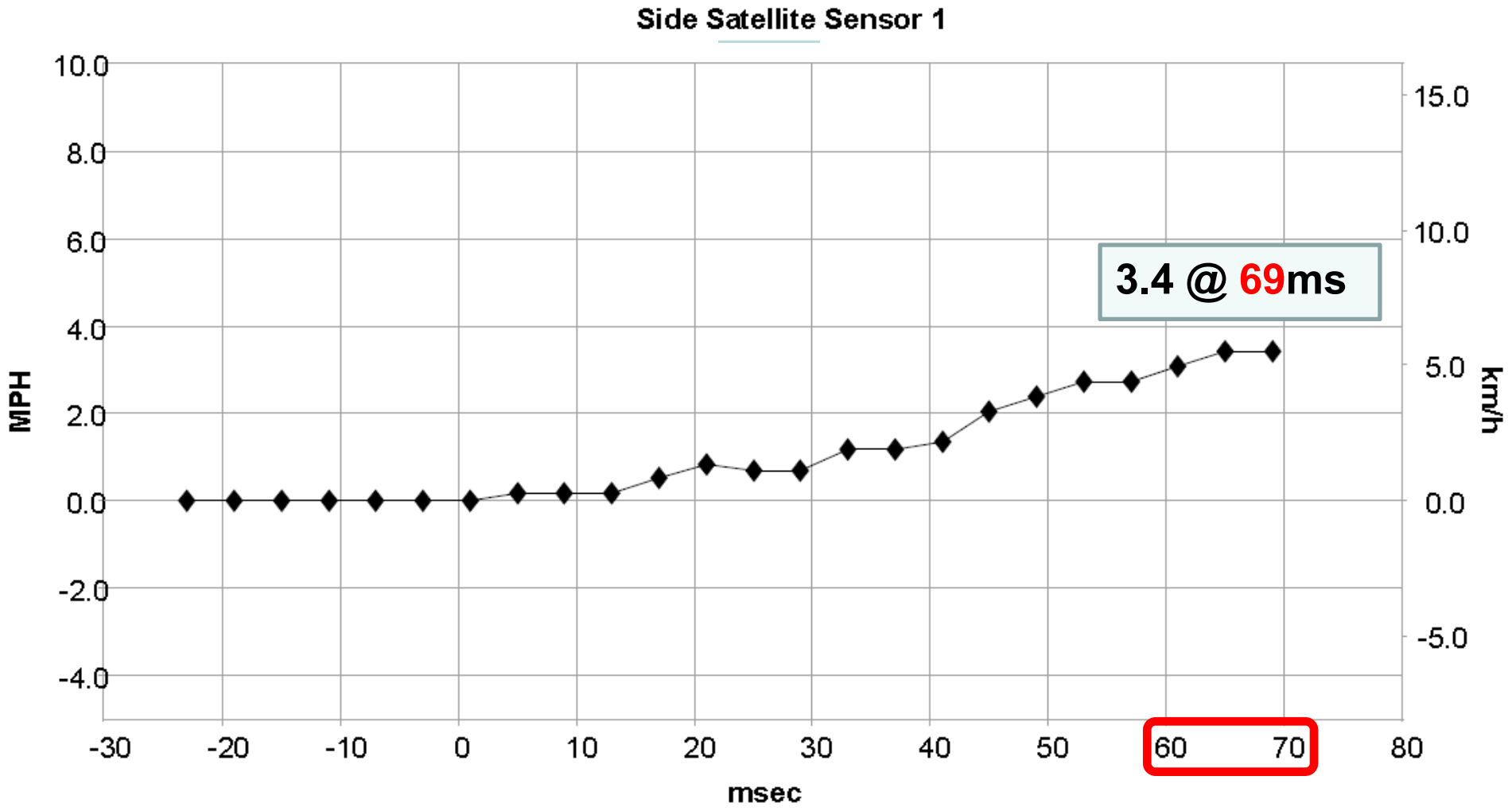
# Delta V analysis – What's the PDOF? The Total?? 12 EDR

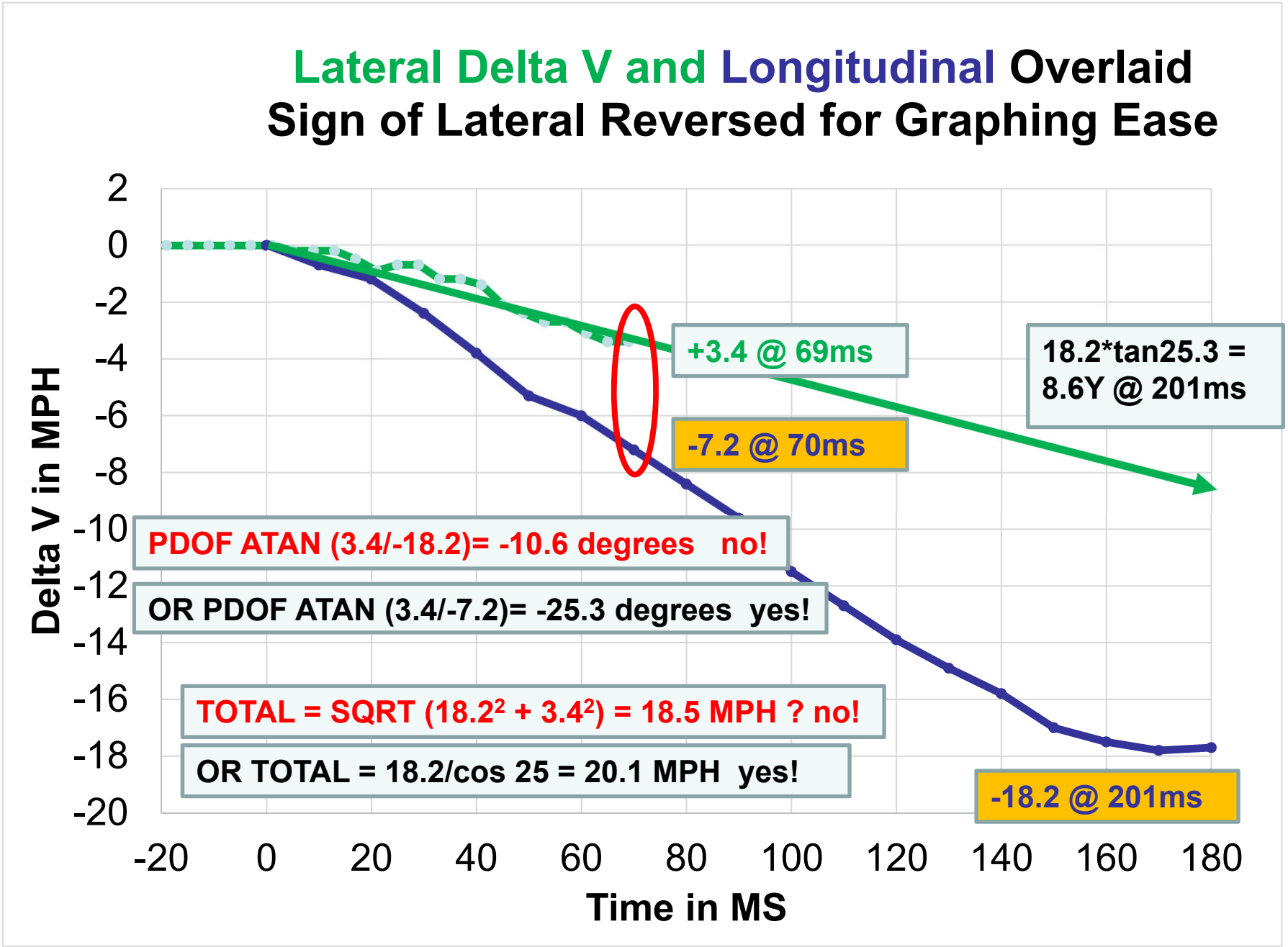
**Longitudinal Crash Pulse (1st Prior Event, TRG 2 - table 1 of 2)**

| | |
|---|---|
| Recording Status, Time Series Data | Complete |
| Time from Time Zero to TRG (msec) | 2.0 |
| Length of Delta-V (msec) | 200 |
| Max. Longitudinal Delta-V (MPH [km/h]) | -18.2 [-29.2] |
| Time, Maximum Delta-V, Longitudinal (msec) | 201.0 |
| Power Supply Status at Max. Delta-V | ON |



# 12 EDR Lateral Delta V -




Lateral Delta V and Longitudinal Overlaid
Sign of Lateral Reversed for Graphing Ease
Delta V in MPH
2
0
-2
-4
-6
-8
-10
-12
-14
-16
-18
-20
-20
0
20
40
60
80
100
120
140
160
180
Time in MS
+3.4 @ 69ms
18.2*tan25.3 =
8.6Y @ 201ms
-7.2 @ 70ms
PDOF ATAN (3.4/-18.2)= -10.6 degrees no!
OR PDOF ATAN (3.4/-7.2)= -25.3 degrees yes!
TOTAL = SQRT (18.2² + 3.4²) = 18.5 MPH ? no!
OR TOTAL = 18.2/cos 25 = 20.1 MPH yes!
-18.2 @ 201ms

# Additional Freeze Frame Data in 2013+

- New “Vehicle Control History” feature in 2013+ Rav 4 and 2014+ Highlander & Lexus IS, 2015+ Lexus NX & RC, Camry, Corolla, Prius C, Prius V. 2016+ Avalon, Land Cruiser, Mirai, Prius, Tacoma, Scion iM, and Lexus ES, GS, LX, and RX.  Likely in all now!
- Triggers EDR-style recording if accel pedal is rapidly depressed or other conditions that might occur during an alleged sudden or unintended acceleration or lane excursion. 26 triggers including crash avoidance, ESC.  Now Lane Keep Assist, ICS, EDR too.
- Accessible with Toyota Techstream service tool (or Nexiq interface or CDR900) and Toyota service subscription software (days, month, or annual) (TSS 2.0 now out, new undocumented trigger #25 EDR).
- Data exports as CSV to Microsoft excel
- Space in memory for MANY events – can show a history of erratic driver behavior
- Details in SAE 2016-01-1495 Appendix D. New **2019-01-0632** “**Reconstructing Vehicle Dynamics from On-Board Event Data**” **discovered VCS has data 6.6 times per second in ABS events vs 2 times in hard accel events)**

# New “13 EDR” & “15 EDR”

- **13 EDR** Adds Lateral ***Accel*** data from ACM in frontal events for 200ms
  (Fixes biggest drawback of **12 EDR**)

**Longitudinal/Lateral Crash Pulse (Most Recent Event,**

| Time (msec) | Longitudinal Delta-V (MPH [km/h]) | Lateral Acceleration for Frontal/Rear Crash, Floor Sensor (m/sec^2) | F |
|---|---|---|---|
| 0 | 0.0 [0.0] | 0.0 | |
| 10 | 0.3 [0.6] | -1.9 | |
| 20 | 0.5 [0.8] | 0.0 | |

- 10ms interval
- $DV_{10ms}$ = ***Accel***/9.8*(32.2/1.466)*0.01s
- Add up intervals to get ***total*** **Delta V**
- **Template available to calculate it**

# New “17 EDR” for Ped Protect

- “Front/Rear” events also triggered by 5mph Y DV or side deployment
- Y Delta V from ACM recording added to frontal events
- More precrash registers (6 vs 2 prior)
- More front/rear & side registers (4 vs 2)
- Separate side events still reported, but do NOT have precrash associated with them. Side events have detail of satellite sensors

# NISSAN EDR's 17.9 (CDR 46%)

| Model | Location | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Last 3 digits of Bosch CDR cable number | | | | | | | | | | | | | | | |
| **INFINITI** | | | | | | | | | | | | | | | |
| G CONVERTIBLE | TUNNEL BETWEEN SEATS | | | | * | | Jul-12 | | 780 | | | | | | |
| G COUPE | TUNNEL BETWEEN SEATS | | * | | | | | | 780 | | | | | | |
| G SEDAN | TUNNEL BETWEEN SEATS | | * | | | | Jul-12 | | 780 | | | | | | |
| M | TUNNEL BETWEEN SEATS | | | | | | | | 780 | | | | | | |
| Q40 | TUNNEL BETWEEN SEATS | | | | | | | | | | 780 | | | | |
| Q50 | TUNNEL BETWEEN SEATS | | | | | | | | 780 | 780 | 780 | 780 | 780 | 780 | 780 |
| Q60 | TUNNEL BETWEEN SEATS | | | | | | | | 780 | 780 | 780 | | 780 | 780 | 780 |
| Q70 | TUNNEL BETWEEN SEATS | | | | | | | | 780 | 780 | 780 | 780 | 780 | 780 | 780 |
| EX | TUNNEL BETWEEN SEATS | | | | | | | | 780 | | | | | | |
| FX | TUNNEL BETWEEN SEATS | | | | | | | | 780 | | | | | | |
| JX35 | TUNNEL BETWEEN SEATS | | | | | | | | 780 | | | | | | |
| QX50-80 | TUNNEL BETWEEN SEATS | | | | | | | | 780 | 780 | 780 | 780 | 780 | QX60& 80 | QX50,60& |
| QX30 | TUNNEL BETWEEN SEATS | | | | | | | | | | | | 598 | 598 | |
| **NISSAN** | | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2018 |
| 370Z | TUNNEL BETWEEN SEATS | | * | | | | Aug-12 | | 780 | 780 | 780 | 780 | 780 | 780 | 780 |
| ALTIMA COUPE | TUNNEL BETWEEN SEATS | | | * | | | | | 780 | | | | | | |
| ALTIMA SEDAN | TUNNEL BETWEEN SEATS | | * | | | | | | 780 | 780 | 780 | 780 | 780 | 780 | |
| ARMADA | TUNNEL BETWEEN SEATS | * | | | | | | | 780 | 780 | 780 | ? | 780 | 780 | |
| CUBE | TUNNEL BETWEEN SEATS | | | | * | | | | 780 | 780 | | | | | |
| FRONTIER | TUNNEL BETWEEN SEATS | * | | | | | Oct-12 | | 780 | 780 | 780 | 780 | 780 | 780 | |
| GT-R | TUNNEL BETWEEN SEATS | | | | * | | | | | 780 | 780 | 780 | 780 | 780 | |
| JUKE | TUNNEL BETWEEN SEATS | | | | | | | | 780 | 780 | 780 | 780 | 780 | | |
| LEAF | TUNNEL BETWEEN SEATS | | | | | | | | 780 | 780 | 780 | 780 | 780 | | |
| MAXIMA SEDAN | TUNNEL BETWEEN SEATS | | * | | | | | | 780 | 780 | 780 | 780 | 780 | 780 | |
| MICRA | Check Manual | | | | | | | | | | 780 | 780 | 780 | 780 | |
| MURANO | TUNNEL BETWEEN SEATS | | * | | * | | | | 780 | 780 | 780 | 780 | 780 | 780 | |
| MURANO CROSS C | TUNNEL BETWEEN SEATS | | | | | | | | included in above | | | | | | |
| NV200 | TUNNEL BETWEEN SEATS | | | | | | | | 780 | 780 | 780 | 780 | 780 | 780 | |
| PATHFINDER | TUNNEL BETWEEN SEATS | * | | | | | | | 780 | 780 | 780 | 780 | 780 | 780 | |
| QUEST | CTR CONSOLE NEAR FRONT | | * | | | | | | 780 | 780 | 780 | 780 | 780 | | |
| ROGUE | Check Manual | | | * | | | | | 780 | 780 | 780 | 780 | 780 | 780 | 780 |
| SENTRA | TUNNEL BETWEEN SEATS | | * | | | | | | 780 | 780 | 780 | 780 | 780 | 780 | 780 |
| TITAN | TUNNEL BETWEEN SEATS | * | | | | | | | 780 | 780 | 780 | 780 | 780 | 780 | |
| VERSA SEDAN | TUNNEL BETWEEN SEATS | | * | | | | | | 780 | 780 | 780 | 780 | | 780 | |
| VERSA NOTE HB | TUNNEL BETWEEN SEATS | | | | | | Sep-12 | | 780 | 780 | 780 | 780 | 780 | 780 | |
| XTERRA | TUNNEL BETWEEN SEATS | * | | | | | Oct-12 | | 780 | 780 | 780 | | | | |

Yellow = Proprietary Consult Tool Only
No direct to module cables
Dealers do NOT have special software required to read EDR

Yellow area not covered by CDR

563

# Nissan Pre-Crash Data

**Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1)**

(the most recent sampled values are recorded prior to the event)

| Time Stamp (sec) | Speed, Vehicle Indicated (MPH [km/h]) | Accelerator Pedal, % full | Engine RPM | Motor RPM | Service Brake (On, Off) | Steering Input (deg) |
|---|---|---|---|---|---|---|
| -5.0 | 24 [ 39] | 16 | 2650 | 0 | Off (Brake Not Activated) | -32 |
| -4.5 | 23 [ 37] | 11.5 | 1750 | 0 | Off (Brake Not Activated) | -2 |
| -4.0 | 21 [ 34] | 9 | 1200 | 0 | On (Brake Activated) | 28 |
| -3.5 | 15 [ 24] | 9 | 1150 | 0 | On (Brake Activated) | 14 |
| -3.0 | 13 [ 21] | 4 | 950 | 0 | On (Brake Activated) | 4 |
| -2.5 | 11 [ 17] | 4 | 950 | 0 | On (Brake Activated) | 4 |
| -2.0 | 5 [ 8] | 4 | 900 | 0 | On (Brake Activated) | -48 |
| -1.5 | 3 [ 5] | 4 | 850 | 0 | On (Brake Activated) | -98 |
| -1.0 | 1 [ 2] | 4 | 850 | 0 | On (Brake Activated) | -160 |
| -0.5 | 3 [ 5] | 10.5 | 1100 | 0 | Off (Brake Not Activated) | -194 |
| 0.0 | 6 [ 9] | 16 | 1150 | 0 | Off (Brake Not Activated) | -218 |

Hybrid/Electric or Trans Input

Trans input tells you convertere slip

Table 2 – not rq'd

At AE

# Nissan Long ΔV Polarity Fixed?

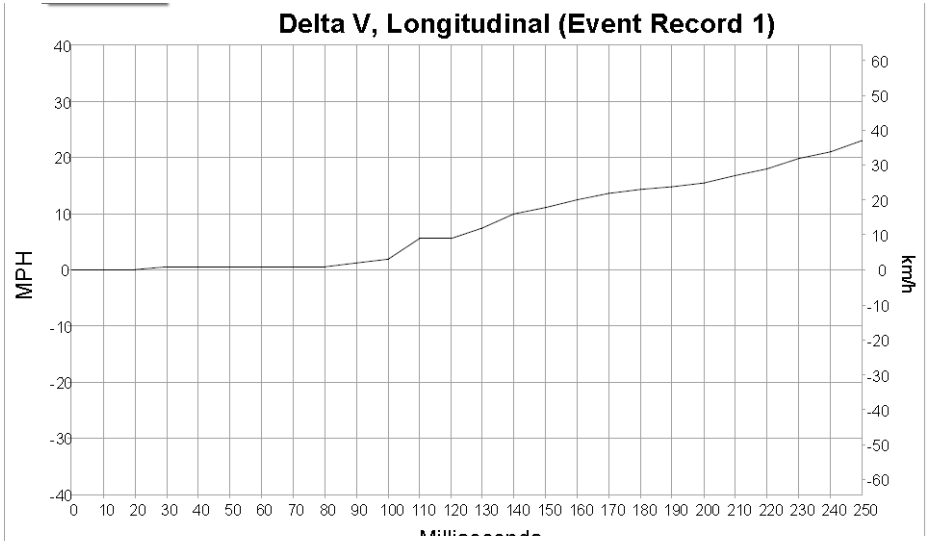
## Front hit *was* positive - ***Some exceptions still in 19.1!***

### 2013 Altima in CDR 12.2

**System Status at Event (Event Record 1)**

| | |
|---|---|
| Frontal Air Bag Suppression Switch Status | S airbag inhibit) |
| Maximum Delta-V, Longitudinal (MPH [km/h]) | 29 [ 47] |
| Time, Maximum Delta-V, Longitudinal (msec) | 300 |
| Maximum Delta-V, Lateral (MPH [km/h]) | 5 [ 8] |
| Time, Maximum Delta-V, Lateral (msec) | 210 |
| Maximum Acceleration, Longitudinal (g) | 32.5 |
| Time, Maximum Acceleration, Longitudinal (msec) | 102.5 |
| Maximum Acceleration, Lateral (g) | 36 |
| Time, Maximum Acceleration, Lateral (msec) | 112.5 |

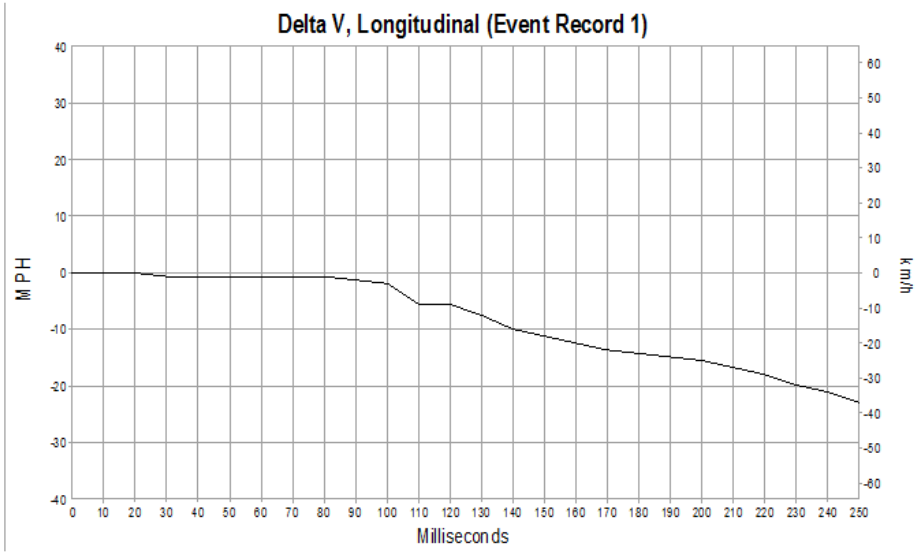
+29 at 300ms




### 2013 Altima in CDR 16.5

| | |
|---|---|
| S airbag inhibit) | |
| Maximum Delta-V, Longitudinal (MPH [km/h]) | -29 [-47] |
| Time, Maximum Delta-V, Longitudinal (msec) | 300 |
| Maximum Delta-V, Lateral (MPH [km/h]) | 5 [ 8] |
| Time, Maximum Delta-V, Lateral (msec) | 210 |
| Maximum Acceleration, Longitudinal (g) | -32.5 |
| Time, Maximum Acceleration, Longitudinal (msec) | 102.5 |
| Maximum Acceleration, Lateral (g) | 36 |
| Time, Maximum Acceleration, Lateral (msec) | 112.5 |

-29 at 300ms



# Mazda CDR 17.10 Coverage 44%

| Model | Location | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Last 3 digits of Bosch CDR cable number shown | | | | | | | | | | | | |
| **MAZDA** | | | | All Mazda use 387 adapter for direct to module | | | | | | | | |
| CX-3 | TUNNEL BELOW IP | | | | | | | | 812 | | | |
| Tribute/CX-5 | TUNNEL AFT SHIFTER | | | | | 779 | | | 811 | 812 | | |
| CX-9 | TUNNEL BELOW IP | | | 778 | | | | | 811 | | | |
| Mazda 2 | TUNNEL AFT SHIFTER | | | 792 | | | | not listed | 812 | not listed | | |
| Mazda 3 | TUNNEL AFT SHIFTER | | | | 779 | | | | 811 | | | |
| Mazda 5 | TUNNEL BELOW IP | | | | 778 | | | | | not listed | | |
| Mazda 6 | TUNNEL BELOW IP | | | 792 | | | 779 | | 811 | | 831 | |
| MX-5 Miatta | TUNNEL BELOW IP | | | | | no coverage | | | 812 | | | |

Law Enforcement Contact Your Instructors - Data available from pre-CDR ACM's from Mazda or Mazda suppliers. Mostly Delta V.

Memo: The Mazda Tribute was sister to the Ford Escape thru 2011. A 2011 Tribute was read using a 2011 Escape VIN.

The source data for this chart is the Bosch CDR help file for Version 17.10
Always consult the latest help file for the most accurate coverage

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# Typical Mazda Precrash Data

**Pre-Crash Data -5 to 0 sec [2 samples/sec] (Event Record 1)**

(the most recent sampled values are recorded prior to the event)

| Time Stamp (sec) | Speed, Vehicle Indicated (MPH [km/h]) | Engine Throttle, % full | Service Brake (On, Off) |
|---|---|---|---|
| -5.0 | 19 [31] | 0 | On |
| -4.5 | 17 [27] | 0 | On |
| -4.0 | 14 [23] | 0 | On |
| -3.5 | 12 [19] | 0 | On |
| -3.0 | 11 [17] | 25 | Off |
| -2.5 | 10 [16] | 61 | Off |
| -2.0 | 11 [17] | 63 | Off |
| -1.5 | 13 [21] | 64 | Off |
| -1.0 | 16 [26] | 64 | Off |
| -0.5 | 17 [27] | 0 | Off |
| 0.0 | 4 [6] | 0 | Off |

Minimum Regulatory Content

Recording threshold <5mph

This is a second event – last data pt at impact

# Algorithm woke up early, missed later event
# Noted in newer Fords, Mazda & Kia/Hyundai



**System Status at Event (Event Record 1)**

| | |
|---|---|
| Safety Belt Status, Driver | Belted |
| Safety Belt Status, Right Front Passenger | Unbelted |
| Occupant Size Classification, Front Passenger | Adult |
| Frontal Air Bag Warning Lamp (On, Off) | Off |
| Ignition Cycle, Crash | 1892 |
| Multi-Event, Number of Events (1, 2) | No. 1 |
| Complete File Recorded (Yes/No) | Yes |
| Ignition Cycle, Download | 1893 |
| Maximum Delta-V, Longitudinal (MPH [km/h]) | 1.2 [2] |
| Time, Maximum Delta-V, Longitudinal (msec) | 297.5 |
| Maximum Delta-V, Lateral (MPH [km/h]) | -1.2 [-2] |
| Time, Maximum Delta-V, Lateral (msec) | 297.5 |
| Time, Maximum Delta-V, Resultant (msec) | 297.5 |

**Deployment Command Data (Event Record 1)**

| | |
|---|---|
| Pretensioner Deployment, Time to Fire, Driver (msec) | 250 |
| Pretensioner Deployment, Time to Fire, Right Front Passenger (msec) | 250 |
| Frontal Air Bag Deployment, Time to Deploy/First Stage, Driver (msec) | 250 |
| Frontal Air Bag Deployment, Time to Deploy/First Stage, Right Front Passenger (msec) | 250 |

Max Value
Mazda

# Volvo CDR 18.0 Coverage 28%

| Model | Location | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Last 3 digits of Bosch CDR cable number sh | | | | | | | | | | | |
| VOLVO | | All Volvo use 387 Adapter for DTM images | | | | | | | | | |
| S40 | | | | | | | | | | | |
| S60 | CTR TUNNEL UNDER CONSOLE | | 799 | | | | | | | 500/830 | |
| S80, S80L | CTR TUNNEL UNDER CONSOLE | | | | | | | | | | |
| S90 | CTR TUNNEL UNDER CONSOLE | | | | | | | | 816 | | |
| V40 | CTR TUNNEL UNDER CONSOLE | | | | | | | | 816 Euro | | |
| V60 | CTR TUNNEL UNDER CONSOLE | | 799 | | | | | | | 500/830 | |
| V70 | CTR TUNNEL UNDER CONSOLE | | | | 799 | | | | | | |
| V90 | CTR TUNNEL UNDER CONSOLE | | | | | | | | 816 | | |
| XC40 | CTR TUNNEL UNDER CONSOLE | | | | | | | | | 500/830 | |
| XC60 | CTR TUNNEL UNDER CONSOLE | | | | 799 | | | | | | |
| XC70 | CTR TUNNEL UNDER CONSOLE | | | | 799 | | | | | | |
| XC90 | CTR TUNNEL UNDER CONSOLE | | | | | | | 816 | | | |

Old Volvo DV only Law Enf only call Volvo

# 2016+ XC90 Sample Volvo File

## (vs. 2013 Volvo with minimum data)

### Pre-Crash Data -1 Sec (Event Record 1)

| | |
|---|---|
| Ignition Cycle, Crash | 2,764 |
| Safety Belt Status, Driver | On, Belted |
| Safety Belt Status, Passenger | On, Belted |
| Frontal Airbag Warning Lamp | Off |
| Frontal Airbag Suppression Switch Status, Front Passenger | On |
| Seat Track Position Switch, Foremost, Status, Driver | Not Equipped |
| Seat Track Position Switch, Foremost, Status, Front Passenger | Not Equipped |
| Occupant Size Right Front Passenger Child | Not Equipped |

### Pre-Crash -5 to 0 sec (Event Record 1)

Added Data for XC90

| Time (sec) | Speed, Vehicle Indicated (MPH [km/h]) | Accelerator Pedal, % Full (%) | Service Brake (On, Off) | Steering input (%) | ABS Activity | Stability control status |
|---|---|---|---|---|---|---|
| -5.0 | 43.5 [70.0] | 100.0 | Off | 0.0 | Off | On |
| -4.5 | 47.8 [77.0] | 100.0 | Off | 0.0 | Off | On |
| -4.0 | 52.2 [84.0] | 100.0 | Off | 2.0 | Off | On |
| -3.5 | 55.9 [90.0] | 100.0 | Off | 3.0 | Off | On |
| -3.0 | 58.4 [94.0] | 100.0 | Off | 3.0 | Off | On |
| -2.5 | 61.5 [99.0] | 100.0 | Off | 11.0 | Off | Engaged |
| -2.0 | 62.8 [101.0] | 100.0 | Off | 6.0 | Off | Engaged |
| -1.5 | 65.2 [105.0] | 100.0 | Off | -9.0 | Off | Engaged |
| -1.0 | 67.1 [108.0] | 100.0 | Off | -8.0 | Off | Engaged |
| -0.5 | 69.6 [112.0] | 100.0 | Off | -9.0 | Off | Engaged |
| 0.0 | 70.2 [113.0] | 100.0 | Off | -13.0 | Off | Engaged |

# 2016+ XC90 Sample Volvo File

## System Status at Retrieval

| | |
|---|---|
| Vehicle Identification Number | YV1LF |
| Application Diagnostic Database Part Number | 31387255 AL |
| Ignition Cycle, Download | 2,765 |
| Lifetime Operating Timer (sec) | 2,154,311 |

## System Status at Event (Event Record 1)

| | |
|---|---|
| Data Area Status, Event Record 1 | Locked, Data Stored |
| Data Area Read Status, Event Record 1 | Data Not Read |
| Complete File Recorded (Yes/No) | Yes |
| Multi-Event, Number of Events (1,2) | Event Number 1 |
| Time from Preceding Event (sec) | Written but No Data Available |
| Maximum Delta-V, Longitudinal (MPH [km/h]) | < -62 [-100] |
| Time, Maximum Delta-V, Longitudinal (msec) | 100 |
| Maximum Delta-V, Lateral (MPH [km/h]) | -19.9 [-32.0] |
| Time, Maximum Delta-V, Lateral (msec) | >300 |

Clipped Value (at Max)

Not present in 2013, Added for XC90

[redacted]

# BMW CDR US 18.0 Coverage 31%

See help file for Rolls Royce

| Model | Location | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 |
|---|---|---|---|---|---|---|---|---|---|
| Last 3 digits of Bosch CDR adapter/cable number shown | | | | | | | | | |
| **BMW** | **All BMW require 387 Adapter for DTM** | | | | | | | | |
| **1 Series** | | | | | | | | | |
| **2 Series** | Driver side, behind IP, rt of strg | | | 796 | | | | | |
| **3 Series** | Driver side, behind IP, rt of strg | | 796 | | | | | | |
| **Active Hybrid 3** | Passenger side, behind the glove box | | 796 | | | | | | |
| **4 Series** | Driver side, behind IP, rt of strg | | | 796 | | | | | |
| **5 Series** | Passenger side, behind the glove box | | 796 | | | | | 500/822 | |
| **6 Series** | Passenger side, behind the glove box | | 796 | | | | | | |
| **6 Series GT** | See Service Manual | | | | | | | | 500/822 |
| **7 Series** | Passenger side, behind the glove box | | 796 | | | 500/796 | 500/822 | | |
| **8 Series** | Passenger side, behind the glove box | | | | | | | | 500/822 |
| **X3 xDrive** | Passenger side, behind the glove box | | 796 | | | | | | |
| **X5** | Center Tunnel, Between Seats | | | 796 | | | | | |
| **X1, X2** | Center Tunnel, Between Seats | | | | 798 | 500/798 | 500/807 | | also X2 |
| **X3, X4** | See Help File - Various | | | | 796 | | | 500/822 | |
| **X6** | Center Tunnel, Between Seats | | | | 796 | | | | |
| **X7** | See Service Manual | | | | | | | | 500/822 |
| **Z4** | Center Tunnel, Under Dash | | | | 798 | | | | 500/822 |
| **i3, i8** | See Service Manual | | | 500/807 | | | | | |

Europe only not covered

No “back model” coverage

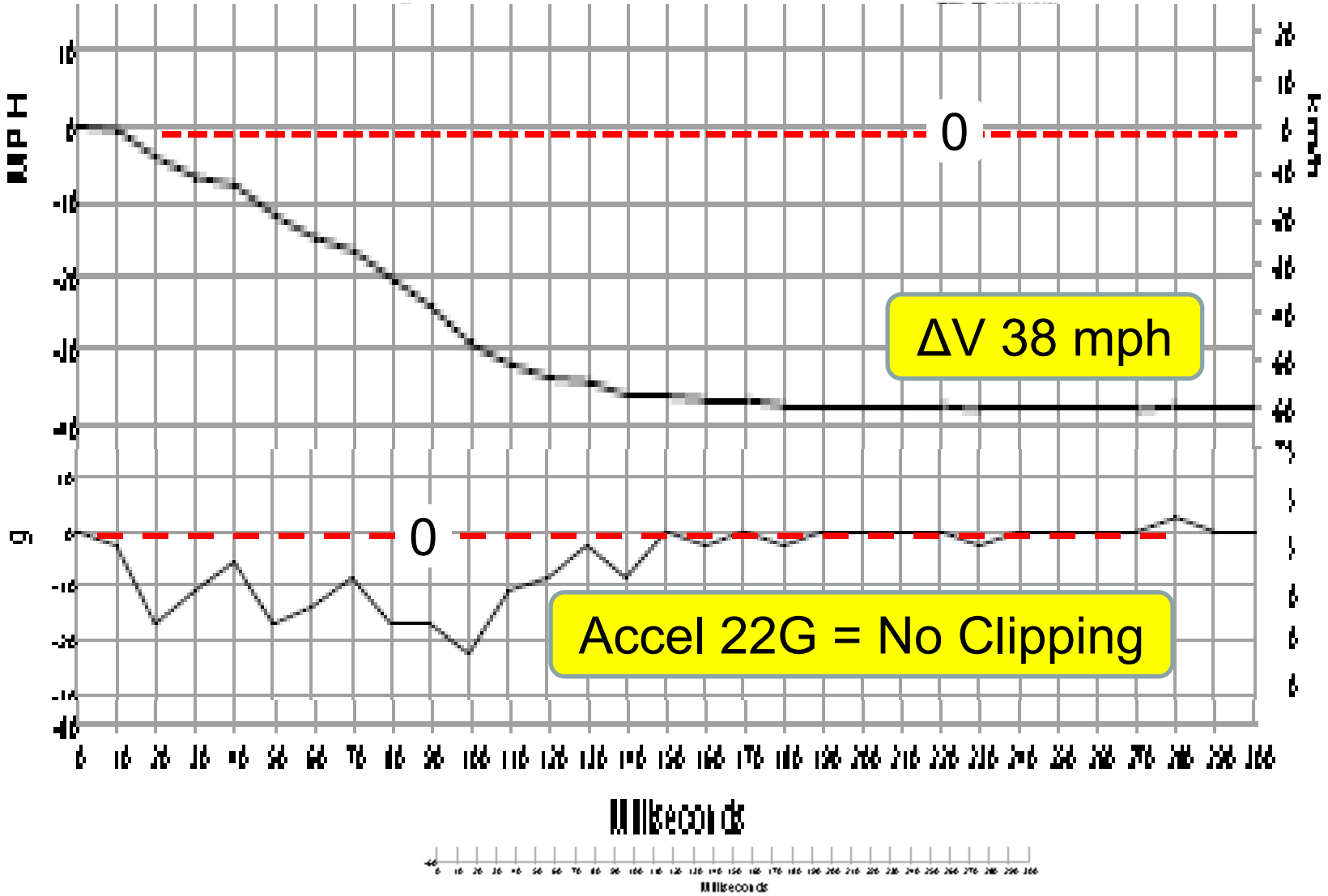
500=flex ray

# BMW Sample Pre-Crash

**Pre-Crash Data -5 to 0 sec (Record 1, Most Recent)**

| Time (sec) | Speed, Vehicle Indicated (MPH [km/h]) | Accelerator Pedal, % Full (%) | Engine RPM | Steering Input (deg) | Service Brake, On/Off | ABS Activity (Engaged, Non-engaged) | Stability Control (On Engaged, Non-engaged) |
|---|---|---|---|---|---|---|---|
| -5.0 | 55 [89] | 16 | 1500 | 0 | Off | No ABS Activity | Non-engaged |
| -4.5 | 55 [89] | 17 | 1500 | 0 | Off | No ABS Activity | Non-engaged |
| -4.0 | 55 [89] | 19 | 1500 | 0 | Off | No ABS Activity | Non-engaged |
| -3.5 | 55 [89] | 19 | 1500 | 0 | Off | No ABS Activity | Non-engaged |
| -3.0 | 55 [89] | 19 | 1500 | 0 | Off | No ABS Activity | Non-engaged |
| -2.5 | 55 [89] | 8 | 1500 | 0 | Off | No ABS Activity | Non-engaged |
| -2.0 | 55 [89] | 7 | 1500 | 0 | Off | No ABS Activity | Non-engaged |
| -1.5 | 55 [89] | 14 | 1500 | 0 | Off | No ABS Activity | Non-engaged |
| -1.0 | 55 [88] | 19 | 1500 | 0 | Off | No ABS Activity | Non-engaged |
| -0.5 | 55 [89] | 0 | 1500 | 0 | Off | No ABS Activity | Non-engaged |
| 0.0 | 45 [73] | 0 | 1300 | 10 | On | ABS Activity | Non-engaged |

# BMW – Both ΔV & Accel



# BMW EDR Evolution

- Basic EDR 2013
- “Extension” 2014
- Some driver assist mid-2015 - timing
- Auto Steering (autonomous vehicle)

  Cameras all views 4 fps

  Not intended for CDR,

  only BMW can access

# BMW Data before 2013?

- Text files in early 2000's BMW were retrieved by BMW w/proprietary tools (in German).
- One speed somewhere <2 sec to impact.
- No Delta V or Acceleration data.
- Time to airbag deployment(s) & software level
- System configuration & component serial #'s
- Diagnostics w/resistance checks of circuits
- Occupancy of seats and seatbelt use.

# Mercedes CDR Coverage 18.0 31%

| Model | Start of Sales | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 |
|---|---|---|---|---|---|---|---|---|
| Last 3 digits of Bosch CDR DTM cable number | | | | | | | | |
| Mercedes | EDR Start | All Mercedes use 387 adapter for direct to module downloads | | | | | | |
| A-Class Limo | | | | | | | | 832 |
| AMG-GT | | | | | 800 | | | |
| B-Class EV | after 4/1/2014 | | 598 | | | | | |
| B-Class | after 6/2/2014 | | | 598 | | | | |
| C Class Sedan | after 9/1/2014 | | | 801 | | | | |
| C Class Coupe | tart of 2015 M.Y. | | | 800 | 801 | | | |
| CLA Class | after 6/2/2014 | | | 598 | | | | |
| CLS-Class | | | | 800 | | | | |
| E Class (includes coupe) | after 4/1/2013 | | 800 | | | | 801 | |
| G Class | after 11/1/2014 | | | 598 | | | | 801 |
| GL Class | after 7/11/2014 | | | 598 | ? | ? | | |
| GLA Class | after 6/2/2014 | | | 598 | | | | |
| GLC Class | | | | | 598 | | 801 | |
| GLE Class | | | | | 598 | | | |
| GLK | after 12/2013 | | 800 | | ? | ? | | |
| GLS Class | | | | | 598 | | | |
| GT | | | | | | | | |
| Maybach S600 | | | | | 801 | | | |
| ML-Class | after 7/11/2014 | | | 598 | ? | ? | ? | |
| S-class Sedan | after 9/2013 | | 801 | | | | | |
| S-Class Coupe | after 9/1/2014 | | | 801 | | | | |
| SL | after 12/2013 | | 800 | | | | | |
| SLK | after12/2013 | | 800 | | | | SLC 800 | |
| Sprinter | | | | | | | | |
| Vito | | | | | 598 | | | |
| Smart & 2016 with "E" | after 9/1/2014 | | | 809 | | | | |
| Smart with VIN 4th "F" | | | | | 820 | | | |
| Smart for two | | | | | | | 820 | |
| Metris | | | | | | | 598 | |

No “back model” coverage

# Mercedes Details

- Mercedes phased in EDR in anticipation of the NPRM requiring EDR's for Sept. 2014
- Precrash speed, brake, throttle & longitudinal ΔV, 563 table 1 items only.
- Can hold 5 events, D's lock, ND's FIFO
- Recording threshold is 5 mph over 150ms
- Speed is avg. of 4 wheel speed sensors
- The new 2018 E Class is reported to have Steering data. This marks a departure from their philosophy to record only the minimum required by law.

# Any EDR data in older Mercedes? NO

- Some diagnostic codes may include “freeze frame” data, but airbag codes don’t record speed. Some codes do.
- By the time those other codes get set after a crash, the speed is no longer the precrash speed.
- Mercedes will assist law enforcement. Call 1-88-4Mercedes.

# VW/Audi Brands EDR

- EDR in 2014 EOS, most 2015+ VW/Audi.
- EDR was installed in anticipation of the US NPRM requiring EDR's by 9/1/2014, which was never implemented.
- **Bentley** has EDR 2016+.
- **Lamborghini** has EDR in Adventadore/Hurracan
- **Porsche** NEVER implemented EDR.
- US only 2015-17, Worldwide 2018+ in transverse engine platforms

# VW/Audi 18.0 US Coverage 28%

| Model | Location | 2009-13 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 |
|---|---|---|---|---|---|---|---|---|
| Last 3 digits of Bosch CDR adapter/cable number | | | Gold = US only (Canada not covered) | | | | Yellow=Worldwide | |
| All VW/Audi Require the 387 Adapter to read DTM | | | | | | | | |
| **VW** | | | | | | | | |
| Arteon | UNDER CENTER STACK | | | | | | 805 | |
| Atlas | UNDER CENTER STACK | | | | | | 805 | |
| Beetle | UNDER CENTER STACK | | | 804 | | | Canada added | |
| CC | UNDER CENTER STACK | | | 804 | | | | |
| Crafter | UNDER CENTER STACK | | | | | | 805 | |
| EOS | UNDER CENTER STACK | | 806 | | | | | |
| Golf, eGolf | UNDER CENTER STACK | | | 805 | | | Worldwide | |
| Jetta | UNDER CENTER STACK | | | 804 | | | 805 | |
| Passat | UNDER CENTER STACK | | | 804 | | | | 500/826 |
| Polo | UNDER CENTER STACK | | | | | | 805 | |
| Routan | UNDER CENTER STACK | 785 | | | | | | |
| Sportwagen + Alltrack | | | | Golf platform, not called out separtely in help file | | | | |
| T-Roc | UNDER CENTER STACK | | | | | | | 805 |
| T-Cross | UNDER CENTER STACK | | | | | | 805 | |
| Tiguan | UNDER CENTER STACK | | | 804 | | 805 | | |
| Touareg | UNDER GEAR SELECTOR | | | 804 | | | | |
| Touran | UNDER CENTER STACK | | | | | | 805 | |
| **AUDI** | | | | 2015 | 2016 | 2017 | 2018 | 2019 |
| A1 | Under Center Stack | | | | | | | 805 |
| A3,S3 | Under Center Stack | | | 805 | | | | |
| A4, allroad | Tunnel between 1st/2nd row seats | | | 804 | | 500/813 | | |
| A5, S5, RS5 | Tunnel between 1st/2nd row seats | | | 804 | | | 500/813 | |
| A6, S6 | Tunnel between 1st/2nd row seats | | | 804 | | | 500/826 | |
| A7, S7, RS7 | Tunnel between 1st/2nd row seats | | | 804 | | | 500/826 | |
| A8, S8 | Tunnel between 1st/2nd row seats | | | 804 | | | 500/826 | |
| Q2 | Under Center Stack | | | | | | 805 | |
| Q3 | Under Center Stack | | | 804 | | | | 805 |
| Q5, SQ5 | Tunnel between 1st/2nd row seats | | | 804 | | | 500/813 | |
| Q7 | Under Center Stack | | | | | 500/813 | | |
| Q8 | Under Center Stack | | | | | | 500/826 | |
| R8 | Under Center Stack | | | | | 804 | | |
| TT, TTS | | | | not covered at this time | | | | |
| eTron | Under Center Stack | | | | | | | 500/826 |

The source data for this chart is the Bosch CDR help file for Version 17.10 Always consult the latest help file for the most accurate coverage

Chrysler Minivan Sister

No “back model” coverage

# VW/Audi Details

- Same data set for all event types
- Time from last precrash to AE noted
- Recorder holds up to 6 ND events in Chronological order.
- New modules will use Flex Ray
- Date/Time of Crash from clock (GPS if clock not set) in 2014+ Audi models, phasing in to VW models

# VW/Audi Details

- Modules live for 15 sec after power out
- Image time 1-5 minutes
- Pushbutton ignition – only 1 push to ACC
- Battery disconnect after serious crashes may force backpowering
- Must power ACM **and** gateway module

# TESLA



- Remember the definition of an EDR is something that records a time series of data **as a result of a crash**.
- Tesla may record information as a result of turning the key on, but that is not an event data recorder. More on this later.
- Tesla did not have any EDR in the ACM of its original 2008 Roadster models.
- Tesla has installed EDR in The 2012+ Model S, the 2015+ Model X, and the 2017 Model 3.
- Effective March 2018, Tesla announced they will sell a publicly available tool to download the Tesla EDR

# Tesla EDR

- Kit includes:
  - TIV-144:  In vehicle cable for Tesla Model S and X
  - TIV-145:  In vehicle cable for Tesla legacy Model S
  - TIV-996:  In vehicle cable for Tesla Model 3
  - TD2M-601:  Direct to Module cable for Tesla Model X and 3
  - TD2M-602:  Direct to Module cable for Tesla Model S
  - TD2M-139:  Direct to Module cable for Tesla Model S (NEW CABLE)
  - AC Power Adapter
  - PCAN-USB adapter
  - Hard Shell Case

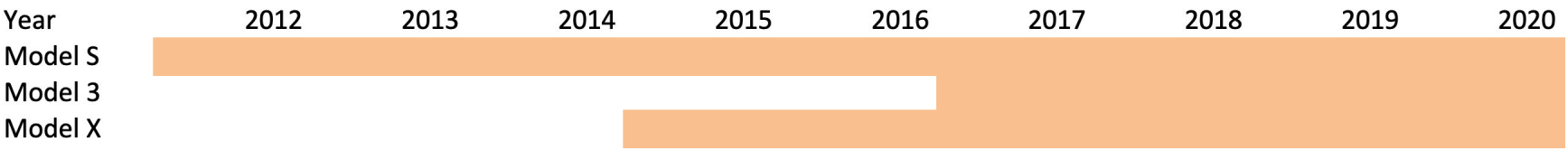


COST:  $1200.00

# Tesla EDR

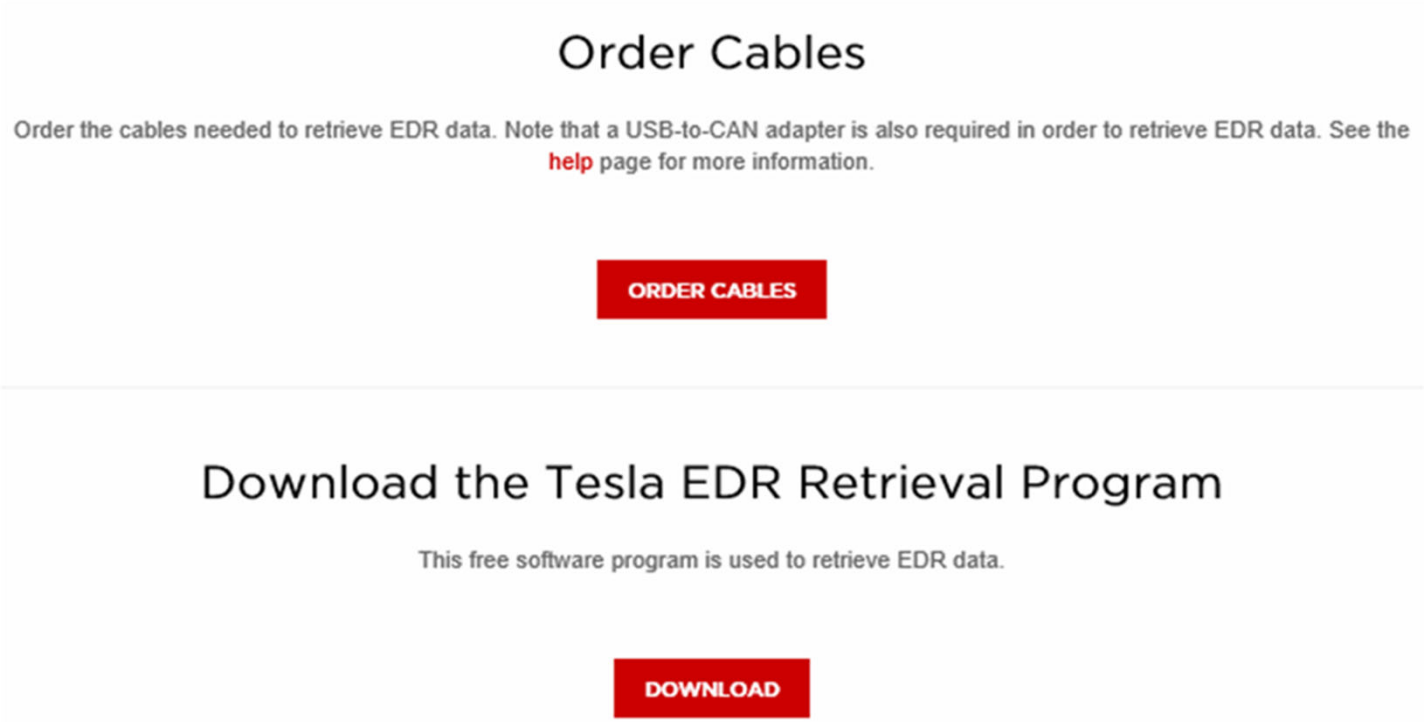
## Coverage

563



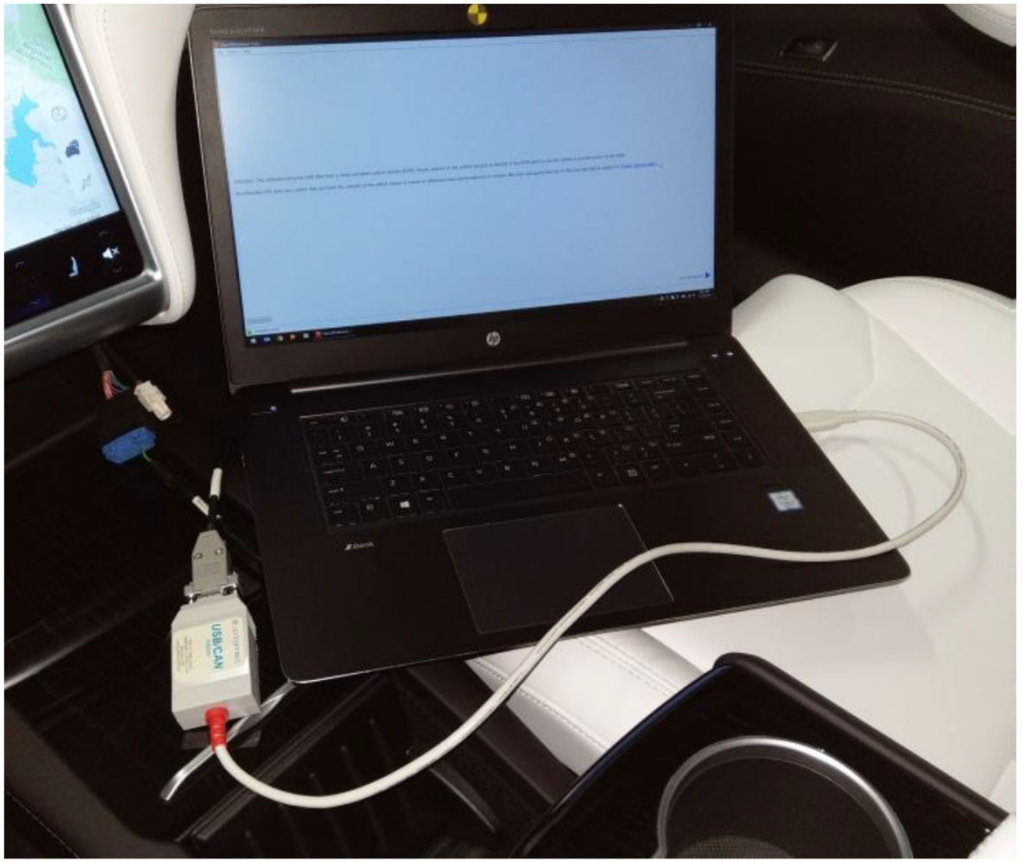
# TESLA

- More information is available on their website https://edr.tesla.com
- See “Other sources” later



# The Tesla EDR Tool

- Tesla provides the software free of charge on their website that installs on your laptop. No special computer is needed.
- The hardware kit sells for a list price of $1200.00.
- The equivalent of the Bosch CDR interface manual is the “PCAN-USB” adapter. It is very small, and reminds you more of a USB to serial adapter, just a slight bit larger.

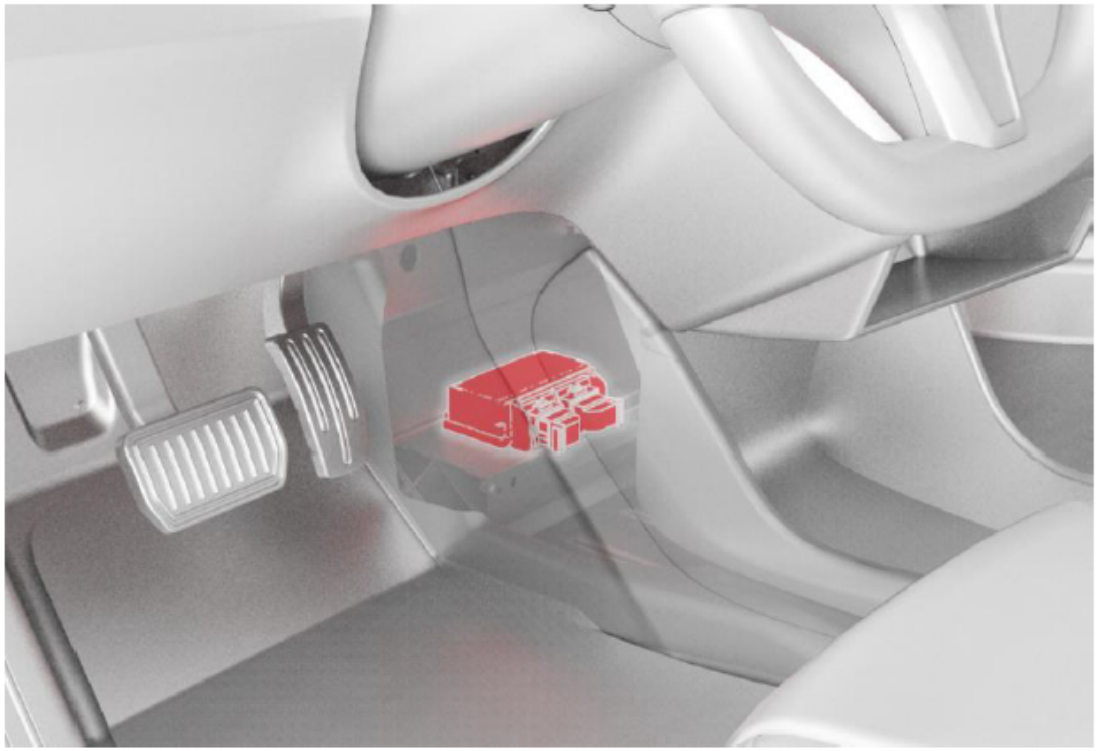


Plug in the PCAN adapter to the Body Harness. Model X and Model 3 are in other places & use a different connector described in Tesla's 27-page EDR guide at **edr.tesla.com/help**.



# TESLA also has 3 direct to module connectors

If the In-Vehicle Connection procedure is unsuccessful, you might be able to connect to the RCM directly, and establish a direct-to-module connection. The RCM is located forward of the center console, below the instrument panel and center screen.



Some 2018 Model 3 Cables must be replaced

# External Power must be used for DTM readouts

·. Connect 12V power to the Tesla Model S Direct-To-Module EDR Retrieval Harness.

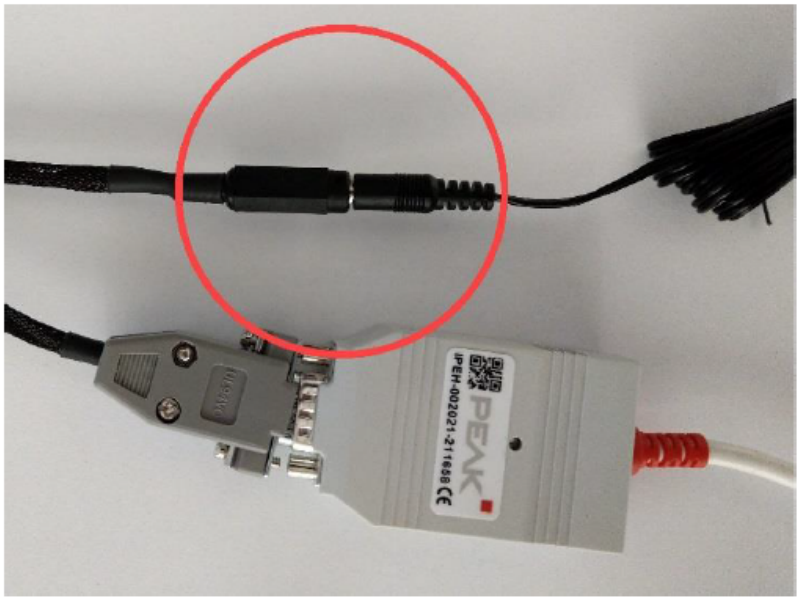



Figure 20. Connecting 12V power to the EDR retrieval harness

·. Proceed to “Retrieve Data” on page 26 to complete the data retrieval procedure.

# The Tesla 27-page Guide

- The Guide is VERY thorough and helpful
- It gives step by step instructions for the hardware assembly and to navigate the EDR program
- There are photographs of connector and module locations
- There are instructions for disassembly to get to the connectors and modules

# Tesla EDR

- Data varies by model

| | Model | | |
|---|---|---|---|
| Model | S | 3 | X |
| **Event Data** | | | |
| Max X Δv | X | X | X |
| Time to X Max ΔV | X | X | X |
| Max Y Δv | X | X | X |
| Time to Y Max ΔV | X | X | X |
| Time to Max Resultant Δv | X | X | X |
| Ignition cylcle Event | X | X | X |
| Ignition cycle runtime | X | X | X |
| Odomter at event | X | X | X |
| Driver Seat Track Posiition | X | X | X |
| Driver belt | X | X | X |
| Passenger Belt/Occupant | X | X | X |
| 2nd row Belt/Occupant | X | | X |
| 3rd row Belt/Occupant | | | X |
| File Complete | X | X | X |

# Tesla EDR

- Data varies by model

| | Model S | Model 3 | Model X |
|---|---|---|---|
| **PreCrash Data** | | | |
| Vehicle Speed | 2Hz | 5Hz | 2Hz |
| Accelerator Pedal | 2Hz | 5Hz | 2Hz |
| Rear Motor Speed | 2Hz | 5Hz | 2Hz |
| Service Brake | 2Hz | 2Hz | 2Hz |
| Stavility Control | 2Hz | 2Hz | 2Hz |
| ABS | 2Hz | 2Hz | 2Hz |
| Steering Wheel Angle | 2Hz | 10 Hz | 10 Hz |
| Lateral Acceleration | Not Reported | 10 Hz | 10 Hz |
| Longitudinal Acceleration | Not Reported | 10 Hz | 10 Hz |
| Roll Rate | Not Reported | 10 Hz | 10 Hz |
| Yaw Rate | Not Reported | 10 Hz | 10 Hz |

# Tesla EDR

- Data varies by model

| | Model S | Model 3 | Model X |
|---|---|---|---|
| **Crash Data** | | | |
| Longitudinal Δv | 300ms | 300ms | 250ms |
| Lateral Δv | 300ms | 300ms | 250ms |
| Longitudinal Acceleration | 500Hz (300ms) | 2000Hz (60ms) | 2000Hz (60ms) |
| Lateral Acceleration | 500Hz (300ms) | 2000Hz (60ms) | 2000Hz (60ms) |
| Normal Acceleration | Not Reported | Not Reported | 50Hz (1.4s) |
| Roll Angle | 100Hz (6s) | Not Reported | Not Reported |

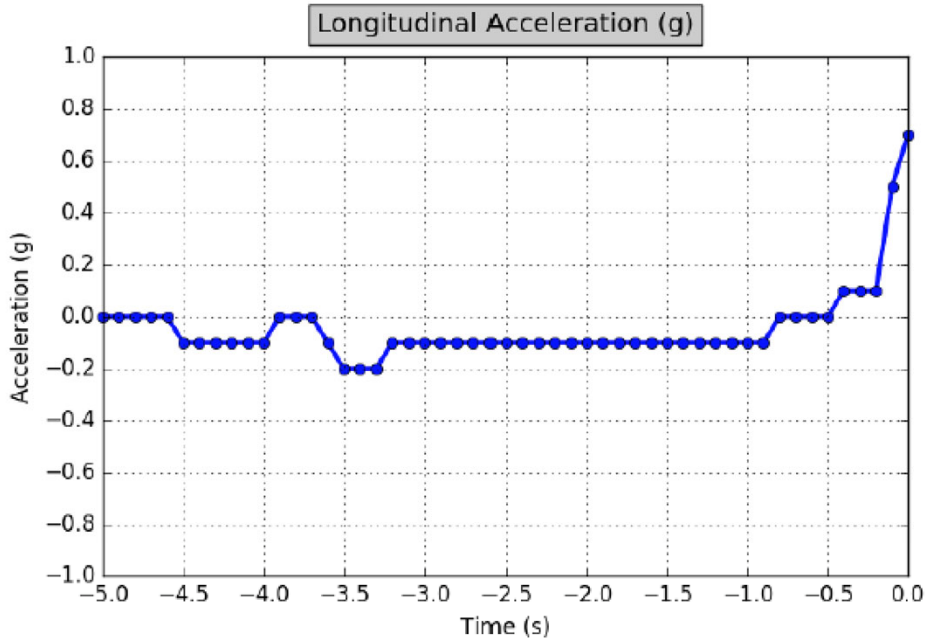
# Tesla EDR Sample File

- Model 3 Data

| Time (sec) | Service Brake | Stability Control | ABS Activity |
|---|---|---|---|
| -5.0 | Off | Off | Off |
| -4.5 | Off | Off | Off |
| -4.0 | Off | Off | Off |
| -3.5 | Off | Off | Off |
| -3.0 | Off | Off | Off |
| -2.5 | Off | Off | Off |
| -2.0 | Off | Off | Off |
| -1.5 | Off | Off | Off |
| -1.0 | Off | Off | Off |
| -0.5 | Off | Off | Off |
| 0.0 | On | Off | Off |
| -2.6 | 6.0 | 10.8 | 714 |
| -2.4 | 6.0 | 13.2 | 746 |
| -2.2 | 6.0 | 14.8 | 784 |
| -2.0 | 7.0 | 12.4 | 807 |
| -1.8 | 7.0 | 12.4 | 845 |
| -1.6 | 7.0 | 12.4 | 883 |
| -1.4 | 7.0 | 11.2 | 905 |
| -1.2 | 8.0 | 12.0 | 922 |
| -1.0 | 8.0 | 10.4 | 941 |
| -0.8 | 8.0 | 4.8 | 932 |
| -0.6 | 8.0 | 0.0 | 916 |
| -0.4 | 8.0 | 0.0 | 888 |
| -0.2 | 8.0 | 0.0 | 843 |
| 0.0 | 7.0 | 0.0 | 749 |

# Tesla EDR Sample File

- Model 3 Data




| Time (s) | Acceleration (g) | Time (s) | Acceleration (g) | Time (s) | Acceleration (g) |
|---|---|---|---|---|---|
| -5.0 | 0.0 | -3.2 | -0.1 | -1.4 | -0.1 |
| -4.9 | 0.0 | -3.1 | -0.1 | -1.3 | -0.1 |
| -4.8 | 0.0 | -3.0 | -0.1 | -1.2 | -0.1 |
| -4.7 | 0.0 | -2.9 | -0.1 | -1.1 | -0.1 |
| -4.6 | 0.0 | -2.8 | -0.1 | -1.0 | -0.1 |
| -4.5 | -0.1 | -2.7 | -0.1 | -0.9 | -0.1 |
| -4.4 | -0.1 | -2.6 | -0.1 | -0.8 | 0.0 |
| -4.3 | -0.1 | -2.5 | -0.1 | -0.7 | 0.0 |
| -4.2 | -0.1 | -2.4 | -0.1 | -0.6 | 0.0 |
| -4.1 | -0.1 | -2.3 | -0.1 | -0.5 | 0.0 |
| -4.0 | -0.1 | -2.2 | -0.1 | -0.4 | 0.1 |
| -3.9 | 0.0 | -2.1 | -0.1 | -0.3 | 0.1 |
| -3.8 | 0.0 | -2.0 | -0.1 | -0.2 | 0.1 |
| -3.7 | 0.0 | -1.9 | -0.1 | -0.1 | 0.5 |
| -3.6 | -0.1 | -1.8 | -0.1 | 0.0 | 0.7 |
| -3.5 | -0.2 | -1.7 | -0.1 | | |
| -3.4 | -0.2 | -1.6 | -0.1 | | |
| -3.3 | -0.2 | -1.5 | -0.1 | | |

# Troubleshooting

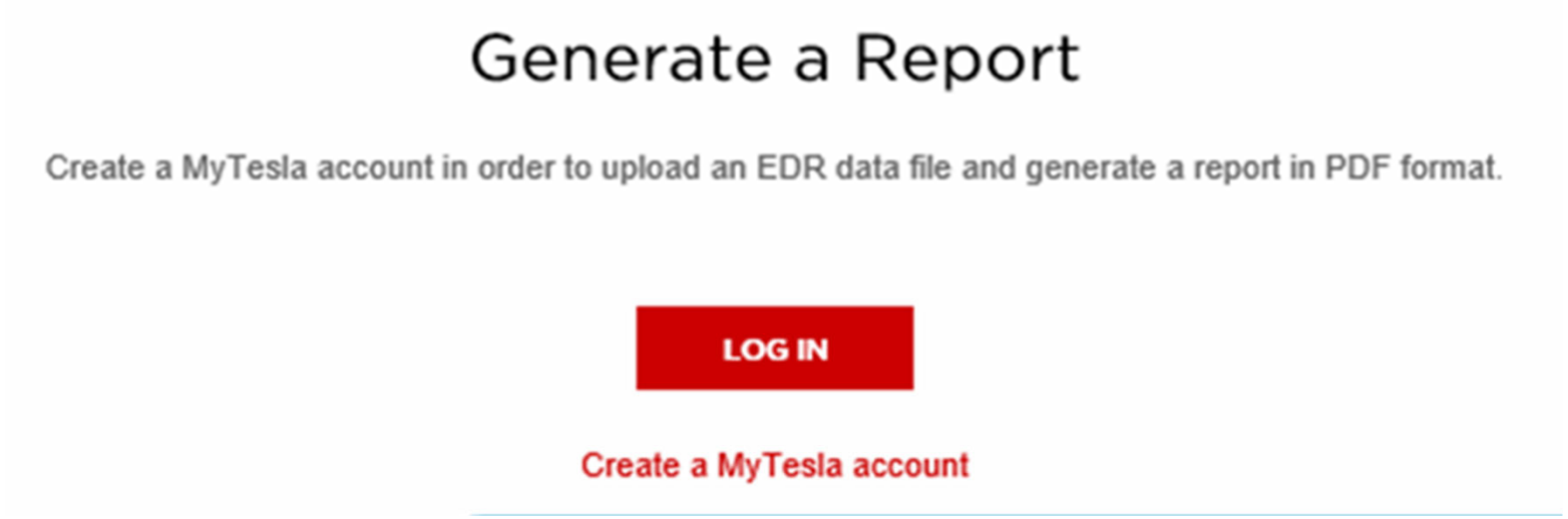
- Some users report a “time out” error when trying to read data. The vehicle is going back to sleep. **Putting your foot on the brake pedal** and keeping it there for the whole readout keeps the vehicle alive.
- Low battery power accentuates the time out problem. Tesla NEMA 5-15 adapter?





# Converting Raw Data to a Report

The Tesla software extracts a raw data file.  To convert the raw data to a report, you must upload the raw data to the Tesla Server. A report is immediately returned in the latest version of software.



# Worried about report conversion being biased by Tesla?

- Data is Encrypted.
- The report is returned IMMEDIATELY. There is no time for Tesla to alter any data to Tesla's benefit.
- However, some clients insist that no one has a copy of the data except them.
- There is a "Save without VIN sequence" option to de-identify data
- There is no alternative.

# What about OTHER data Tesla Data Records?

- Most of the time you will NOT NEED to access the other data Tesla records, the EDR should be sufficient. But if you need a longer time history, Tesla records MANY data parameters every one second plus, when anything electronically monitored happens. It gets uploaded to the cloud.
- It is NOT an "event" data recorder, it is a continuous recorder (does **NOT** meet Part 563 definition of a data recorder) (so **NO** PUBLICLY AVAILABLE tool to read it)
- Law Enforcement contact Tesla and demonstrate proper legal authority
- Tell them what you are interested in (they have so much you have to narrow it down)

# Other sources for Tesla equipment

- www.edrfinder.com has developed their own independent Tesla hardware kit. (They are affiliated with one of the Bosch CDR distributors for Europe, and are located in Germany
- It utilizes the same interface PCAN-USB adapter
- They have made an adapter from the PCAN adapter to a Bosch CDR DLC cable and the Bosch 783 direct-to-module cable
- They still have to make a special cable for the Delphi DTM cable
- Assuming you have a CDR system, this system is lower cost because it has fewer parts
- Introductory pricing is $695

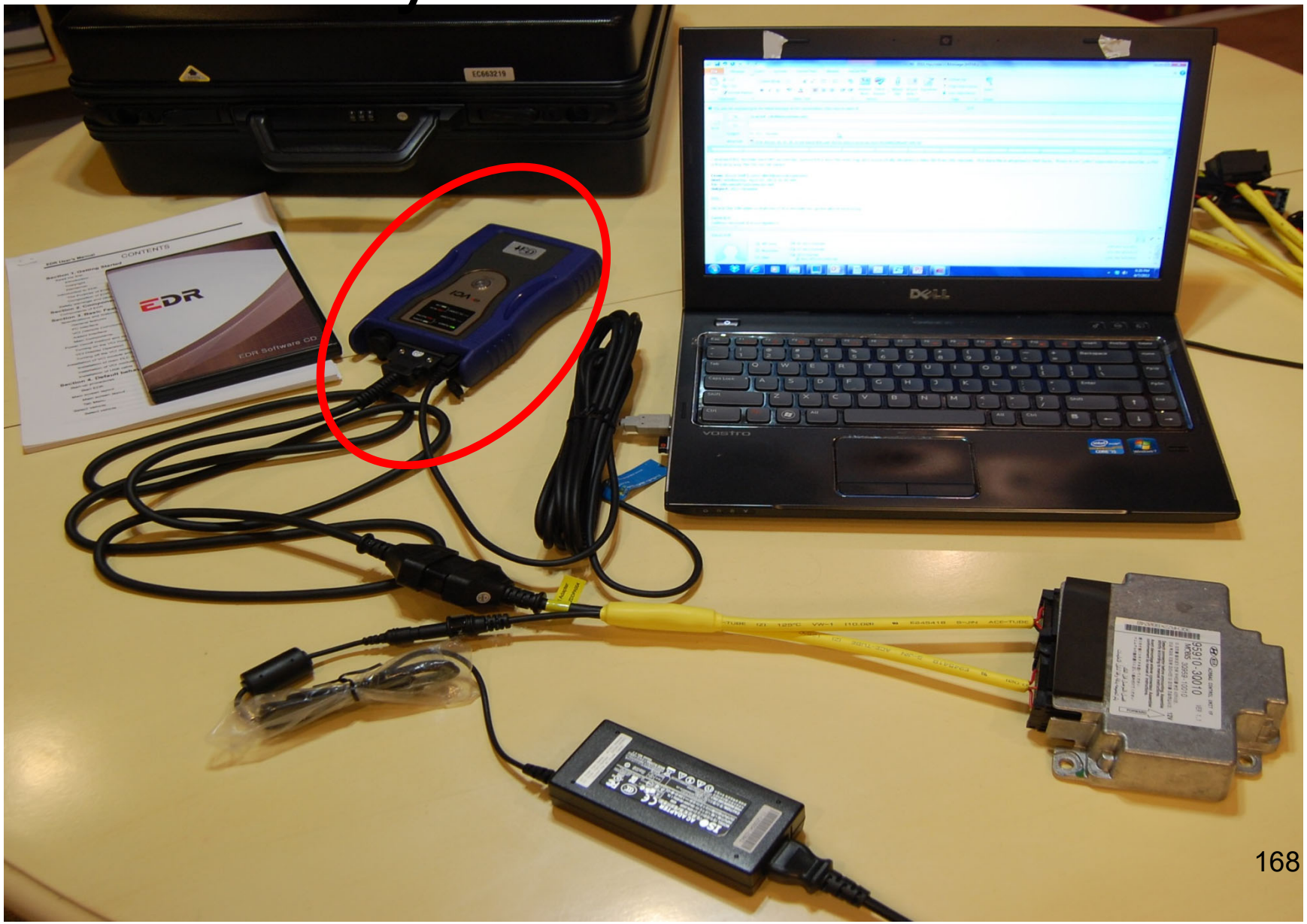
# Hyundai/Kia Tools

- There is no DLC only kit. Hyundai kit comes with 13 DTM cables thru #24, Kia with 15 DTM cables thru #23. 10 Cables are in both. New cables cost $250. #25-28 for 2019's not yet included.
- Hyundai: Call GIT America Tool Company (714)433-2180 order Part G0Z**H**DMN001 ***$6450 does not include software.***
- GIT now also sells Kia tool, (was Snap On),order part GIT0Z**K**DMN001 ***$6950 + tax+ship***
- Software updates **$495/yr** per brand from GIT Tool Company. Software V#31 Hyundia/ V#30 Kia released 4/12/19.
- Different version of tool for Korea and non-US markets.

# Additional Info

- K/H Interface modules are identical, but you must buy both to get software for both. Software updates won't download unless the correct interface module is attached to your laptop.
- I read K/H modules for others for $500. More people are buying Hyundai  kits now that we have 6+ model years on the road, only a few own Kia tool.

# The Hyundai EDR Reader



# Hyundai/Kia Tool Info (not new)

- The tool “officially” only reads 2013 and later models – software requires VIN input and locks out 2010-2012.
- But it communicates with several 2010-2012 models.
- However, your report will say you told the software it is a 2013.
- Admissibility must be presented skillfully (get help). **Data successfully admitted in Virginia, Florida, and California.**

# Kia/Hyundai GIT Tool 12/18

| Model | Vehicle Size, Descrip | Model Years 2010 | 2011 | 2012 | 2013* | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **KIA** | | Cable # | Cable # | Cable # | Cable # | Cable # | Cable # | Cable # | Cable # | Cable # | Cable # |
| CADENZA (VG, 2014) (Azera sister) | 112.0" WB | YG requires cable 13 per software | | | | 19(VG) | 19 | 19 | 13(YG) | 13 (YG) | 13 (YG) |
| FORTE (TD 2013, YD 2014)) | 104.3"WB, compact | 10 | 10 | 10 | 10 (TD) | 6/12 (YD) | 6/12 (YD) | 6/12 (YD) | 6/12 (YD) | 12(YDM) | 12(YDM) |
| K900 (KH 2015) | 119.9"WB | | | | | | 11 (KH) | 11 | 11 | 11 | 11 |
| NIRO | | | | | | | | | 21(DE HEV) | 21 | 21 |
| OPTIMA (2013 TF QF, 2016 JF, JFA) | 110.0" WB sedan | No | 4 | 4 | 4 (TF/QF) | 4 | 4 | 14 (JF,JFA)* | 14 | 14 | 14 |
| OPTIMA HYBRID | 110.0" WB sedan | No | 4 | 4 | (TF HEV) | 4 | 4 | 4 | 23(JFHEV) | 23 | 23 |
| RIO (UB) | 94.5"WB, compact | No | No | 8 | 8(UB) | 8 | 8 | 8 | 8 | 21 (SC) | 21 (SC) |
| RONDO (Canada only 2011 MY+) | | No | No | 9 | (not officially supported since non-US, read as a Kia Rio) | | | | | | |
| SEDONA (VQ '14, YP '15) | 113.8/118.9" WB miniv | No | No | No | No | 10(VQ) | 12(YP) | 12 | 12 | 12 | 12 |
| SORENTO (XM to 2015)(UMA 2016) | 106.3" WB, crossover | No | 3 | 3 | 3(XM) | 7(XM) | 7 | 13 (UMA) | 13 | 13 | 13 |
| SOUL (AM 2013, PS 2014+) | 100.4" WB | 1 see note 1 | 1 see note 1 | 1 | 1(AM) | 12(PS) | 12(PSEV) | 12 | 12 | 12 | 12 |
| SPORTAGE (SL) | 103.9" WB, crossover | No | Yes | 2 | 2(SL) | 2 | 2 | 2 | 13(QL) | 13 | 13 |
| STINGER | | | | | | | | | | 21 | 21 |
| **HYUNDAI** | | | | | | | | | | | |
| ACCENT (RB) | 94.5"WB, compact | No 10 | No? | 5 see note 2 | 5(RB) | 5 | 5 | 5 | 5 | 21(HC) | 21(HC) |
| AZERA (HG) | 109.4"WB, sedan | | likely | Yes | 7(HG) | 7 | 7 | 7 | 7 | 7 | 7 |
| ELANTRA Coupe(JK), Sedan(UD/MD) | 104.3"WB, compact | No | Yes | 8 | 8(JK/UD/MD) | 8 | 8 | 8 | 21(AD, ADA) | 21 | 21 |
| ELANTRA GT (GD) (aka "i30") | former Elantra Touring | old model | | new model | 1 | 6/12(GD) | 6/12(GD) | 6/12(GD) | 12 | 21(DT) | 21(DT) |
| EQUUS (VI)(stretch genesis) | 119.9" WB, luxury seda | ? | yes? | yes? | 9(VI) | 11(VI) | 11 | 11 | | | |
| GENESIS (Coupe BK, Sedan BH/DH)) | 115.6" WB, luxury seda | 9 shave keyway | 9 shave keyway | ? | 9(BK, BH) | 9 | 13(DH) | 13 | | | |
| GENESIS ***BRAND*** G80, G90 MODELS | | | | | | | | | 13(HI) | 13 | 13 |
| IONIQ EV and HEV, **2019 G70(IK)** | | | | | | | | | 21(AE) | 21 | 21 |
| KONA (New very small SUV) | | | | | | | | | | 24 (OS) | 27 (OS EV) |
| NEXO | | | | | | | | | | | 26 (FE) |
| SANTE FE (Sport DMA, NC) | 106.3" WB, crossover | 9 shave keyway note 3 | 9 shave keyway | 9 shave keyway | 7(NC,DMA) | 7 | 7 | 7 | 7 | 7 | 25 (TMA) |
| SONATA (YF sedan, YF HEV) | 110.0" WB sedan | | 4 | ? | 4(YF) | 4 | 14 (LF) 4 (HEV) | 14 (LF HEV) | 14 | 14 | 14 |
| TUCSON (LM to 2015)(TL in 2016) | 103.9" WB, crossover | Yes | Yes | 2 | 2(LM) | 2 | 2 | 13(TL) | 13 | 13 | 13 |
| VELOSTER coupe (FS) | 104.3"WB, compact | out of productior | | ? | 5(FS) | 5 | 5 | 5 | 5 | 5 | 14 (JS) |
| | | | | | | | | | | | 28 (JSN) |

Gold = 2013 Part 563 Compliant | Pink = Some data may be inacc | K/H Common Cables

**Yes** = Successful report of DLC readout, **No =** No data using current tool **?** = No information on whether model has EDR

# New GENESIS brand

- The Luxury brand of Hyundai
- G80 and G90 models. G70 new for 2019.
- Read with Hyundai data recorder tool

# Kia/Hyundai Tool Issues

- You input VIN, it prints what you entered in report. VIN is not stored in the ACM. Do sufficient documentation to prove your readout came from your vehicle.
- The K/H report is a PDF file. There is no raw data file. If K/H ever discovers an error and updates software, you will have to re-read the module. The software version at time of readout prints in the report.
- Recommended: ***Secure module*** as evidence

# K/H 2010-2012 data reliability?

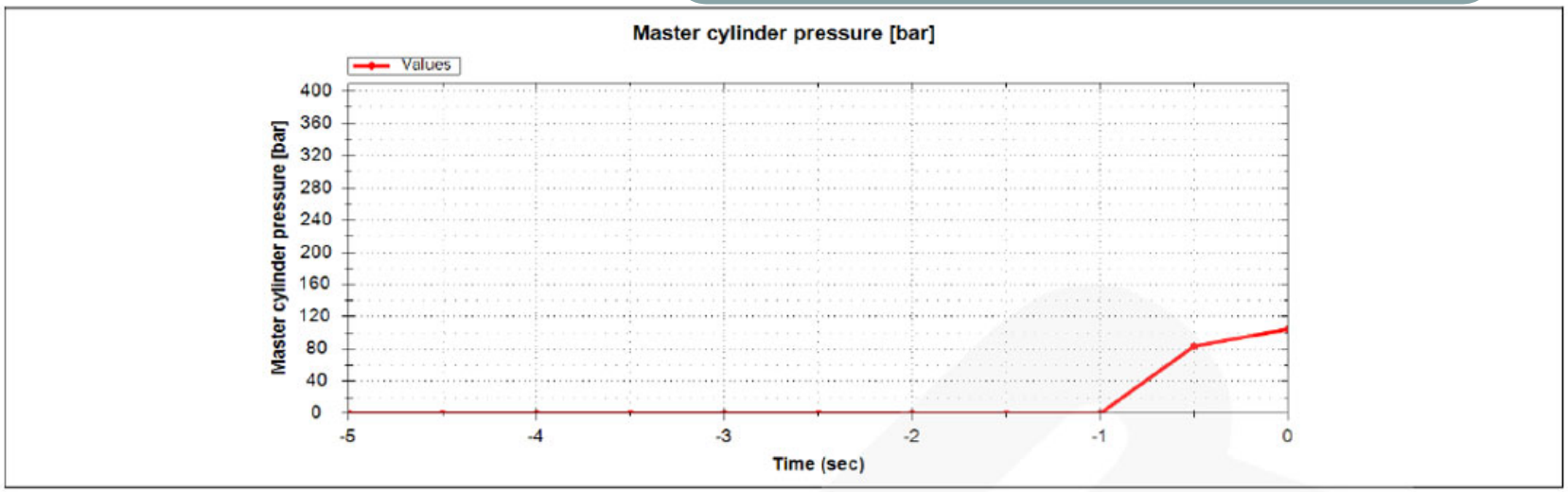
- In 50+ NHTSA crashes, speed, delta V and belt use appear OK (one exception)
- Steering angle,  and accel data appear implausible on some
- Key cycles at download not in all models
- SAE 2014-01-0502 has details.
- Additional testing on 2012 Soul reported in SAE 2015-01-1445.

# Brake Pres. added to 2019 Forte

‹ Event 1 ›

Master cylinder pressure

**Units in Bars. 1 Mpa=10 bars**
**Toyota max scale is 12 Mpa=120 bar**




| Num | Time (sec) | Master cylinder pressure [bar] |
|---|---|---|
| 1 | -5.0 | 0.2 |
| 2 | -4.5 | 0.2 |
| 3 | -4.0 | 0.2 |
| 4 | -3.5 | 0.2 |
| 5 | -3.0 | 0.2 |
| 6 | -2.5 | 0.2 |
| 7 | -2.0 | 0.2 |
| 8 | -1.5 | 0.2 |
| 9 | -1.0 | 0.2 |
| 10 | -0.5 | 83.7 |
| 11 | 0.0 | 104.6 |

# Other Non-Bosch CDR Tools

Law Enforcement: Ask Subaru dealer to help you with my tutorial

| | 2012 Nov YTD SALES | Nov YTD US Market SHARE | 2013 EDR? | Tool? |
|---|---|---|---|---|
| SUBARU (Bosch coverage starts) | 299,788 | 2.3% | YES | Denso DST-i |
| Dealer scan tool $2835 plus software $1835/mfr (905)669-4812 | | | | |
| New Denso tool from Nuspire required for 2016+, $6K, $2700 software/yr | | | | |
| MITSUBISHI (Bosch coverage starts) | 53,677 | 0.4% | YES | OTC tool co |
| Panasonic Touchpad + software $6395, annual $1200/yr, 888 727 6672 | | | | |
| JAG/LAND ROVER (Tata Motors India) | 49,452 | 0.4% | YES | OTC/Bosch SPX |
| Full dealer scan tool, Toughbook & software approx $5000 | | | | |
| SAAB | 480 | 0.0% | YES? | Hitachi HDS-3000? |
| Hitachi HDS-3000, software $1835 for Saab | | | | |

# Subaru EDR Availability

CDR900

Hitachi HDS-3000

Denso DSTi Interface

SSM3 Software

SSM4

CDR 19.1 on some

| Model / Year | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 |
|---|---|---|---|---|---|---|---|---|---|---|
| SUBARU | | <----Yellow=Old Hitachi SSM3 or new Denso w SSM4 | | | | Blue = SSM4 ONLY New Denso Tool | | | | |
| Ascent | | | | | | | | | CDR 835 | |
| BRZ (Scion FRS sister) | | Japan | Yes SSM3 | Yes | Yes | Yes | SSM4 | Yes | Yes | |
| | | | For BRZ you can use CDR - select Toyota as Brand but enter Subaru VIN - it will work | | | | | | | |
| Impreza | No | Yes SSM3 | Yes | Yes | Yes | SSM4 | Yes | Yes | Yes | |
| WRX/STI | No | No | No | No | Yes SSM3 | SSM4 | Yes | Yes | Yes | |
| Legacy | No | No | Yes SSM3 | Yes | CDR 836 Bosch ECU | | | | | |
| Outback | No | No | Yes SSM3 | Yes | CDR 836 Bosch ECU | | | | | |
| Forester | No | No | Japan | Yes SSM3 | Yes | Yes | SSM4 | Yes | CDR 835 | |
| Tribeca | No | No | No | No | Discontinued | | | | | |
| Crosstrek | n/a | n/a | Yes SSM3 | Yes | Yes | SSM4 | Yes | Yes | Yes | |

NOTE: The Subaru tool is a dealership fault code tester made to work in thru the DLC in vehicles with functioning electrical systems only.

Instructors may be able to fabricate a direct to module cable from a vehicle side connector



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# Subaru EDR Tools

- Bosch CDR 900 reads 2015+ Legacy/Outback and 2019+Forester/Ascender. Others expected to be covered in 2020.
- 2016+ Subaru went to a new Denso DSTi diagnostic interface with SSM**4** software from Nuspire. Tool + 1st year software sub now $5.8K/yr, is compatible to ‘04. Software sub $2700/yr.
- Denso is reverse compatible with 2012-2015 Models but uses the older SSM3 software.
- SSM3 allows data to be saved as a CSV file. There is no PDF report and no data limitations

# Subaru

- The Dealer tool (DSTi) is sold as DLC only.  The electrical system must be working, or you must backpower both ACM and DLC fuses, or put ACM in surrogate vehicle.
- Tutorial on Subaru available my website and your reference thumb drive
- 2016+ SSM4 software does not have a PDF or CSV output – must be viewed in software (Requires MANY screen shots to document readout – or a good screen video capture software on laptop)

# Subaru DTM Cables Available

**ATTENTION:**
GENERAL MANAGER ❑
PARTS MANAGER ❑
CLAIMS PERSONNEL ❑
SERVICE MANAGER ❑

IMPORTANT - All Service Personnel Should Read and Initial in the boxes provided, right.

SUBARU
QUALITY DRIVEN® SERVICE

## SERVICE BULLETIN

**APPLICABILITY:** 2013MY and Later Legacy and Outback
2014MY and Later Forester
2012MY and Later Impreza 2.0L NA
2013MY and Later XV Crosstrek
2013MY and Later BRZ
2015MY and Later WRX and WRX STI

**NUMBER:** 17-17-13R
**DATE:** 03/18/13
**REVISED:** 12/15/17

**SUBJECT:** Event Data Recorder (EDR) Function of Airbag Module

**INTRODUCTION**

**NOTE:** The Airbag systems used on previous and current WRX, STI and Tribeca models are NOT equipped with EDR functionality.

CDR cable 835 for 2019 Ascender & 836 for 2015+ Outback now available

Test harnesses required to access EDR data from the airbag control module along with current pricing are available from SPX / Kent-Moore (1-800-345-2233). Use the new harness applicability table below.

| Description | Part Number | Applicability |
|---|---|---|
| EDR HARNESS | 98299FJ040 | 2013-14MY Legacy / Outback |
| | | 2012MY and Later Impreza |
| | | 2013MY and Later Crosstrek |
| | | 2015MY and Later WRX and STI |
| | | 2013MY and Later BRZ |
| | 98299AL040 | 2015MY and Later Legacy / Outback |

2 in 1 2013 cable $1060, new 2015 Outback cable is $3K

# Subaru BR-Z

- Is a sister/clone of a Scion FS-R and Toyota 86
- Data is same as other Toyotas but Subaru presents in different format in Subaru SSM
- Data is very different than other Subaru's
- **Bosch CDR accepts Subaru BRZ VIN under Toyota brand.**
- **Bosch CDR used on a 2017 BRZ gave steering data, the official Subaru tool did not!**

# MITSUBISHI 0.5% market share

- Bosch CDR now covers 2011+ Outlander Sport and 2014+ Outlander.
- Pre-Bosch Mitsubishi EDR only tool based on dealership a Panasonic Touchpad. Cost to $6380.
- The kit is made to work only on vehicles with a functioning electrical system thru the **DLC**.
- Some Mitsubishi have an **"ETACS" gateway module (interior fuse box)**that must be included in circuit to make it work DTM (need to build a DTM circuit/cable).
- Some incomplete files retrieved (no data).
- Bug in V33 of software prevented converting .csv file into report format – now fixed

# Mitsubishi EDR's
# Some use CDR, some use private tool

| Model | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 |
|---|---|---|---|---|---|---|---|---|---|
| Eclipse (PS Platform) | | Production ends 8/2011 | | | | | | | |
| Eclipse Spyder Conv. | | | | | | | | | |
| Endeavor SUV (PS Platform) | | Production ends 8/2011 replaced by Outlander | | | | | | | |
| Galant (PS Platform) | | | Production ends 8/2012 | | | | | | |
| iMiEV | | | **No gateway module** | | | | | ? | |
| Lancer | Part 288 11-12 read w CDR as Caliber | | **ETACS gateway. Part A307-308 thru 2015** | | | | | Production Ends | |
| Lancer Evolution | | | | | | | | | |
| Lancer Sportback | | | | | | | | | |
| Mirage (subcompact) | | | **No gateway module** | | | | | | |
| Outlander SUV (GF platform) | A208/A304 mix. A208 5 sec@0.1. Deploy only. Read as 09 Caliber. | | **ETACS g** | **838 Bosch CDR Supported** | | | | | |
| Outlander Sport, RVR, ASX | **838 Bosch CDR Supporte** | | | | | | | | |
| Raider pickup (Dodge Dakota sister) | Not in production | | | | | | | | |

# Typical Mitsubishi CDR Data

**Pre-Crash Data -5 to 0 Sec (Record 1)**

| Time (sec) | Speed, Vehicle Indicated (MPH [km/h]) | Accelerator Pedal, % Full (%) | Service Brake, On/Off | Engine RPM (RPM) | Steering Input (deg) |
|---|---|---|---|---|---|
| -5.0 | 3.1 [5] | 0 | On | 797 | 4.5 |
| -4.5 | 2.5 [4] | 0 | On | 751 | 4.5 |
| -4.0 | 1.9 [3] | 0 | On | 648 | 4.0 |
| -3.5 | 1.2 [2] | 0 | On | 593 | 4.0 |
| -3.0 | 0.6 [1] | 0 | On | 641 | 4.0 |
| -2.5 | 0.6 [1] | 60 | Off | 719 | 4.5 |
| -2.0 | 1.2 [2] | 71 | Off | 1,297 | 14.0 |
| -1.5 | 4.3 [7] | 75 | Off | 1,840 | 46.5 |
| -1.0 | 7.5 [12] | 77 | Off | 2,010 | 90.0 |
| -0.5 | 9.9 [16] | 76 | Off | 2,180 | 124.5 |
| 0.0 | 11.2 [18] | 72 | Off | 2,206 | 146.0 |

- +Long & Lat Delta V and Acceleration
- +Roll Angle -1 to +5 sec @ 0.1 sec interval

# Jaguar/Land Rover

- Jag/LR have EDR's, they are read with a dedicated laptop and software. Tool only retrieves hex data, must be sent to England for interp.  NOBODY HAS TOOL.
- Jag/LR have 2 engineers in New Jersey HQ as EDR contacts.
- **They are now referring requesters to <u>suppliers.</u>** (call to find out who supplier is)
- Some Mid 2000's have ***PCM data*** (when Jag/LR was owned by Ford) but *not spoofable* with CDR.

# Newly published research

- Every year at the SAE World Congress in April new research is presented.
- A very brief summary follows, I encourage you to read these papers.
- Other good sources of EDR info include Collision Magazine and Accident Reconstruction Journal.

# EDR SAE Papers 2019

**2019-01-0633 Performance of Event Data Recorders Found in Toyota Airbag Control Modules in High Severity Frontal Oblique Offset Crash Tests** by William Bortles, Kineticorp et al (finds Delta V is understated due to offset).

**2019-01-0634 Event Data Recorder Performance during High Speed Yaw Testing Subsequent to a Simulated Tire Tread Separation Event** by William Bortles et al

(finds tires balloon after tread sep causing speed under rerporting)

# EDR SAE Papers 2019

**2019-01-0635 Laboratory Experiments Using a 2007 Toyota Auris Event Data Recorder and Additional Data from CAN Bus by Miguel Tabone et al, University of Malta (took electrical system out of car into lab, injected ABS wheel speed sensor signals to simulate moving vehicle as events were set by hammer blows)**

**2019-01-0632 Reconstructing Vehicle Dynamics from On-Board Event Data by Brandon Tsuge et al, MEA Forensic (read out Vehicle Control History, discovered VCS has data 6.6 times per second in ABS events vs 2 times in hard accel events)**

# EDR SAE Papers 2019

**2019-01-0416 A Study of the Performance of Automatic Emergency Braking (AEB) Systems Equipped on Passenger Vehicles for Model Years 2013 to 2018**

**By Djordje Miholjcic et al, 30 Forensic Engineering (good summary)**

**2019-01-0412 Brake Vacuum Booster Characterization by James Walker, Carr Engineering et al**

**Memo Heavy Truck Papers**

**2019-01-0636** Evaluation of the Heavy Vehicle Event Data Recorder for the Freightliner New Cascadia with Detroit Diesel Engines by David Plant, D P Plant & Associates et al (found numerous anomalies in time base)

# Regulatory Update

- NHTSA issued NPRM in Dec 2012 to **require** 563-EDR's by Sept 2014 (past!)
- NPRM also proposed EDR be made an FMVSS 405 (Safety Standard), OEM's would have to self certify EDR's. Strong OEM objections were filed.
- Officially withdrawn Feb 2019 (Fed Reg'tr)

https://www.transportation.gov/regulations/significant-rulemaking-report-archive (click on "November 2018", scroll to #53

- Mfrs are NOT expected to remove EDR's

# Regulatory Update

- NHTSA/IIHS got 10 auto mfrs to agree on 9/11/2015 to making **A**uto **E**mergency **B**raking **(AEB)** standard (timeline 2022). Voluntary commitment is a relatively new strategy by NHTSA. Now 20 have agreed.
- Next big issue is Autonomous Vehicles (self driving cars).
- States are drafting conflicting regulations on type of certification required – could hinder development.

# EDR admissibility update

- "Every year EDR evidence is admitted at the trial level in hundreds of cases and we never hear about them." - Prosecutor Bill Melkonian at the 2014 CDR User's Conference Legal Update. **No change.**
- Warrants now required, admissibility shifts to Probable Cause for getting that warrant.
- Smaller battles on 2010-2012 Kia/Hyundai, Nissan 2006-2012, unsupported Ford PCM's.
- Delaware Daubert hearing accepted EDR!
- See new Ruth presentation for prosecutors

# EDR read before Privacy Act?

- Florida vs Worsham recent 2016 decision holds that warrant was still required before Driver Privacy Act. Court likened EDR to cell phones or tablets (we disagree). Went to appellate level.
- Worsham circumstances were unusual, police read module without warrant then applied for one. Judge denied the warrant since readout was already complete.

# Crash Data Retrieval Update Oct 2019




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